

DEVELOPMENT OF A COUNT PERFORMANCE EVALUATION PROCEDURE FOR ON-LINE PARTICLE COUNTERS USED IN DRINKING WATER TREATMENT

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FORWARD

The AWWA Research Foundation is a nonprofit corporation that is dedicated to the implementation of a research effort to help utilities respond to regulatory requirements and traditional high-priority concerns of the industry. The research agenda is developed through a process of consultation with subscribers and drinking water professionals. Under the umbrella of the Strategic Research Plan, the Research Advisory Council prioritizes the suggested projects based upon current and future needs, applicability, and past work; the recommendations are forwarded to the Board of Trustees for final selection. The foundation also sponsors research projects through the unsolicited proposal process; the Collaborative Research, Research Application, and Tailored Collaboration programs; and various joint research efforts with organizations such as the U.S. Environmental Protection Agency, the U.S. Bureau of Reclamation, and the Association of California Water Agencies.

This publication is a result of one of these sponsored studies, and it is hoped that its findings will be applied in communities throughout the world. The following report serves not only as a means of communicating the results of the water industry's centralized research program but also as a tool to enlist the further support of the nonmember utilities and individuals.

Projects are managed closely from their inception to the final report by the Foundation's staff and large cadre of volunteers who willingly contribute their time and expertise. The foundation serves a planning and management function and awards contracts to other institutions such as water utilities, universities, and engineering firms. The funding for this research effort comes primarily from the Subscription Program, through which water utilities subscribe to the research program and make an annual payment proportionate to the volume of water they deliver and consultants and manufacturers subscribe based on their annual billings. The program offers a cost-effective and fair method for funding research in the public interest.

A broad spectrum of water supply issues is addressed by the Foundation's research agenda: resources, treatment and operation, distribution and storage, water quality and analysis, toxicology, economics, and management. The ultimate purpose of the coordinated effort is to assist water suppliers to provide the highest possible quality of water economically and reliably. The true benefits are realized when the results are implemented at the utility level. The foundation trustees are pleased to offer this publication as a contribution toward that end. Particle counters have already become an integral and, even essential, part of treatment optimization and

regular process monitoring programs for many utilities and more utilities are considering purchasing this instrumentation. These utilities range from small installations producing less than 20 MLD (5 mgd) and serving populations of less than 10,000 to very large utilities producing more than 1,000 MLD (270 mgd) and serving millions of people.

Research sponsored by AWWARF has shown that number concentration (count) measurements made by particle counters of different makes and models frequently do not agree and count measurements made with particle counters do not agree with counts measured by referee methods such as scanning electron microscopes with computerized image analysis and by electro sensing zone (“Coulter Counter”) instruments.

This study developed a practical method for testing the count performance of on-line particle counters. The method uses aqueous suspensions of a polydispersed dust. This standard dust is characterized and sold by the National Institute of Standards and Technology to support count verification methods used in the fluid power industry. The results covered by this report increase our understanding of particle counting in water utilities and the value of this already important measurement.

George W. Johnstone
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The field study work could not have been done without the assistance of personnel from the Onondaga County Water Authority (Syracuse, New York), including Anthony Geiss, (Deputy Administrator for Operations), and Mark Murphy (Plant Superintendent) and Bob Rossin (Chief Chemist) of the Otisco Lake Water Treatment Plant.

Chris E. Johnson, Associate Professor of Civil and Environmental Engineering at Syracuse University, generously helped us develop the statistical analysis plan.

EXECUTIVE SUMMARY

RESEARCH OBJECTIVES

The principal objective of this study was to select materials and develop a procedure for the count performance evaluation of on-line particle counters. The count performance evaluation (CPE) procedure that was developed can be used for several purposes including testing the agreement between the particle size distribution measured with the particle counter and the size distribution measured with a reference method such as visible light or scanning electron microscopy. It can also be used to establish if two or more particle counters will be in acceptable agreement when they are counting a real filtered water suspension and to determine if this agreement is constant with time. A relatively stable and reproducible test suspension made with particles that resemble, to some extent, the particles in the filtered water suspension is a requisite part of the CPE procedure. Part of the study involved finding and testing a well-characterized particle that can be used to prepare test suspensions for the CPE procedure.

RESEARCH APPROACH

The study determined the best type of suspension for count performance evaluation by first testing the count and size performance of five counters made by four different manufacturers. This testing, called the instrument performance analysis (IPA) in this report, was done using two types of suspensions: near mono-disperse polystyrene latex (PSL) suspensions and poly-disperse National Institute of Standards and Technology (NIST) ISO medium test dust. The IPA tests were not conducted to determine which counters were better or more efficient in counting particles but simply to understand the characteristics that a suspension must have to be useful for count performance evaluation. Based on the IPA results, a count performance evaluation procedure that uses suspensions of NIST ISO medium test dust fed to the on-line counters using a gear pump system was developed and tested in the laboratory and in the field at a full-scale water treatment plant.

RESEARCH FINDINGS

The results of the IPA part of this study are in agreement with the observation that light obscuration particle counters do not count all the particles that pass through the sensor. This has been known for a significant period of time and has been described in a number of reports and other publications (See Chapter 1). According to our results and the literature, undercounting is most significant at smaller particles sizes ($<5\ \mu\text{m}$), and has been observed with different types of particles including polystyrene latex micro-spheres, mineral dusts and particles in filtered drinking water. Undercounting is not caused exclusively by poor or inappropriate calibration, or poor resolution, or any other single instrumental parameter; the particle counters simply do not register a “count” for each and every particle of theoretically measurable size that passes through the light beam in the sensor. Because of this, it is not meaningful or useful to calibrate a light obscuration particle counters using the basis of count. One can only use an appropriate “standard” suspension and compare the instrument measured size distribution with the size distribution measured by a reference method such as scanning electron microscopy (or some other method that is acceptable to the standard setters).

Size calibration of a particle-counting instrument with particles that are appropriate for this purpose (such as PSL micro-spheres) is useful because it gives the millivolt thresholds of the counting electronics at least approximate physical meaning. Instead of stating that all the particles were larger than the 25 millivolt threshold one can say that all the particles counted were larger than an equivalent sphere diameter of $5.5\ \mu\text{m}$ based on calibration with PSL micro-spheres. For most users the $5.5\text{-}\mu\text{m}$ threshold setting label, even though it can be difficult to interpret when counting non-PSL particles, is better than the 25 millivolt label. In this study there was no evidence that polystyrene latex suspensions are inappropriate for the size calibration of counters. Size calibration verification experiments showed that measured diameters with different PSL suspensions were within $\pm 10\%$ of the manufacturer’s certified diameters.

Counting Mono-disperse Polystyrene Latex Micro-spheres

Mono-disperse PSL suspensions are widely used for the size calibration of on-line particle counters (ASTM Method F658 – 87). Five size-certified PSL suspensions were tested and the measured size distributions and particle number concentrations were compared with

expected values derived from measurements and other information in the manufacturer's literature. The results indicate that the counters do not measure all the PSL particles in mono-disperse suspensions and the efficiency of counting varies with particle size and from instrument to instrument. (This should not, however, adversely affect the use of PSL particles for the size calibration of counters.)

Our experiments with the on-line particle counters using PSL micro-spheres showed that the lowest count efficiencies (less than 77%) were measured with the smallest mean diameter particles, 3 μm . Instruments A and B, with relatively low resolution ($R > 10\%$), had higher average count efficiencies (88 – 108%) and instruments C and D with relatively high resolution ($R < 10\%$) had lower average count efficiencies (66 – 76%). Model system calculations with Gaussian (near mono-disperse) particle size distributions, which took into effect the sensor resolution and an error of 10 % associated with the threshold setting, predicted count efficiencies of $100 \pm 2 \%$ with essentially all the PSL suspensions. From this it was concluded that the low count efficiencies measured with the PSL suspensions were not due to contributions from sensor resolution and the error associated with threshold factors but simply to the inability of the instrument to detect and/or count all the particles that passed through the sensor.

Counting Particles of NIST ISO Medium Test Dust

The performance of the on-line instruments was also analyzed using NIST's ISO medium test dust. This powder is a reference material supplied by NIST in Gaithersburg, MD [Reference Material (RM) 8631]. It has been characterized by NIST using scanning electron microscopy and image analysis (after filtration from hydraulic fluid) and the fluid power industry uses it suspended in hydraulic oil as a primary count calibration standard [ISO 11171:1999]. The suspension in hydraulic fluid is sold by NIST as a Standard Reference Material, SRM 2806.

Using suspensions of NIST ISO medium test dust in low particle water and a threshold setting of 2 μm (based on size calibration with PSL microspheres), all the counters gave count efficiencies of less than 50 %. Model system calculations with poly-disperse suspensions (with size distributions that resemble the test dust) included the effect of sensor resolution at the threshold and assumed a threshold setting error of 10 %, predicted count efficiencies of $100 \pm 20 \%$. It was concluded that sensor resolution and errors associated with the threshold setting do not give a complete explanation of the low count efficiencies (less than 50 %), which characterized

all the counters. These results have important implications for the application of particle counting in drinking water treatment and point to the need for a count performance evaluation suspension and procedure.

According to the results of this study the count performance of on-line particle counters should always be determined with a well-characterized poly-disperse suspension that has particles with a size distribution similar to that of the particles that will be measured in the water treatment application of the particle counter. A mono-disperse suspension such as PSL micro-spheres is not suited to this purpose. The model system calculations for these suspensions showed that, if the threshold is set well below the mean diameter of the particles, the ability of the counter to detect each particle will be the only significant factor; differences in sensor resolution and errors associated with threshold settings will not influence the count measurements. If PSL micro-spheres were used to evaluate count performance, instruments with equal abilities to detect particles but with different resolutions and/or threshold setting errors would tend to give the same count results, but because of their resolution and threshold error differences they could give very different count results when the particles in the suspension were poly-disperse. On the other hand, when a poly-disperse suspension is used to evaluate count performance, sensor resolution, errors associated with threshold settings and the ability to detect the particles, will all tend to have an effect on the measured count performance and all the relevant inter-instrument differences will be revealed.

The size distribution for NIST ISO medium test dust obtained from a HIAC-Royco counter is similar to filtered water particle size distributions measured at treatment plants from around the country. NIST dust, therefore, seems to be a reasonable alternative for the count performance evaluation of particle counters used to monitor filtered water quality.

The Count Performance Evaluation Protocol

The count performance evaluation (CPE) protocol developed in this study includes four essential parts; preparing an initial stock suspension of NIST ISO medium test dust, diluting this suspension to make working suspensions that have a dust concentration that is appropriate for counter evaluation, feeding the working suspension to the counter and collecting and analyzing the data. The statistical analysis of the data (by analysis of variance, ANOVA) gives information about inter-instrument agreement and trends in agreement with time.

In the CPE protocol a 20-gram sample of ISO medium test dust is purchased from NIST. The dust is shipped with NIST's measured particle size distribution. The entire 20 grams is divided into approximately 310 mg portions using a micro-riffler apparatus to minimize segregation by particle size. Each ~310 mg portion is carefully weighed and stored in a water-soluble cellulose gelcap. Stock suspensions of dust are prepared by combining 100 mL of low-particle water with the gelcap and dust in a 120 mL plastic container. Measurements suggest that the stock suspension can be stored for at least 90 days. A gravimetric procedure was tested for verifying the stock suspension dust concentration. The stock suspension is shaken and ultrasonicated before portions are withdrawn to prepare working suspensions for particle counter testing. All suspensions are prepared with low-particle water from a laboratory reverse osmosis unit.

The working suspensions are prepared using an adjustable-volume microdispenser to add 2 mL of stock suspension to 20 L of low particle water in a polyethylene container. The dust concentration in this suspension is approximately 0.3 mg/L and the approximate particle count is $1000 > 2 \mu\text{m/mL}$. Tests suggest that a coefficient of variance of 5 % or less for 5 replicate count measurements can be consistently achieved if the working suspension count is greater than about $500 > 2 \mu\text{m/mL}$.

Each of the 5 replicate measurements in a particle counter test is made using one 2 L aliquot of working suspension. The working suspension in the 20 L container is continuously mixed at low speed with a mechanical mixer while the 2 L aliquots are withdrawn; the volume of each is measured with a graduated cylinder. The 2 L aliquots are stored (for a short time) in 2 L HDPE containers. Before a particle count measurement is made each 2 L quantity of working suspension is inverted several times to distribute the suspension. Mechanical mixing is not recommended for this step.

The working suspension is drawn through the sensor of the particle counter (after it has been disconnected from its flow control device) using a gear pump. The gear pump is installed after the sensor. The measured particle concentration in counts per mL (the average of 5 to 15, 1 – 2 minute measurements by the counter) is divided by the concentration (in $\mu\text{g/mL}$) of dust in the working suspension. The final result of each measurement is “counts $> X \mu\text{m}/\mu\text{g}$ ”, where X is the instrument's threshold setting, e.g., $2 \mu\text{m}$. For each test the 5 replicate count measurements are used to compute an average and standard deviation. Statistical tests including analysis of

variance (ANOVA) and a *post hoc* analysis are then used to determine, for example, if the different instruments in a comparison group are giving comparable test results or if measured counts are varying with time.

Chapter 5 presents the experimental results used to make the choices that led to the steps proposed for the CPE protocol. The protocol was tested at the Otisco Lake plant of the Onondaga County Water Authority and these results are discussed at the end of Chapter 5.

RECOMMENDATIONS FOR ADDITIONAL WORK

The following are recommendations for additional work on the development and application of the count performance evaluation protocol and the NIST ISO medium test dust suspension upon which the protocol depends:

1. Collaborate with the National Institute of Standards and Technology (NIST) on a project to develop an ISO medium test dust reference material specifically for the drinking water industry. The use of cellulosic gelatin capsules (gel-caps) for containing the riffled and weighed portions of dry dust should be considered and evaluated. The experiments of this project suggest that gel-caps are a useful option. The co-principal investigators met with NIST personnel at their laboratories in January 1999 and started discussions on the feasibility of this collaboration. The NIST researchers expressed strong interest. The minutes of this meeting are presented in Appendix G.

Rationale: NIST prepared a detailed and accurate characterization of ISO medium test dust for the fluid power industry. However, the dust suspensions were prepared in a standard hydraulic fluid. The slides used to measure the particle size distribution with a scanning electron microscope and image analysis system were prepared by filtering the particles onto a membrane filter and removing residual hydraulic fluid with solvents. For the drinking water industry the slides would be prepared using dust in water suspensions. The gel-cap method seems to facilitate the storage of dust samples and the preparation of suspensions but more work needs to be done on factors such as storage time and the type of capsule.

2. It is possible that particle counter manufacturers will modify or redesign their instruments and the “instrument response” factor detection observed inefficiency issue of this and other studies will become insignificant. In this case, it would be useful to collaborate with

the manufacturers to develop a primary count calibration procedure using NIST ISO medium test dust. It is likely a count calibration procedure would involve making adjustments in the threshold setting to compensate for differences in instrument resolution.

Rationale: A particle counter that counts essentially all the particles larger than the threshold setting would be a very useful tool in water treatment practice and might make particle counting a regulatory option to the turbidity measurement. It is possible that affordable particle counters that perform at high count efficiency are about to become available. A count performance standard will be needed to test these instruments.

3. The voluntary standards system should be used to bring together particle counter manufacturers, particle counter users, regulators, and other particle counting researchers to evaluate and refine the count performance evaluation protocol and the NIST ISO medium test dust suspension strategy. It is possible that the count performance evaluation protocol could become the basis of an AWWA or Standard Methods standard for particle counting in the drinking water.

Rationale: There is no single, absolutely correct way to evaluate the performance of particle counters. The best way to bring together all the stakeholders and to make the necessary choices through discussion and compromise is the voluntary standards system.

4. Field studies are needed to beta test the proposed count performance evaluation protocol at utilities. The fieldwork could include testing an alternative scheme in which a lab-based or portable “master counter” is used in conjunction with the NIST medium test dust and the CPE protocol. It seems logical to do this in conjunction with the work that is done with stakeholders through the voluntary standards system. This would help ensure that all interested parties have a voice in the design of the tests and the interpretation of the results and it should facilitate the development of a standard that uses the CPE protocol in an effective and fair way.

Rationale: The field work completed in the current study gave essentially positive and encouraging results about the efficacy of the CPE protocol but it is clear that more work needs to be done.

CHAPTER 1

INTRODUCTION AND BACKGROUND

INTRODUCTION

Over the last 20 years, particle counters have been used increasingly to monitor the operation of particle removal processes. Their use in the drinking water industry has been limited to some extent by particle counter performance. One of their shortcomings is inconsistency in the data collected from different counters. For instance, Routt et al. (1996) and Routt et al. (1997) have shown that even with the same traceable particle size standards running concurrently through calibrated particle counters, different counters measured significantly different counts. Also, researchers (Cleasby et al. 1989; Fletcher et al. 1998; Chowdhury, et al. 2000) have questioned the use of particle count data for defining absolute number of particles, or for defining absolute particle size distributions. These investigators provide evidence that light obscuration particles counters do not “see” all the particles that are measured with visible light and electron microscopes and with electrical resistance-type particle counters.

Gilbert-Snyder (1998) used an assortment of particle counters (batch and on-line) to monitor a broad spectrum of California water systems. He observed that total counts varied by as much as 50% from one sensor to another and count comparisons by discreet size range varied even more significantly. While recognizing the value of particle counting for monitoring the performance of treatment plants, Gilbert-Snyder concluded that poor inter-instrument count agreement limited the usefulness of counters as a regulatory tool and that industry-wide calibration and verification standards are needed. Table 1.1 summarizes the results obtained by Gilbert-Snyder and other researchers.

The principal objective of this study was to develop materials and a procedure for the count performance evaluation (CPE) of on-line particle counters. To determine the required attributes of the CPE suspension the first part of the study was an instrument performance analysis (IPA) of four on-line particle counters that had been size calibrated using polystyrene latex (PSL) suspensions. The IPA tests were not conducted to determine which counters were better or more efficient in counting particles but simply to find the characteristics that a suspension must have to be useful for count performance evaluation. Two suspensions were used in this process, near mono-disperse PSL and a poly-disperse test dust purchased from the

National Institute for Standards and Technology (NIST). The factors that affect count performance are identified and discussed in this report. Recommendations are made regarding suitable particles and suspensions for size calibration and count performance evaluation for on-line particle counters used in water treatment applications. A count performance evaluation protocol based on the use of a standard test dust is developed and the method is presented.

Table 1.1
List of studies that found significant discrepancies between particle counter measurements and counts made by reference methods such as light and electron microscopy and electrical sensing zone instruments

Reference	Summary	Results										
Cleasby et al. (1989)	Results obtained from different particle counters for the same standard suspension were different; counts obtained from particle counters were significantly lower than counts obtained using microscope analyzer system.	<table><tr><th>Instrument</th><th>Counts/mL > 2 μm</th></tr><tr><td>HIAC</td><td>7632</td></tr><tr><td>60 μm sensor</td><td>7200</td></tr><tr><td>Microscope analyzer</td><td>59,000</td></tr></table>	Instrument	Counts/mL > 2 μm	HIAC	7632	60 μm sensor	7200	Microscope analyzer	59,000		
Instrument	Counts/mL > 2 μm											
HIAC	7632											
60 μm sensor	7200											
Microscope analyzer	59,000											
Fletcher et al. (1998)	Counts obtained by NIST for medium test dust from SEM and image analysis were much higher than counts measured using a HIAC Royco batch particle counter.	<table><tr><th>Instrument</th><th>Counts/mL > 2 μm</th></tr><tr><td>SEM and image analysis</td><td>27,035</td></tr><tr><td>HIAC Royco</td><td>4000</td></tr></table>	Instrument	Counts/mL > 2 μm	SEM and image analysis	27,035	HIAC Royco	4000				
Instrument	Counts/mL > 2 μm											
SEM and image analysis	27,035											
HIAC Royco	4000											
Gilbert-Snyder (1998)	CA Department of Health Services Study – Median filtered water results showed inter-instrument differences as great as factor of 2.	<table><tr><th>Instrument</th><th>Counts/mL > 2 μm</th></tr><tr><td>23</td><td></td></tr><tr><td>30</td><td></td></tr><tr><td>46</td><td></td></tr></table>	Instrument	Counts/mL > 2 μm	23		30		46			
Instrument	Counts/mL > 2 μm											
23												
30												
46												
Routt et al. (1996)	Counts obtained from different counters simultaneously counting a 3 μm standard suspension were different and lower than estimated counts.	<table><tr><th>Instrument</th><th>Count efficiency (%)</th></tr><tr><td>1</td><td>49</td></tr><tr><td>2</td><td>80</td></tr><tr><td>3</td><td>63</td></tr><tr><td>4</td><td>66</td></tr></table>	Instrument	Count efficiency (%)	1	49	2	80	3	63	4	66
Instrument	Count efficiency (%)											
1	49											
2	80											
3	63											
4	66											
Routt et al. (1997)	Counts obtained from different counters simultaneously counting a 5 μm standard suspension were different.	<table><tr><th>Instrument</th><th>Counts/mL > 2 μm</th></tr><tr><td>1</td><td>1200</td></tr><tr><td>2</td><td>1000</td></tr><tr><td>3</td><td>1500</td></tr><tr><td>4</td><td>3800</td></tr></table>	Instrument	Counts/mL > 2 μm	1	1200	2	1000	3	1500	4	3800
Instrument	Counts/mL > 2 μm											
1	1200											
2	1000											
3	1500											
4	3800											
Van Gelder et al. (1999) and Chowdhury et al. (2000)	Light-obscurtion (L-O) particle counters consistently measured fewer particles in the 2 to 5 μm size range than electrical sensing zone (ESZ or “Coulter Counter”) instrument.	<table><tr><th>Instrument</th><th>Counts/mL > 2 μm</th></tr><tr><td>ESZ</td><td>2965</td></tr><tr><td>L-O PC(1)</td><td>175</td></tr><tr><td>L-O PC(2)</td><td>161</td></tr></table>	Instrument	Counts/mL > 2 μm	ESZ	2965	L-O PC(1)	175	L-O PC(2)	161		
Instrument	Counts/mL > 2 μm											
ESZ	2965											
L-O PC(1)	175											
L-O PC(2)	161											

LITERATURE REVIEW

This section includes background information and a discussion of recent research and other literature on the design and performance of the optical particle counters used in drinking water treatment applications. Important terminology associated with these instruments has been identified and discussed. Consensus standards for particle counters and recommended standard materials used in other application areas such as the fluid power industry have been reviewed. Standards from other application areas are an important source of information for the design and operation of counters used in drinking water applications. This section begins with a brief description of the components of a typical optical particle counter and an overview of how the units work. Information on the different types and classifications of particle counters is also provided. More detailed information can be found in an AWWARF manual on particle counting by Hargesheimer et al. (1999).

TYPICAL OPTICAL PARTICLE COUNTER CONFIGURATION

Figure 1.1 shows the typical components of an on-line optical particle counter used in the drinking water industry. The essential components are a light-based sensor, counting electronics and an overflow weir. Hargesheimer et al. (1992a) and Hargesheimer et al. (1992b) describe the basic configurations of particle counters used in the drinking water industry.

A sensor in a typical particle counter consists of a light beam, a photo-detector and two transparent “windows”. The light beam is directed through both the windows and on to the detector. The detector converts light energy into an electrical (voltage) signal. The overflow weir is used to maintain constant sample flow rate through the sensor. Flow rate control is critical since particle count data is reported based on sample volume while the instrument counts using elapsed time. The weir is adjusted to a particular height such that the manufacturer specified flow rate is achieved through the sensor and a constant overflow is maintained. The sample flow passes between the two windows so that particles in the sample pass through the beam. The percentage of incident light scattered or blocked as each particle passes through the beam is detected by the photodiode. This results in a change in the electrical output of the photo detector and a short voltage pulse (or plateau) is produced at the output of the detector. The amplitude of the pulse is proportional to the projected area of the particle. In the case of nearly spherical

particles, e.g., the polystyrene latex micro-spheres frequently used for size calibration, simple geometry relates the diameter of the particle to the projected area. The sensor output is fed to the counter electronics, which sorts the pulses according to their amplitude and counts them.

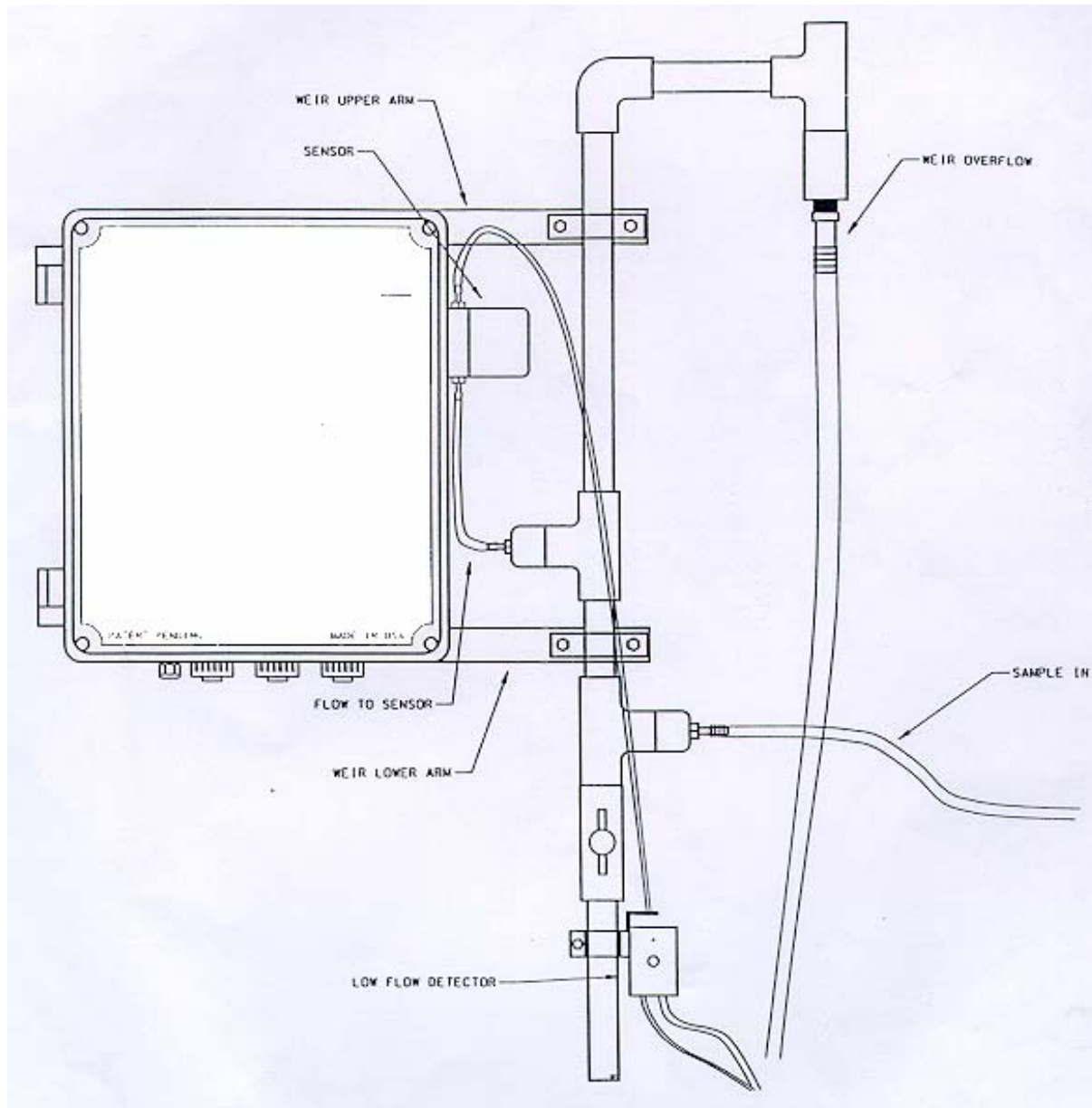


Figure 1.1 Typical components of an optical particle counter (Source: Chemtrac Systems Inc., 1996)

TYPES AND CLASSIFICATION OF OPTICAL PARTICLE COUNTERS

Optical particle counters can be classified into different types based on the type of sensor technology and the type of sampling configuration. Hargesheimer et al. (1992a), Lewis et al. (1992) and Hargesheimer and Lewis (1995) present a comprehensive description of sensor technologies and sampling configurations. Particle counters are classified into two categories based on the type of sensor technology: light scattering and light-blockage.

Light-scattering Sensors

When a light beam illuminates a particle, the energy of the radiation source is redirected or absorbed. Redirection of the energy is called scattering (Sommer et. al. 1993b). In light-scattering sensors, the amount of light scattered (at a specific angle from the incident light beam) when a particle passes through a light beam is measured by the photodiode. The electric signal from the photodiode is analyzed to give the diameter of the particle, based on a calibration relationship. Figure 1.2, shows the basic configuration of a light-scattering sensor.

Light scattering is highly dependent on the refractive index of the particle and particle morphology (Allen T. 1997).

Light-blockage Sensors

In light-blockage sensors, (also called light obscuration sensors) the percentage of incident light that is blocked when a particle passes through a light beam is measured by the photodiode. The cross-section of a typical light extinction sensor is shown in Figure 1.3.

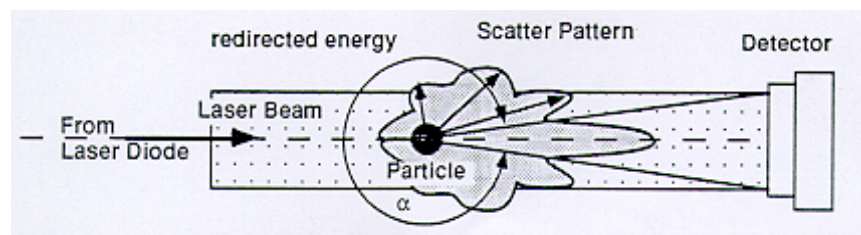


Figure 1.2 A typical light scattering sensor (Source: Sommer et al. 1993a)

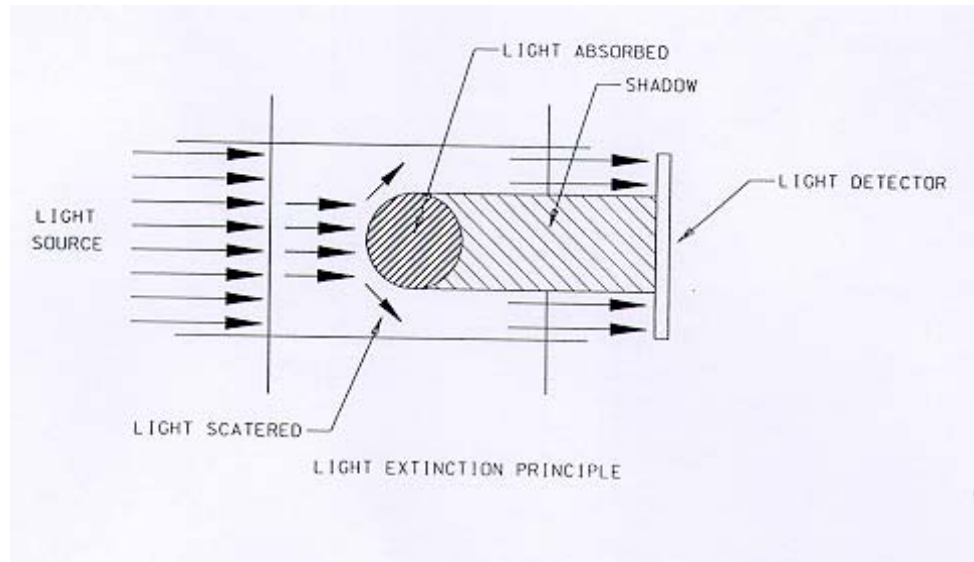


Figure 1.3 A typical light blockage sensor (Source: Chemtrac Systems Inc. 1996)

Light blockage counters typically have a narrow area of uniform illumination established across the flow channel of the sensor. The passage of a small particle through the beam causes an amount of light proportional to the cross-sectional area of the particle to be blocked. These sensors are less affected by changes in particle refractive index and shape than light scattering sensors (Allen T. 1997). This is one reason why light blockage sensors are used in drinking water applications since contaminants in drinking water may consist of different materials with different refractive indices and shapes.

The light blockage counters used in the water industry have either a volumetric or an in-situ sensor. A volumetric sensor is assumed to count the particles in the entire flow through the sample cell. In this type of sensor the laser beam is a sheet of light that is equal in extent to the size of the flow conduit.

An in-situ sensor counts the particles in some fraction of the total flow passing through the sample cell. The aperture is an important component in an in-situ sensor. The aperture is the area of the sample cell through which the laser is directed and where the particles are counted. The size of the aperture relates the raw counts to the corrected counts as given by the following equation (Barsotti et al. 1998):

$$\text{Corrected count} = \frac{\text{Raw count}}{(\text{Flow}) (\text{Sample period}) (\text{Aperture factor})} \quad (1.1)$$

The magnitude of the aperture factor is typically estimated by comparing the in-situ sensor count with a volumetric sensor count when a near mono-disperse PSL suspension is flowing through both sensors. The magnitude of the factor is set in the instruments' software (Barsotti et al. 1998).

Type of Sampling Configuration

The particles counters found in water treatment applications typically use one of two sampling configurations, batch or on-line. Batch counters are sometimes called grab samplers and are used to measure particles in discrete volumes of suspension. These instruments do not have the flow control weir component of Figure 1.1, instead a gear or vacuum pump is used to move the sample and maintain a constant flow rate through the sensor.

On-line counters are used in continuous flow-monitoring applications and measure particles in a continuous stream of fluid. They typically use adjustable overflow weirs to maintain a constant flow through the sensor (See Figure 1.1).

TERMINOLOGY ASSOCIATED WITH PARTICLE COUNTER DESIGN AND PERFORMANCE

This section discusses some of the important terms that are commonly associated with particle counter design, operation and performance. These terms include resolution, threshold setting, calibration, calibration-verification and count performance evaluation.

Resolution

The resolution of an instrument is determined by the relative amount it adds to the standard deviation (the spread) of a measured particle size distribution. Sommer (1995) describes it as the broadening of a mono-disperse particle challenge of PSL spheres due to imperfections in the instrument optics and signal processing system. When particles are counted that are close in size to a threshold setting the resolution indicates the extent to which particles smaller than the threshold are counted when they shouldn't be and the extent to which particles larger than the threshold are not counted when they would be in a perfect (resolution = 0 %) counter.

The significance of resolution is illustrated (schematically) by Figure 1.4. The dashed curve represents the counting response curve from a high (good) resolution sensor and the bold curve represents the response curve from a low (poor) resolution sensor. The particle size distribution obtained from microscopic analysis is also shown. For a narrow (near mono-disperse) particle size distribution, a high-resolution sensor gives less broadening of the size distribution when compared to a sensor with low resolution. The effect of the resolution on count performance at a size threshold is discussed in detail in Chapter 4.

The magnitude of the resolution, R , for the points to the left and right of the mean diameter is given (as a percent) by the following equation (*USP 23/NF 18 (788)*)

$$R = \frac{100 \times \sqrt{\sigma_m^2 - \sigma_p^2}}{\bar{d}} \quad (1.2)$$

where σ_m is the measured standard deviation (left or right of the mean), σ_p is the standard deviation of the particle diameter and $\bar{d}$ is the measured mean particle diameter. The quantities $\bar{d}$ and σ_p are determined using measurements from a reference sizing method such as

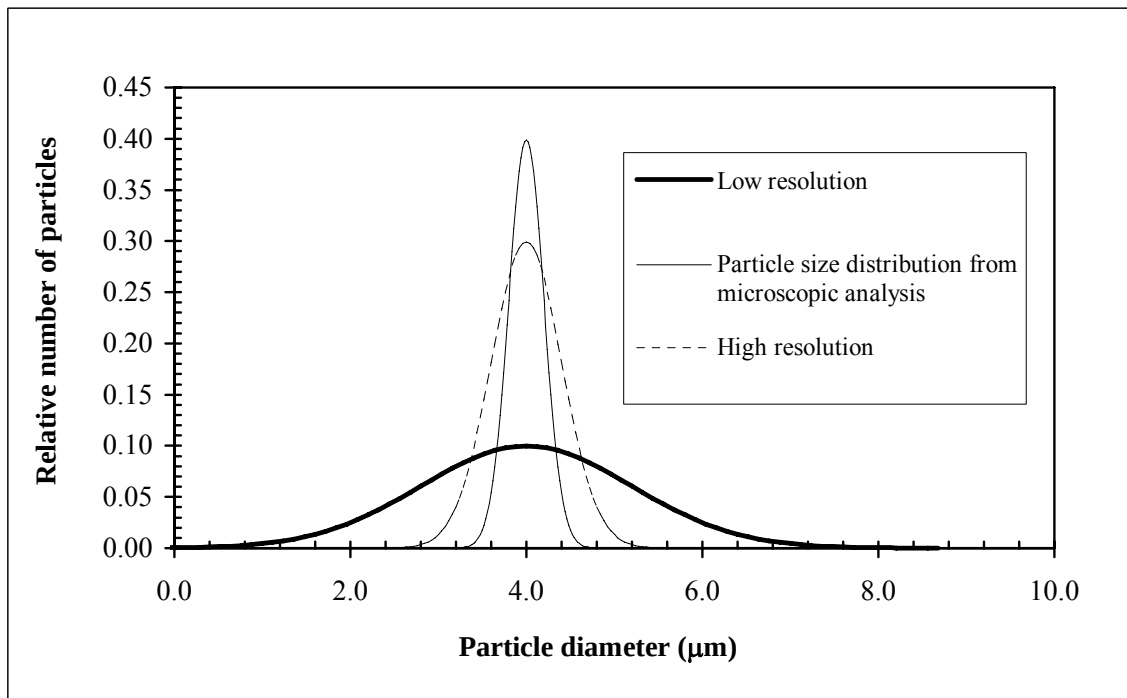


Figure 1.4: The relationship between the particle size distribution and resolution

optical microscopy. A low value of R indicates a high (good) resolution sensor and a high value of R indicates a low (poor) resolution sensor. When the resolution is used to characterize the performance of a sensor at a given threshold setting the quantity d is replaced by the magnitude of the threshold setting (See Chapter 4).

Threshold Settings

A voltage pulse or signal is produced as each particle passes through the instrument's light beam. Threshold settings are the “labels” that characterize the magnitude of these pulses. They are used to provide information about the size (and, in some cases, the number concentration) of the particles in a sample. Instead of an output that says a particle produced a pulse that exceeded a 23 mV threshold the calibrated output says the particle produced a pulse that, for example, exceeded the pulse produced by a 5- μm polystyrene latex microsphere. The 5- μm label for the 23 mV pulse height is a threshold setting. This “labeling” is relatively unambiguous when the optical properties (e.g., the refractive index) and morphology of the measured particles are the same as those of the calibration particles. Two particles of identical size and shape but different refractive indices will tend to produce different shadows and may give significantly different responses from the counting instrument's photo-diode. The millivolt signal corresponding to, for example, the 2- μm diameter size threshold will depend on the refractive index of the particles used for calibration.

Calibration

Calibration refers to the process of establishing a relationship between the millivolt signal thresholds (the “threshold settings”) used by the particle counter and the size (and/or number concentration) of the particles in a standard suspension. The threshold settings are given specific “size” values during calibration depending on the standard suspension. Usually, the user can set different size bins (bins are defined by upper and lower threshold settings) in the counter software with certain limitations that are instrument-specific. There are two types of particle counter calibration: size calibration and count calibration.

Size Calibration

The usual procedure in the drinking water industry is to calibrate particle counters for particle size. Size calibration is done (typically by the instrument manufacturer) by challenging the instrument with a series of near mono-disperse PSL microspheres. The “true” size or size distribution of each microsphere suspension is established using a generally accepted (i.e., standardized) sizing method such as optical microscopy. As particles move with the fluid through the sensor, a millivolt threshold setting is adjusted until half the counts fall above the setting and half fall below the setting. This setting is labeled with the mean (i.e., the “true”) PSL particle diameter measured by the standard method. The process is repeated with PSL spheres of a number of mean diameters (usually in the 3 to 15 μm range) and the set of calibration values is entered into the instrument software. It is not necessary for the counter to register all the particles of each size for this procedure to work with reasonable effectiveness.

The relationship established between the millivolt signal produced by the instrument and the calibration particle size (usually the diameter for PSL microspheres) is known as a size calibration curve. A typical size calibration curve is shown in Figure 1.5 where the instrument response (in millivolts) is plotted as a function of the log of the calibration particle mean diameter.

When a size calibration curve is used to make particle size measurements it is necessary to assume that the particles in the sample suspension have optical and other physical properties that are similar to the calibration spheres. If the particles in the sample suspension and the calibration spheres appear to have the same “size” under the microscope but produce dissimilar shadows on the photo-diode as they move through the light beam, then the calibration curve may not give the “correct” results, i.e., the counter measured size may not agree with the optical microscope measurements. Determining whether or not the calibration spheres and the particles in the suspension are compatible is difficult.

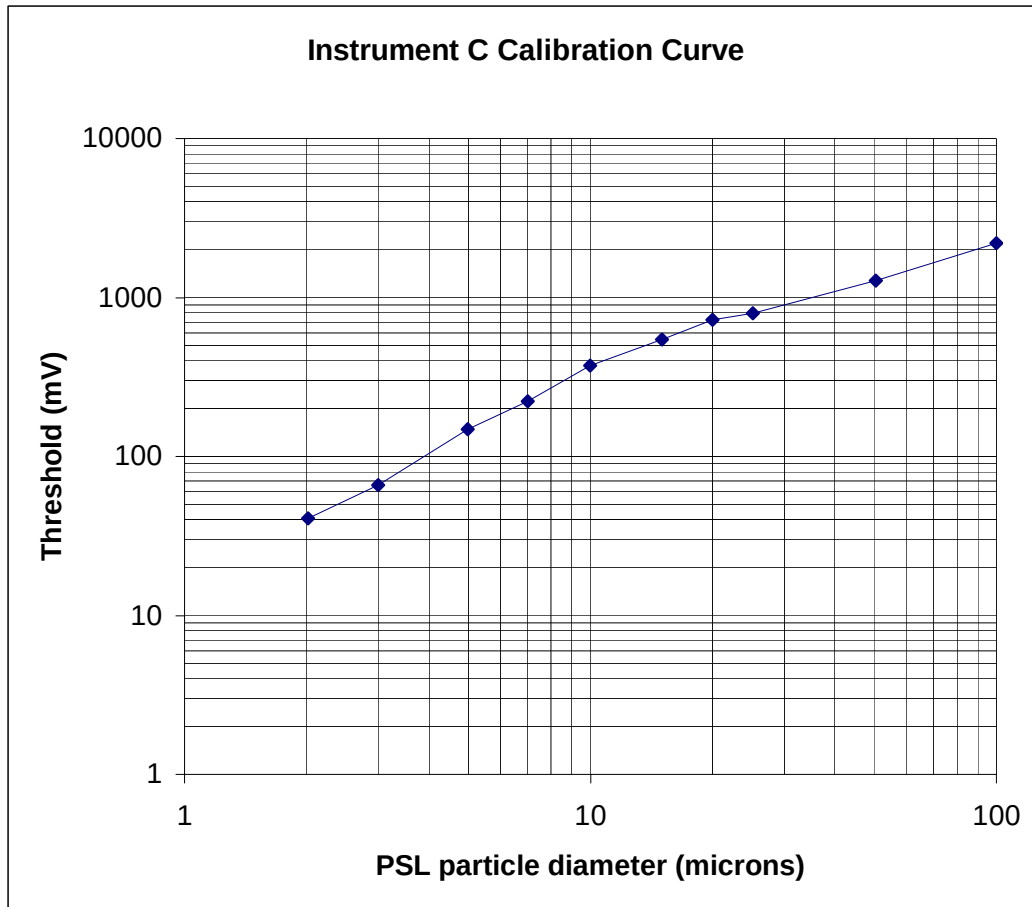


Figure 1.5 Typical size calibration curve

Count Calibration

Count calibration is usually done using a poly-disperse suspension that has been characterized with a standard size measurement technique such as light or electron microscopy. In some cases the standard size measurement device is a special particle counter. The microscopic analysis yields a relationship between the characteristic size of the particles (e.g., the diameter of a circle of equivalent area) and the particle number concentration. The number of particles greater than a certain particle size (e.g., a diameter) in this suspension is estimated using the size-number concentration relationship from the standard analysis. The particle counter is challenged with the suspension and the threshold setting is found that gives the desired number concentration of particles. This threshold setting is then labeled with the target size value. For example, if the standard size-number concentration relationship says that for a given test

suspension concentration 9542 particles/mL are larger than a 3- μ m diameter then the threshold setting that gives 9542 particles/mL will be labeled as the 3- μ m particle diameter threshold.

Count calibration is currently used in the fluid power industry (ISO 11171:1999) where the standard count calibration suspension is ISO medium test dust (MTD) suspended in hydraulic fluid [NIST Standard Reference Material SRM 2806]. This suspension has been well characterized by NIST using a scanning electron microscope (SEM) coupled with image analysis software and it has been found that 1 microgram of NIST MTD suspended in 1mL of hydraulic fluid has 9655 particles greater than 2 μ m (Fletcher et al. 1996). The entire size distribution is published by NIST. The “size” used by NIST is the diameter of a circle that has the same area as the particle. Count calibration is only valid when the counter’s ability to detect particles is 100% across all intervals of particle size above the instruments size detection limit.

Size Calibration Verification

Size calibration is typically done using well-characterized, and typically expensive, “size-certified” suspensions of near mono-disperse PSL particles. To verify that the size calibration of an instrument is not drifting with time, counters are challenged periodically with less expensive calibration verification suspensions. These suspensions have typically been characterized using a recently calibrated particle counter.

Count Performance Evaluation

Count performance evaluation is used to determine how well the particle size distribution (expressed as number concentration greater than or equal to each particle size) measured by a counter agrees with the size distribution measured by a generally accepted reference method such as optical (or scanning electron) microscopy. One way to measure the count performance is to challenge the particle counter with an appropriate poly-disperse suspension and to record the particle number concentration for particle sizes that are equal to or greater than one or more threshold settings. The particle counter reading for each threshold setting is compared to the corresponding count measured with the reference method. The readings are not used to calibrate the instrument, i.e., to adjust settings to obtain size or count agreement.

The optical properties of the particles in the CPE suspension do not have to be the same as the particles used to calibrate the instrument. However, if the count performance of several instruments is to be compared, all the instruments should have been calibrated in the same way with the same type of particle, e.g., size calibration with PSL micro-spheres.

Count performance evaluation is especially significant and useful for counters that have been size calibrated with near mono-disperse PSL particles but are used to measure the number concentrations of particles in filtered water. When an instrument that has been size calibrated with PSL spheres is used to count “real” particles in actual samples, the instrument converts the millivolt responses to equivalent latex spherical diameter. When sample particles have refractive indices that are different from PSL spheres or when particles have complex shapes and block light differently, it may be difficult to interpret their size in terms of equivalent latex sphere diameters (Van Gelder et al. 1999). Therefore, the instrument manufacturer’s calibration with PSL particles may not provide a satisfactory interpretation of size, i.e., a size that gives accurate particle count results in real water. A count performance evaluation must be done before number concentration versus particle size data from instruments in the field is interpreted. Count performance evaluation can be used to determine if instrument output is changing with time and if different instruments are in reasonable (effective) agreement.

FACTORS AFFECTING PARTICLE COUNTER PERFORMANCE

The ability of a particle counter to count and size particles is influenced by a number of factors. The design parameters of the instrument, such as the width of the sample cell, the energy spectrum of the light beam, the consistency of the light beam intensity across the sample cell, the bandwidth of the photo-detection circuitry and the accuracy and speed of the counting electronics, affect sizing and count ability (Vasiliou et al. 1997; Chowdhury et al. 2000). Instrument resolution and threshold settings are important design/operation-related factors.

The coincidence limit of an instrument can affect its performance. It is assumed that when a sample is being analyzed, there is only one particle in the sensing volume at a time. If there is more than one particle in the sensing volume, the individual pulses cannot be distinguished and the particles are said to be coincident. The coincidence limit depends on flow rate, speed of the instrument’s electronics and the sensing volume. To minimize problems associated with coincidence, instruments are operated at the specified flow rates and the total

concentration of particles in the sample is not allowed to exceed a specified upper limit (Chowdhury et al. 2000). Gilbert-Snyder and Milea (1996) showed that coincidence counting typically generates a lower overall count and skews the particle size distribution toward the larger diameter particles.

As noted above, particle counter response also depends on the particle size, shape, and orientation (in the light beam), and the relative refractive index of the particle and the surrounding fluid (Sommer et al. 1993a). Also, if a counter has been calibrated with a particle that differs in shape and/or optical properties from the particle that is being measured, the size and count results obtained may be inaccurate.

Sommer and Hart (1991) studied the effects of the optical properties of particle materials. Three instruments (two light extinction sensors and one light scattering sensor) were calibrated in three different ways: size calibration with PSL particles in water, size calibration with PSL particles in hydraulic oil and number calibration with ACFTD (air cleaner fine test dust) in hydraulic oil. Three series of experiments were conducted to investigate the significance of the calibration method. The sensors were challenged with suspensions of five different particles in two fluids – water and hydraulic oil. The five particle materials were aluminum (spherical), stainless steel (spherical), glass (spherical), carbonyl-iron (clusters of individual spheres) and air cleaner fine test dust, ACFTD (irregular). The results showed that the index of refraction and absorption characteristics of the particle contribute significantly to instrument performance. For example, sensors calibrated using PSL/water undersized glass particles (index of refraction = 1.51) when compared to the PSL calibration spheres (index of refraction = 1.59). With this difference in refractive index a larger glass bead is needed to produce the same signal as a smaller PSL sphere.

CONSENSUS STANDARDS ON PARTICLE COUNTER CALIBRATION

A literature review was conducted on consensus standards available for particle counting in other applications such as pharmaceuticals and hydraulic fluids in order to evaluate and select the appropriate particles for count performance evaluation of particle counters in the water industry. Table 1.2 lists relevant and currently available standards from an assortment of areas where particle counters are used.

The American Society for Testing Materials (ASTM) provides a standard, *ASTM F-658 87*, for the size calibration, resolution determination and measurement of particles by optical particle counters. This standard recommends the use of mono-sized PSL micro-spheres for size calibration and for checking the accuracy of count measurements. Resolution should be determined, it suggests, using latex spheres with particle size distributions that have low standard deviations.

The United States Pharmacopoeia specifies a standard, *USP 23/NF 18 (788)*, (USP 1992) for determining the particulate matter in injections using particle counters. The size calibration of particle counters used in these applications is to be done using near mono-disperse PSL spheres. Near mono-sized PSL spheres are also recommended for determining resolution and counting accuracy. The standard also states that the measured resolution should not exceed 10 %.

Table 1.2
Current standards from different industries that use particle counter calibration and performance evaluation

Standard	Summary
<i>ASTM F-658 87</i> : Standard practice for defining size calibration, resolution and counting accuracy of a liquid-borne particle counter using near-mono-disperse spherical particulate matter (ASTM 1987)	Calibration, resolution determination and counting accuracy accomplished using near mono-disperse PSL suspensions.
<i>USP 23/NF 18 (788)</i> : Particulate matter in injections (USP 1992)	Calibration, resolution determination and counting accuracy using near mono-disperse PSL spheres; Instrument resolution not to exceed 10%.
<i>ISO 11171:1999</i> : Hydraulic Fluid Power – Calibration of automatic particle counters for liquids (ISO 1999)	Primary count calibration using SRM 2806 (ISO medium test dust in hydraulic oil); secondary calibration by preparing suspensions using RM 8631; resolution determination using PSL spheres
<i>Designation JIS B 9925-1991</i> : Light Scattering Automatic Particle Counter for Liquids, Japanese Standards Association, Tokyo, Japan. (JAS 1991).	Counting efficiency determined with suspensions of near mono-disperse PSL particles. Counting efficiency should be 20 % or higher for smallest particles (near the minimum measurable size).

Standards prepared by the National Fluid Power Association (NFPA) have undergone a number of changes involving calibration methods and instrument performance standards. Until 1990, ACFTD was used as the standard calibration material in the fluid power industry (*ANSI/(NFPA) T2.9.6 R1-1990*). ACFTD is a poly-disperse dust that has a large number of particles (as documented by ISO 4402:1991) within the 1-20 micron size range. However, the use of ACFTD was found to be problematic. Fletcher et al. (1996) note that there was an on-going concern among researchers and users of the standard that the particle size distribution was not accurate in the small particle regime of the distribution (< 10 microns). For example, Johnston and Swanson (1982) and Masbaum (1981) noted that there were more sub-10 μm particles in ACFTD than was reported by the ISO 4402:1991 standard. Also, the production of ACFTD was discontinued by the supplier, a division of the General Motors Corporation (Fletcher et al. 1996).

In 1991, the NFPA published *ANSI/(NFPA) T2.9.6 R1-1990*. The procedure in this standard used mono-disperse latex spheres suspended in hydraulic fluid for primary size calibration. Calibration verification was done using ACFTD suspended in hydraulic fluid.

The NFPA replaced their 1990 *T2.9.6 R1-1990* standard with one that eventually became *ISO 11171:1999*. The 1999 ISO method uses NIST's Standard Reference Material (SRM) 2806 (ISO medium test dust in hydraulic oil) for primary count calibration. Count calibration verification is to be done using ISO medium test dust purchased from NIST as a dry powder (RM 8631) and dispersed in standard hydraulic fluid. This standard also specifies the use of 10- μm diameter near mono-disperse PSL micro-spheres for measuring the counter resolution.

CONCENTRATIONS STANDARDS IN EARLIER AWWAF SPONSORED RESEARCH

In a recently completed study sponsored by the AWWA Research Foundation, Chowdhury et al. (2000) investigated the development of a concentration standard¹ for on-line particle counters. Two types of suspensions were studied, mixtures of near mono-disperse polystyrene latex (PSL) microspheres and a poly-disperse alumino-silicate dust (ACFTD a.k.a.

¹Chowdhury et al. (2000) call it a concentration standard, however, based on how it is used and described it is really a count verification suspension. The measurements made with the suspension are not used to adjust instrument settings but just to check the instrument count performance.

ISO fine test dust). The particles in ACFTD are similar in shape and mineral composition to the particles in the NIST medium test dust used in this study. The manufacturer-prepared PSL suspensions used by Chowdhury et al. were reported by the manufacturer to contain 2,300 counts/mL of 3- μ m microspheres and 210 counts/mL of 10- μ m microspheres.

Chowdhury et al. (2000) observed significant variation in the counting performance of the instruments tested. For the PSL suspensions the inherent variation of the concentration standard could not be distinguished from the variation caused by the instruments. The light-obscuration particle counters undercounted the smaller (3- μ m) PSL particles by a significant amount. The measured counts were well below those measured with an electrical sensing zone (Coulter-counter-type) instrument. This is significant because the light-obscuration counters had been size calibrated with PSL particles that had the same nearly spherical shape and refractive index as the particles counted in the test suspensions. Counts of the larger (10- μ m) particles compared well between instruments but all the instruments' counts were significantly higher than the manufacturer's estimate of the particle concentration.

Chowdhury et al. (2000) noted that ACFTD suspensions have two advantages over PSL suspensions. The particle size distribution is more like that of the particles in suspensions from water treatment plants and its continuous size distribution helps "assimilate" different particle size analyzers, even if they have relatively poor resolution.

According to Chowdhury et al (2000) ACFTD has three disadvantages when used as a concentration standard. First, the particle size distribution is given by the manufacturer using a mass (or volume) basis and not a count basis. Consequently, it is not possible to use the material to prepare a concentration standard. Second, it is not possible to determine the instrument resolution with ACFTD suspensions because they are poly-disperse and not near mono-disperse, which they should be to accurately determine the resolution. Third, the particles in the ACFTD suspensions are non-spherical and Chowdhury et al. (2000) conclude that this makes the measured size distribution dependent on the method of measurement, an untenable attribute, they believe, for a concentration standard.

The study by Chowdhury et al. (2000) was conducted before NIST finished developing a scanning electron microscope – image analysis software method for accurately determining the size distribution (count versus projected area, equivalent sphere, diameter) of ISO medium test dust particles. This accomplishment, made for and with input from the fluid power industry,

provided important direction to this AWWARF sponsored study. Before this was accomplished no organization with the expertise and independence of NIST had prepared and characterized a suspension or dry dust reference material for use in the evaluation and calibration of light obscuration particle counters.

CHAPTER 2

EXPERIMENTAL APPARATUS, MATERIALS AND PROCEDURES

INTRODUCTION

This chapter describes the experimental devices, materials and methods that were used in this study. The Apparatus section describes the particle counting instruments and other devices utilized in the experiments. The second section, titled “Materials”, includes information about the particles that were used in the instrument performance analysis and count performance evaluation parts of the study. The third section, “Methods”, provides a description of the experimental methodology.

APPARATUS

This section describes the instruments and other devices used in the experiments. The items discussed include:

1. Particle counting instruments
2. Gravity flow suspension feed system
3. Rotary micro-riffler

Particle Counting Instruments

Four on-line counters and one grab sample counter were used in this study. The grab sample unit was used for some of the QA/QC measurements. The essential characteristics of each of the on-line instruments are given in Table 2.1 below. The four on-line counters were a Metone PCX, an IBR WPCS, a Hach 1900 WPC and a Chemtrac PC2400D. The grab sample unit was a Metone WGS. The sensors in all the counters used the light blockage (i.e., light obscuration) principal. (Note: the Hach Company is now selling the “Metone” on-line and grab sampler units. The “Hach instrument” is now being sold by GLI, Inc. This instrument was designed by PMS, Inc. of Loveland, CO). When the results of the measurements are discussed in this report a letter code (A, B, C, D and E) is used instead of the manufacturer’s name.

Table 2.1
Characteristics of particle counters¹

Characteristics	Metone	IBR	Hach	Chemtrac	Metone
Model	PCX	WPCS	1900 WPC	PC2400D	WGS-267 DWS
Counter type	On-line	On-line	On-line	On-line	Grab sampler
Sensor type	Light blockage	Light blockage	Light blockage	Light blockage	Light blockage
Sensor configuration	Volumetric	Volumetric	ND	ND	Volumetric
Light source	Laser diode	Laser diode	Laser source, 780 nm (infrared)	Laser diode	Laser diode
Resolution	ND	Better than 10%	10 %	ND	ND
Sample cell size	0.75 x 0.75 mm	0.8 x 0.5 mm	1 x 2 mm	1 x 1 mm	0.75 x 0.75 mm
Sample cell material	ND	ND	Sapphire	ND	ND
Particle size range	2 – 750 microns	2 – 400 microns	2 – 800 microns	2 – 400 microns	2 – 100 microns
Name of software used	Water Quality Software (WQS)	PC Bridge	Aqua View +	Tracware	Universal Utility Software (UUS)
Number of thresholds available	15	8	15	8	Preset: 2, 3, 5, 7, 10, 15 microns
Flow rate through sensor	100 mL/min	60 mL/min	200 mL/min	100 mL/min	100 mL/min
Head across sensor	74 cm H ₂ O	134 cm H ₂ O	31cm H ₂ O	47 cm H ₂ O	NA
Maximum particle concentration	ND	ND	20,000 counts/mL (10 % coincidence)	ND	ND
Coincidence loss	ND	ND	Maximum 10% at sensor limit concentration	ND	Less than 10 % at 17,000 particles/mL
ND = Not Disclosed, NA = Not Applicable					

¹Source: manufacturer's literature

The target sensor flow rate through the Metone counters (on-line and grab) and the Chemtrac counter was 100 mL/min, 60 mL/min through the IBR, and 200 mL/min through the Hach. The four on-line counters used a weir device to control the flow rate and the Metone grab sampler used a gear pump and a manual flow rate controller.

The Met One grab sampler had manufacturer-set particle size thresholds of 2, 3, 5, 7, 10 and 15 microns. These settings could not be easily changed. The Metone and the Hach on-line counters had fifteen user-adjustable threshold settings and the IBR unit had eight. In the Metone,

Hach and IBR units the settings could include decimal values. The Chemtrac unit had eight user-adjustable threshold setting that accommodated only whole numbers.

Each counter used software provided by the manufacturer (listed in Table 2.1) to record the data and store it in files. Usually these files were archived in Microsoft Excel format and later used to present the data and analyze the results.

Gravity Suspension Feed System

The instrument and count performance analysis experiments used a gravity flow feed system that allowed a test suspension to flow simultaneously through four on-line counters. The apparatus is described below and is illustrated schematically in Figure 2.1.

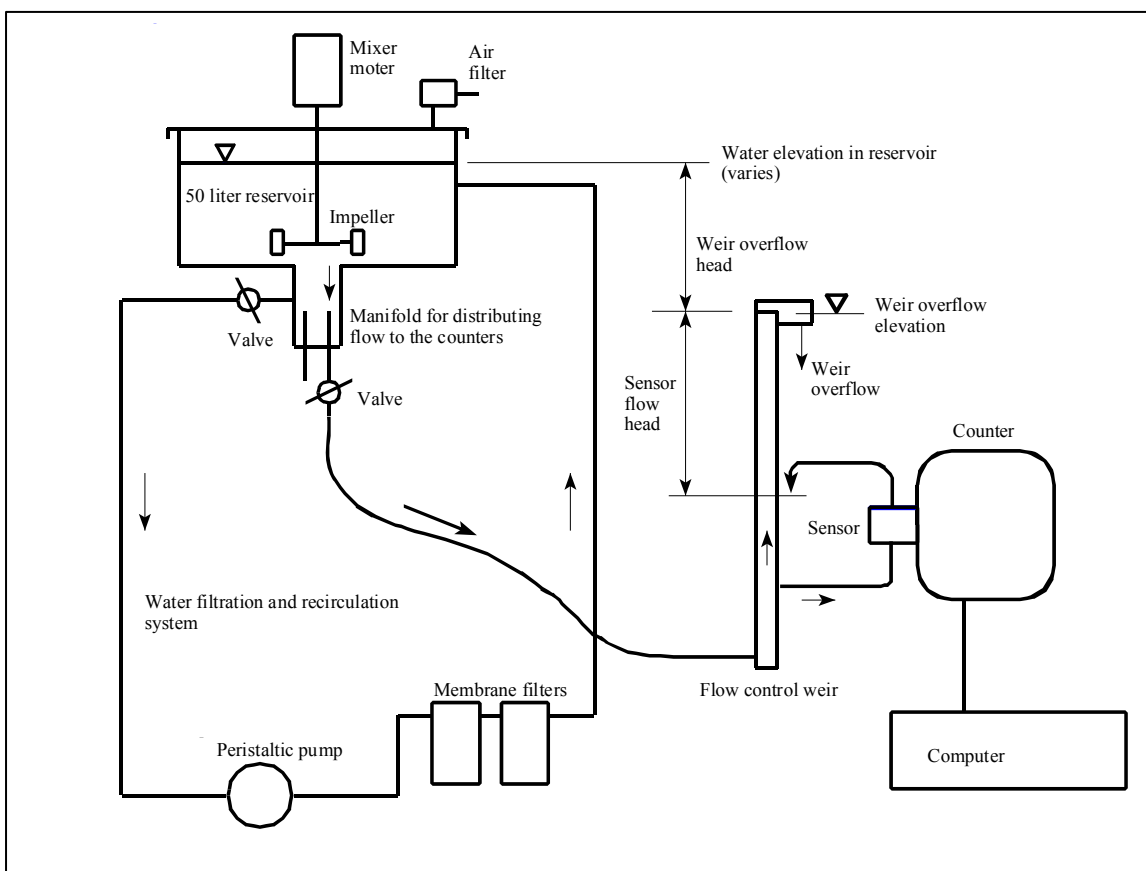


Figure 2.1 Schematic diagram of the suspension feed system

The suspension feed system consisted of a laboratory test stand (bar-rack) with the on-line particle counters mounted on it. A 50-L polyethylene reservoir, graduated in 10-L increments was installed on top of the stand. A tube that supplied low-particle water from a reverse osmosis (RO) unit was fitted into one side of the reservoir. The RO water passed through a pair of cartridge membrane filters before it entered the reservoir. The first membrane filter had a 0.8- μm (nominal) pore size and the second had a 0.2- μm (nominal) pore size. The impeller shaft of an adjustable-speed stainless steel mixer was inserted through a special bearing in the cover of the reservoir. As suspension flowed from the sealed reservoir, air entered through a membrane filter.

The suspension was fed from the 50-liter reservoir to the counters through a 2-inch diameter x 12-inch long PVC manifold located just underneath the mixer impeller. From the manifold the suspension flowed through 2 to 3 feet of 0.25-inch ID tubing to the flow control weir of each sensor. Teflon valves located just below the manifold controlled the flow to the weirs (and sensors). Quick disconnects positioned just below the manifold on each of the four tubes allowed the collection of grab samples in plastic sampling bottles before the suspension flowed to the on-line counters.

Rotary Micro-Riffler

When dusts and powders are shipped and handled, particle size segregation can occur. Therefore, to obtain representative samples from potentially segregated quantities of dust that are too large to be handled easily, premixing must be used. A riffing apparatus is an effective device for doing this (Allen 1997).

One of the materials used to analyze the performance of the instruments was National Institute of Standards and Technology (NIST) ISO medium test dust. In its catalogue, NIST lists this dust as a reference material “RM 8631”. It is supplied as a dry powder in 20 g quantities. A rotary micro-riffler was used to divide the entire 20 g quantity of NIST ISO medium test dust into smaller portions. The smaller portions weighed between 250 and 300 mg.

The micro-riffler was purchased from Quantachrome Corporation, Boynton Beach, FL. In this device the dust is placed in a small stainless steel bowl and an adjustable vibrator moves the particles down a chute and onto a rotating collector head. The collector head is a nickel-plated brass disc with eight collection funnels arranged in a circle. A small test tube is fixed

below each of the funnels. A variable speed drive motor rotates the collector head and test tubes at the selected speed. A small brush, mounted in a fixed position above the collector head with its bristles just touching the funnels, helps distribute the powder into the test tubes.

MATERIALS

This section describes the materials, including the particles and suspensions used for analyzing the performance of the on-line instruments and the reverse-osmosis-treated, low-particle dilution water (RO water).

Particles Used for Instrument Performance Analysis

The size distributions of the particles used in this study fall into two general categories, near mono-disperse and poly-disperse. Instrument performance analysis experiments were conducted with both types of particles. The near mono-disperse particles were also used as a QA/QC procedure to check the size calibration of the instruments.

Near Mono-disperse Suspensions

Near mono-disperse suspensions of polystyrene latex (PSL) particles were used for instrument performance analysis and size calibration checks for the following reasons:

- a. High quality, near mono-disperse PSL suspensions are available from several suppliers. The suspensions used in this study were purchased from Duke Scientific Corporation (Duke) in Palo Alto, CA.
- b. The average particle diameters have been determined with methods covered by consensus standards. The microscopic method that is used by Duke to determine the average diameter includes “calibration” with size-standard PSL particles from NIST. Therefore, Duke states that the average diameters are “certified” as traceable to NIST standard reference material particles.
- c. The diameters of the particles in the “certified” PSL suspensions have low standard deviations (i.e., they are “near” mono-disperse) and are shipped with a certificate that gives the average diameter of the particles. Five certified

suspensions with nominal diameters of 3, 5, 7, 10 and 15 microns were purchased for use in the study.

- d. The number of PSL particles in a given volume of “certified” PSL suspension can be estimated using manufacturer-supplied information that includes an equation presented later. The nominal as well as the “certified” diameters of each suspension along with the coefficient of variation (COV) associated with the particle size distribution are listed in Table 2.2, below. For example, the PSL suspension with a nominal diameter of 3 μm had a “certified” diameter of 3.063 μm with a COV of 1 %. All the suspensions had COVs that were less than 1.2 %. The solids content of each PSL suspension is also listed in the table along with the date of packing. The manufacturer stated that the solids content values are accurate to ± 10 %.

Table 2.2
Characteristics of the “certified” Duke Scientific polystyrene latex particles
(Source: Material data sheets from Duke Scientific Corporation, CA)

Nominal PSL particle diameter (μm)	Certified mean particle diameter (μm)	Coefficient of variation (%)	Date suspension was packed	Solids in the suspension (%)
3	3.063	1.0	5/1/98	0.48
5	4.991	1.2	2/20/98	0.30
7	6.992	1.0	4/7/98	0.28
10	9.975	0.9	2/17/98	0.22
15	15.02	1.0	11/01/97	0.28

Poly-disperse Suspensions

NIST ISO medium test dust was purchased as a candidate poly-disperse material for the count performance evaluation of on-line counters. NIST ISO medium test dust was selected for testing for the following reasons:

- a. It is a reference material readily available in 20 g quantities of dry powder from NIST. NIST ISO medium test dust suspended in hydraulic oil [Standard Reference Material (SRM) 2806] is used for the primary calibration of particle counters in the fluid power industry. This dust replaced General Motors’ Air Cleaner Fine Test Dust (ACFTD) that was used by the fluid power industry for many years. Powder Technology Inc. (PTI), of Burnsville, MN, provides the dust to NIST and NIST characterizes and packages it. NIST, in a study sponsored by

the fluid power industry through the National Fluid Power Association (NFPA), spent three years doing a detailed size characterization of this dust.

- b. An accurate size distribution of the particles in NIST ISO medium test dust is available from NIST (Fletcher et al. 1996). The distribution was determined (at NIST) using scanning electron microscopy (SEM) coupled with computerized image analysis. NIST used a \$300,000 grant from the fluid power industry to develop the measurement method. The particle “diameter” is the diameter of a circle that has the same projected area as the SEM image of the particle. This description of “size” is compatible with light obscuration particle counting when size calibration is done with spherical particles and each particle is detected using the shadow cast by the particle in the sensor’s light beam. The NIST size distribution results are presented below in Table 2.3.

The size distribution results in Table 2.3 are given as the number of particles per microgram of dry dust (number/ μg > particle diameter). For example, there are 9655 particles/ μg greater than the 2- μm diameter. During this study the numbers in this table were compared with particle counter measurements to estimate count efficiency for a poly-disperse suspension.

- c. It will be shown in Chapter 4 that a poly-disperse suspension used for count performance evaluation should have a size distribution that is similar to that of the particle suspensions that will be measured in filtered water. NIST ISO medium test dust was determined by the method discussed below to have this characteristic. According to Lawler et al. (1980), a power law relationship of the form given by Equation 2.1 describes the particle size distribution of many natural and water treatment suspensions:

$$\frac{dN}{dd_p} = A d_p^{-\beta} \quad (2.1)$$

where N is the cumulative number concentration of particles greater in size than the particle diameter d_p , A is a constant with a magnitude determined by the amount of particles (mass or volume) in the suspension and β is constant that describes the shape of the distribution. Lawler et

al. (1980) reported that the exponent (β) is typically between 1 and 5 for particle sizes between 1 and 200 μm with most suspensions being represented by β equal to 4.

Table 2.3
Particle size distribution for NIST ISO medium test dust from SEM and image analysis

Particle Diameter ¹ (μm)	Number/ μg > Particle Diameter	Particle Diameter ¹ (μm)	Number/ μg > Particle Diameter
1	38,714	16	40
2	9655	17	33
3	4003	18	27
4	2177	19	22
5	1335	20	18
6	855	21	15
7	562	22	13
8	377	23	10
9	259	24	9
10	183	25	7
11	134	26	6
12	100	27	5
13	77	28	4
14	61	29	4
15	49	30	3
¹ Diameter of a circle with the same projected area as the SEM image of the particle			

A cumulative distribution function of the form shown below in Equation 2.2 was derived using Equation 2.1 and fitted to the particle size distribution that NIST measured using SEM and image analysis:

$$N = \frac{A}{\beta - 1} d_p^{1-\beta} \quad (2.2)$$

Using this equation, a β value of 3.4 was determined for the NIST ISO medium test dust. This value is within the range reported by Lawler et al. (1980). Figure 2.2 shows the particle size distribution of NIST ISO medium test dust obtained from SEM analysis (Table 2.3 and Fletcher et al. 1996).

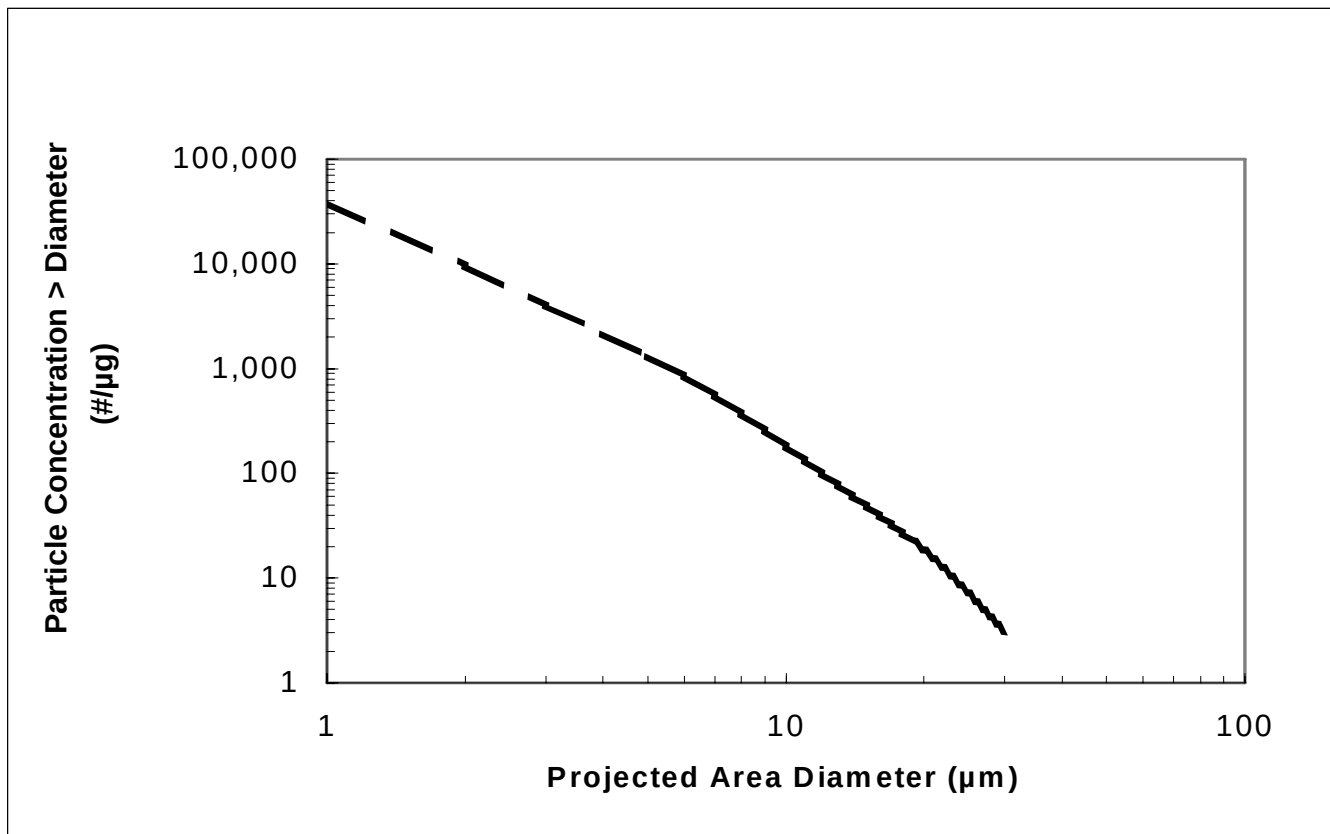


Figure 2.2 NIST ISO medium test dust size cumulative particle size distribution. Measured by NIST using scanning electron microscopy and custom image analysis software.

- d. Cleasby et al. (1989) used a HIAC/Royco particle counter to measure particle size distributions for filtered water samples from 21 filtration plants distributed across the United States. (A typical size distribution from Cleasby's report is given in Table 2.4). A distribution of the form described by Equation 2.2 was fitted to these data and β values between 2 and 4 with an average value of 3.03 were obtained. Table 2.5 lists the β values from the 21 filtration plants along with the coefficients of determination (R^2) obtained from fitting the equation to the particle size distributions. The R^2 values fall between 0.96 and 0.99.

Table 2.4
Filtered water HIAC/Royco results obtained for Tucaloosa, AL (from Cleasby et al. 1989)

Particle diameter (μm)	Particle number conc. ($\#/\text{mL}$ > diameter)
2.11	300
3.06	190
4.43	98
6.43	44
9.33	17
13.54	6
19.64	1
28.50	0
41.35	0
60.00	0

Table 2.5
“ β ” values and coefficients of determination (R^2) from an analysis of data in Cleasby et al. (1989) for filtered water results

Location	R^2	β	Location	R^2	β
Concord	0.991	3.80	Corvallis	0.993	2.98
Glendale	0.959	3.72	Tuscaloosa	0.982	2.96
Durham	0.972	3.54	Winnetka	0.994	2.89
Spartanburg	0.989	3.42	Merrifield	0.993	2.72
Phoenix (I)	0.964	3.22	Los Angeles	0.992	2.68
Phoenix (II)	0.990	3.18	Colorado Springs	0.988	2.55
Oakland	0.988	3.14	Eugene	0.997	2.42
Oceanside	0.976	3.03	Mission	0.981	2.07
Lake Oswego	0.985	3.02	Duluth	0.998	2.00
Las Vegas	0.994	3.01			
			AVERAGE		3.03

As part of their study for the fluid power industry, NIST (Fletcher et al. 1996) used a HIAC/Royco Model HR-LD 150 light extinction particle counter with batch sampler, size calibrated with ISO 4402:1991, to measure the particle size distribution of NIST ISO medium test dust suspended in hydraulic oil. The complete set of results is listed with the ISO medium test dust “certificate” at the NIST web site¹ (http://ois.nist.gov/srmcatalog/certificates/view_cert2gif.cfm?certificate=8631).

When Equation 2.2 was fitted to these results (over the 2 to 12 μm range), a β value of 2.34 was obtained. This is within the range of β values for the 21-filtration plants sampled by Cleasby et al. (1989) (See Table 2.5). Figure 2.3 shows filtered-water, HIAC-measured particle size distributions for 10 of the 21 water utilities monitored by Cleasby et al. (1989). The NIST measured HIAC results for ISO medium test dust in hydraulic oil (Fletcher et al. 1996) are included in Figure 2.3. The dust-in-hydraulic-fluid curve falls within the range of Cleasby's filtered water results.

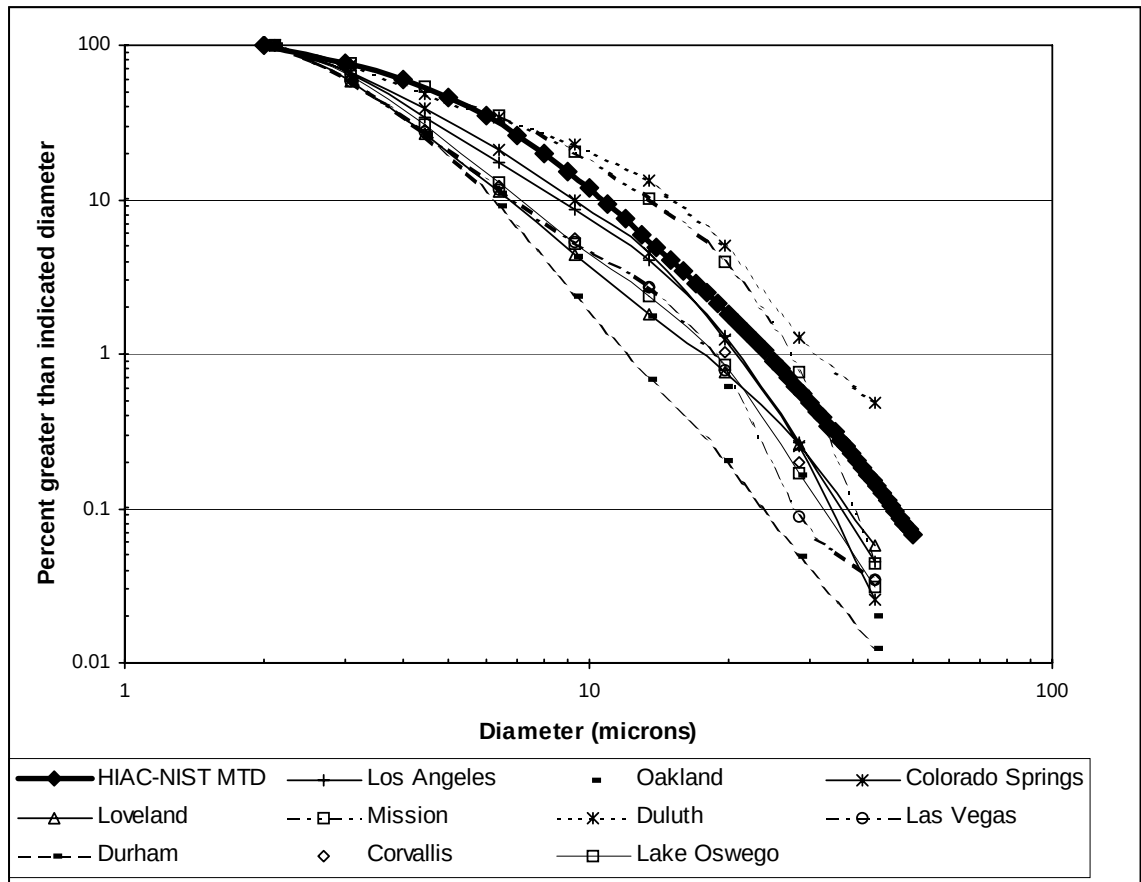


Figure 2.3 Comparison of filtered water results obtained by Cleasby et al. (1989) with results obtained for NIST ISO medium test dust in oil measured by NIST (Fletcher et al. 1998) using a HIAC/Royco particle counter.

The project team visited NIST in January of 1999 (See Appendix G) and described the attributes a poly-disperse dust must have to be effective in a count performance evaluation method for drinking water applications. The NIST experts were asked if there were particles from NIST (or any other source) that might be appropriate for our CPE purpose. The group concluded that NIST ISO medium test dust was the only reasonable option at that time.

Particles Used for QA/QC Analysis

The QA/QC measurements included an initial check of each instrument's size calibration curve and then occasional checks of this size calibration with time. The particles used in the QA/QC measurements are described below.

Size Calibration Verification

Each instrument was supplied with a size calibration curve that had been prepared by the instrument manufacturer using near mono-disperse PSL suspensions. Therefore, it was appropriate to verify this size calibration with near mono-disperse PSL suspensions. The PSL suspensions used for this purpose were the five “certified” PSL suspensions from the instrument performance evaluation experiments. Their characteristics are shown in Table 2.2.

Stability of Size Calibration Over Time

The stability of the size calibration over time for all the on-line counters was tested as a QA/QC measure using “research-grade” PSL suspensions. “Research-grade” PSL suspensions are much less expensive than “certified” PSL suspensions and therefore were more appropriate for repeated, long-term stability check experiments. The COV and standard deviation of the particle size distribution are higher for research grade PSL than for “certified” PSL suspensions.

Two “research grade” PSL suspensions were used. The first suspension (Lot number 19156) had a mean diameter of 8.1 μm and COV of 16 %. The second suspension (Lot number 20363) also had a mean diameter of 8.1 μm and a coefficient of variation of 16 %. Both suspensions had a solids content (reported by the manufacturer) of 10 %.

Low Particle Dilution Water (RO Water)

A small reverse osmosis unit manufactured by US Filter, Lowell, MA was used to produce low particle dilution water (RO water). The flow rate when the unit was operating effectively was 0.3 to 0.4 gallons per minute (0.019 to 0.025 L/s). The RO water particle counts were typically between 0 and 10/mL $> 2 \mu\text{m}$ (measured with the Met One grab sampler using water samples from the reservoir above the counters on the test stand). The water was used in all the experiments to prepare the test (PSL and NIST ISO medium test dust) suspensions. It was also used to wash and rinse, tubing, glassware and sample bottles.

METHODS

This section describes the experimental methodology. It has been divided into four sub-sections:

1. Container and apparatus washing procedures
2. Instrument performance analysis using PSL suspensions
3. Instrument performance analysis using NIST ISO medium test dust.
4. QA/QC experiments

The first section outlines the washing procedures used to obtain clean glassware and other containers. The second part describes the instrument performance evaluation experiments with “certified” PSL suspensions and the third part describes the experiments with NIST ISO medium test dust. The fourth section describes the size calibration verification experiments with the “certified” PSL suspensions and the experiments that checked the stability of the size calibration with time using “research-grade” PSL suspensions.

Container and Apparatus Washing Procedures

Washing procedures outlined by Chowdhury et al. (1997) and Hargesheimer and Lewis (1995) were reviewed before a sample container washing protocol was developed. All the sample containers and glassware used in the experiments were washed according to the following steps. About 0.5 mL of liquid detergent was added to each bottle along with 20 mL of tap water. The bottle was closed and shaken and a clean nylon bristle brush was used to scrub the insides of the bottle. It was rinsed with tap water until all the surfactant was washed away. 10 mL of 0.02 N

NaOH solution was then added to each bottle and the bottle was filled to the top with RO water. The bottle was closed and left undisturbed. After two hours, the bottles were emptied and rinsed thoroughly three times with RO water. Each bottle was then filled with RO water and covered with parafilm until it was used. In some cases the RO water was checked with the Met One grab sampler after storage.

Instrument Performance Analysis Using PSL Suspensions

The instrument performance analysis of the on-line instruments using PSL suspensions involved determining count efficiencies and sensor resolutions for each counter with each of the five “certified” PSL suspensions. Similar experiments were conducted with each of the five suspensions. A typical experiment is described below.

The gravity flow reservoir was filled to the 50-liter mark with RO water. The reservoir mixer was set at 350 rpm and then switched on as the reservoir was filling. The particle count for the RO water was checked with the Met One grab sampler.

The container in which the PSL suspension was supplied was swirled gently 4 -5 times and then placed in the ultrasonic bath for 30 seconds. 2-3 drops of PSL suspension were squeezed from the nozzle attached to the container into a 25-mL clean glass beaker. A micropipette was used to extract the desired volume of the suspension and dispense it into 30 mL of RO water in a 50-mL beaker. The diluted suspension in the beaker was ultra-sonicated for 30 seconds and then poured quickly into the reservoir. Water from the reservoir was used to rinse the container. The suspension was allowed to mix in the reservoir for ten minutes.

The computers and the software that were used to collect data from the four counters were started and the threshold settings were set in the software. The threshold settings are listed in the next section. The software was set to begin collecting data from the counter sensors.

The suspension was mixed in the reservoir for 10 minutes and then allowed to flow to the counters by opening the four valves below the manifold. The readings measured by the four counters were recorded by the software.

The volume of the stock PSL suspension that was pipetted and added to the reservoir was different for each PSL particle size. The volumes used are shown in Table 2.6.

Table 2.6
Volumes of “certified” PSL suspensions

Nominal PSL particle diameter (μm)	Certified mean particle diameter (μm)	Solids in the suspension (%)	Volume of RO water in reservoir (V_r) (L)	Volume of stock suspension added (V_s)(mL)	Estimated concentration (#/mL)
3	3.063	0.48	40	0.280	1902
5	4.991	0.30	50	1.1	2018
7	6.992	0.28	50	12	1872
10	9.975	0.22	50	23.8	2018
15	15.02	0.28	50	60	1903

Threshold Settings

For each counter the threshold settings were varied according to the PSL suspension in use. The settings are listed in Table 2.7. For example, for Counter B, and the measurements with the 3.063 μm PSL suspension, the size thresholds were set at close intervals between 2.0 and 3.5 μm (at 2.0, 2.2, 2.5, 2.7, 2.9, 3.0 and 3.5 μm), whereas for the experiment with the 4.991 μm suspension, the size thresholds were set at close intervals between 3.5 and 5.5 μm (at 3.5, 4, 4.5, 4.7, 5, 5.2 and 5.5 μm).

Flow Rates Through the Sensors

The control of the flow rate through the sensor is important since particle count results are reported on a volume basis and the counter electronics record the counts over a time interval (Hargesheimer and Lewis 1995). To convert, for example, the number of particles counted in 1 minute to a concentration in counts per mL you must know the volume that passed through the sensor in the one-minute counting period. Therefore, the flow rates through all the counters were measured during each size calibration verification experiment. The sensor flow rate was measured using a graduated cylinder and stopwatch before and after the particle concentration measurements were logged. The graduated cylinder was used to time the collection of a measured quantity of water leaving the effluent/outlet tube of the sensor of each counter. In all the experiments the final measured flow rate was within ± 2 mL/min of the manufacturer’s recommended value.

Table 2.7
Threshold settings for PSL experiments – used to determine
mean particle diameter and the instrument resolution

COUNTER A	
Certified/Mean diameter (microns)	Threshold settings (microns)
3.063	2, 3, 4, 5, 6, 7, 8, 9
4.991	2, 3, 4, 5, 6, 7, 9, 11
6.992	2, 4, 5, 6, 7, 8, 9, 10
9.975	2, 5, 7, 8, 9, 10, 11, 12
15.02	2, 10, 12, 13, 14, 15, 16, 18
COUNTER B	
Certified/Mean diameter (microns)	Threshold settings (microns)
3.063	2, 2.2, 2.5, 2.7, 2.9, 3.0, 3.5, 4, 5, 7, 9, 12
4.991	2, 3, 3.5, 4, 4.5, 4.7, 5, 5.2, 5.5, 6, 7, 9, 12
6.992	2, 4, 5, 5.5, 6, 6.5, 7, 7.5, 8, 9, 10, 15
9.975	2, 4, 6, 8, 8.5, 9, 9.2, 9.5, 9.7, 9.9, 10, 10.5, 11, 12, 15
15.02	6, 10, 13, 13.5, 14, 14.5, 14.7, 14.9, 15, 15.5, 16, 17, 19
COUNTER C	
Certified/Mean diameter (microns)	Threshold settings (microns)
3.063	2, 2.2, 2.5, 3, 3.5, 4, 6, 8
4.991	2, 4.5, 5, 5.3, 5.5, 6, 8, 11
6.992	2, 4.5, 5, 5.3, 5.5, 6, 8, 11
9.975	2, 9, 9.5, 9.7, 10, 10.5, 11, 12
15.02	2, 12, 13, 14, 14.5, 15, 16, 18
COUNTER D	
Certified/Mean diameter (microns)	Threshold settings (microns)
3.063	2, 2.2, 2.5, 2.7, 2.9, 3.5, 4, 4.5, 5, 7, 9, 12
4.991	2, 3, 4, 4.5, 4.7, 5, 5.2, 5.5, 6, 6.5, 7, 9, 12
6.992	2, 3, 4, 4.5, 4.7, 5, 5.2, 5.5, 6, 6.5, 7, 9, 12
9.975	2, 4, 6, 8, 8.5, 9, 9.2, 9.5, 9.7, 9.9, 10, 10.5, 11, 12, 15
15.02	2, 5, 8, 11, 12, 13, 14, 15, 16, 17, 18, 20

Reverse Osmosis (RO) Dilution Water Background Counts

RO background counts were checked in all the experiments to verify that the concentration of particles in the dilution water did not have a significant effect on the measured PSL counts.

RO grab samples were obtained from the reservoir before starting the experiment by disconnecting one of the quick disconnects above the counter feed lines and collecting 1 L of RO water in a sample bottle. The grab sampler was used to measure the particles in this sample. Ten

readings were taken using the grab sampler and the average of these readings was calculated to give the average RO background count for each experiment. The average RO counts were recorded and were between 5 and 20 counts/mL > 2 µm in all experiments. The RO counts were always less than 1 % of the PSL or dust suspension counts used in the experiment.

Measurements

Count versus size readings. The basic data obtained from the counters during each experiment consisted of count versus size readings. In all experiments data was recorded for a total time period of 20 minutes. The data recorded in the first 3 minutes was not used in the analysis because it usually took 2 minutes for the readings to stabilize.

The formatted data sheets used for each of the counters are shown in Appendix A. All the data sheets are from an instrument performance analysis experiment conducted on 5/20/98 using a PSL suspension with a “certified” diameter of 6.992 microns.

From these datasheets it can be seen that counters A, B and D recorded counts/mL greater than threshold settings and counter C recorded counts/mL in the “bins” between each threshold setting. The count value for each 20-minute experiment is the average of the 10 to 15 individual particle count readings logged by the instrument software.

Count efficiency. The measured count (counts/mL > 2 µm) for each experiment is the average of the 10 to 15 logged values for each test suspension. For example, from the data sheet obtained for instrument A from the experiment performed on 5/20/98 and shown in Appendix A, the measured count is 1546 particles/mL. The measured counts were used to determine count efficiencies using the following equation:

$$\text{Count efficiency (\%)} = \frac{(\text{Measured count, \#/mL} > 2 \mu\text{m}) (100)}{(\text{Estimated PSL particle conc., \#/mL} > 2 \mu\text{m})} \quad (2.3)$$

The estimated PSL particle concentration (#/mL > 2 µm) was determined using the following relationship from Duke Scientific:

$$EC = \frac{0.06 \cdot S \cdot \rho \cdot V_s}{\pi \cdot \bar{d} \cdot \rho_p \cdot V_r} \quad (2.4)$$

where EC is the estimated concentration of PSL particles in the gravity flow reservoir, V_s is the volume of Duke PSL suspension added to the reservoir, V_r is the volume of RO water in the reservoir, ρ_p is the density (1.05 g/cm^3) of the PSL particles, ρ is the density (1.0 g/cm^3) of the PSL stock suspension, S is the percent solids concentration in the stock suspension and $\bar{d}$ is the certified mean diameter of the PSL particles. The expected value of EC is within the interval $EC \pm 0.15 \times EC$.

Instrument resolution. For each counter two values of the resolution were determined for each “certified” near mono-disperse PSL suspension, one for the measurements to the left of the mean particle diameter and one for those on the right. The calculations used are based on the method employed by the pharmaceutical industry and detailed in USP 788 (USP 1992). The USP method was adapted for use in spreadsheet calculations. The calculations are discussed below.

The cumulative particle count greater than each threshold setting was obtained from the count versus threshold setting data for each counter. From this the fraction of particles greater than each setting, starting with 100 % greater than $2 \mu\text{m}$ was determined. This fraction is labeled “f”. The denominator in this fraction was always the count greater than $2 \mu\text{m}$ because $2 \mu\text{m}$ was the lowest threshold setting for each of the four counters in all the experiments.

The “f” values were used to calculate values of the standard normal deviate (z) and these were plotted versus the measured PSL particle diameter. The magnitude of z is equal to the quantity $(d - \bar{d})/\sigma_p$ where $\bar{d}$ is the measured mean particle diameter (the diameter at $z = 0$) and σ_p is the standard deviation of the particle diameter from the COV in the manufacturer’s literature. Each value of “f” was converted to a value of “z” using the Excel spreadsheet function $z = @normsinv(f)$. A detailed explanation of how “z curves” were obtained is provided in Appendix B. The instrument resolution, R, for the points to the left and right of the mean diameter was determined using the z-curves and the following equation,

$$R = \frac{100 \times \sqrt{\sigma_m^2 - \sigma_p^2}}{\bar{d}} \quad (2.5)$$

where σ_m is the standard deviation of the measured counts (left or right).

In Equation 2.5 the standard deviation (σ_m) for the points that fall to the left of the mean is equal to the mean particle diameter minus the diameter at $z = +1$. The standard deviation for

the points that fall to the right of the mean is the diameter at $z = -1$ minus the mean diameter. A “z curve” obtained for Counter B for an experiment performed on 6/11/98 using the 7 μm diameter (nominal) PSL suspension is shown in Figure 2.4. The measured mean particle diameter, i.e., the particle diameter at $z = 0$, is 6.3 μm . The diameter at $z = +1$ is 4.7 μm and the diameter at $z = -1$ is 6.8 μm . The standard deviation for the points to the left of the mean is therefore $6.3 - 4.7 = 1.6 \mu\text{m}$ and the standard deviation for the points to the right is $6.8 - 6.3 = 0.5 \mu\text{m}$. Since the particle standard deviation (σ_p) for the 7 μm PSL is 1% of 6.992 μm or 0.07 μm (Table 2.2), the resolution (from Eq. 2.5) is 17 % for the points on the left and 10 % for the points on the right.

Instrument Performance Analysis Using NIST ISO Medium Test Dust

NIST ISO medium test dust was also used to determine the counting efficiency for each counter. Measured counts for a known mass concentration of dust were compared with the counts measured by NIST using SEM and image analysis. For a 2 μm threshold setting (based on size calibration with PSL micro-spheres) the count efficiency is given by the following equation:

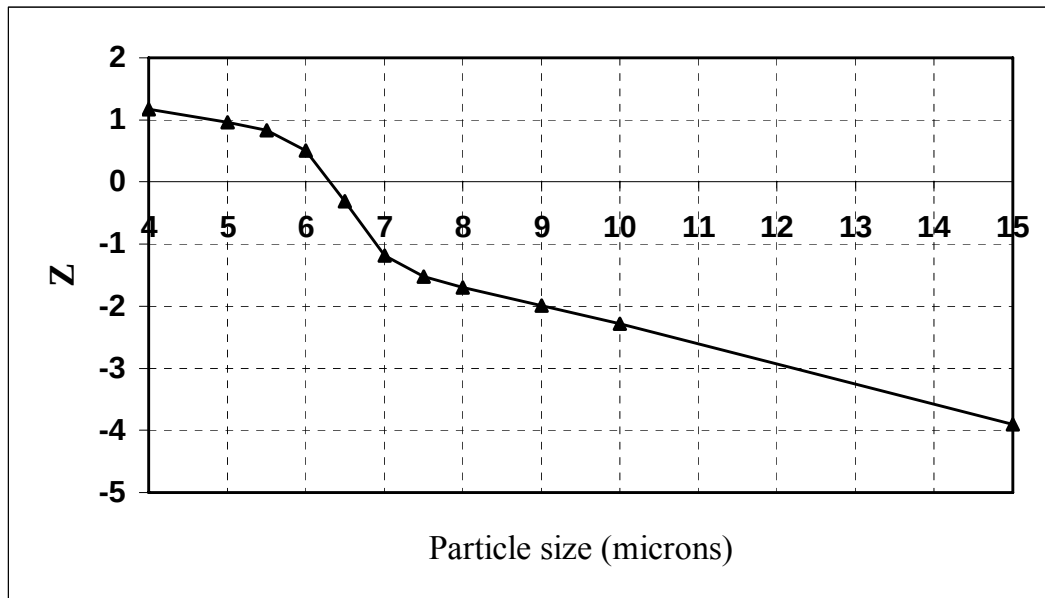


Figure 2.4 Typical "z curve" used to determine the instrument resolution

$$\text{Count efficiency (\%)} > 2 \mu\text{m} = \frac{(\text{Measured counts per unit mass of dust}) (100)}{(9655)} \quad (2.6)$$

where 9655 = number of particles $> 2 \mu\text{m}$ per μg of NIST ISO medium test dust (See Table 2.3).

NIST ISO medium test dust stock suspensions

The NIST ISO medium test dust was purchased as 20 g of dry powder. The 20 g quantity was divided into 300 mg portions using the micro-riffler. The smaller portions facilitated storage and handling and the preparation of useful and economical suspensions. The portions suspended in 100 mL of RO water are called “stock suspensions”. Known volumes of stock suspensions were diluted in the gravity flow reservoir to obtain “working or test suspensions” that were used in the experiments for determining the counting efficiency.

The entire 20 g were placed in the bowl of the rotary micro-riffler and each of the eight collector funnels was fitted with a test tube. The motor that rotated the collector head was switched on. The 20 g sample was thus riffled into eight representative samples of approximately 2.5 g each.

One of the 2.5 g portions from the first step was placed in the bowl of the riffler and riffled into eight clean test tubes. Each test tube contained the bottom half of an empty cellulose capsule. The capsules, top and bottom halves together, had been carefully pre-weighed. After the dust was riffled into the bottom half of the capsule in the test tube, the top half was inserted and the entire capsule with dust was weighed. The weight of dust was determined by subtracting the weight of the empty capsule. Thus, eight representative samples of approximately 300 mg of dust were obtained. Each of these representative samples was suspended in 100 mL of RO water in a leak-proof container by gentle swirling followed by ultrasonification for 20 seconds to give the “stock” suspensions.

NIST ISO Medium Test Dust Working Suspensions

The instrument performance analysis experiments with NIST ISO medium test dust consisted of a set of five gravity flow experiments. In each experiment, a known volume of NIST ISO medium test dust stock suspension was introduced into a known volume of RO water in the

reservoir to prepare the working suspension. The volume of the stock suspension used to prepare the working suspension depended on the number concentration of dust particles desired in the reservoir.

The volume of stock suspension used to prepare the working or test suspension and the volume of RO water in the reservoir for each of the five experiments is shown in Table 2.8 below. All of these experiments were essentially the same except that the volume of the stock suspension added to the reservoir was varied. The experiments are described in detail below.

The counts measured by each counter for that working suspension were recorded. This experiment was repeated with the five different volumes of stock suspension in the reservoir.

Threshold Settings for NIST ISO Medium Test Dust Experiments

The threshold settings were the same for all the performance evaluation experiments with NIST ISO medium test dust and are shown in Table 2.9.

Table 2.8
Volume of NIST dust stock suspension used for each performance evaluation experiment

Experiment No.	Target (Approximate) number concentration in reservoir > 2 microns (PSL diameter) (#/mL)	Volume of RO water in reservoir (L)	Volume of stock suspension used to prepare working suspension (mL)
1	30	40	0.140
2	100	40	0.467
3	300	40	1.400
4	1000	40	5.300
5	3000	40	14.000

Flow Rates Through Sensors

The flow rates through all counter sensors were measured according to the procedures used for the PSL experiments. The measured flow rates were within 1% of the manufacturers recommended flow rates.

RO Background Counts

The RO background counts were recorded by all the counters (before the dust was added to the reservoir) using 10 L of the RO water in the 50 L reservoir. The PSL suspension was added to the remaining 40 L of RO water in the reservoir after the background counts were logged in the instrument software. In these experiments, the measured counts were corrected for the RO background counts as explained below.

Measurements

Count versus size readings. Data obtained from the on-line counters after each performance evaluation experiment consisted of count versus threshold setting. In all of these experiments, as in the PSL experiments, data obtained in the first 3 minutes was discarded.

The data sheets used for the on-line counters were similar to the sheets employed in the PSL experiments. Examples are shown in Appendix C along with the RO background correction that was applied.

Table 2.9
Threshold settings for performance evaluation experiments
using NIST ISO medium test dust

Counter	Threshold Settings (μm)
A	2, 3, 4, 5, 6, 7, 10, 15
B	2, 2.5, 3, 3.5, 4, 5, 6, 7, 8, 9, 10, 12, 15, 20
C	2, 2.5, 3, 5, 7, 9, 12, 16
D	2, 2.5, 3, 3.5, 4, 4, 5, 6, 7, 8, 9, 10, 12, 15, 20

RO background correction and average corrected counts. The average counts/mL $> 2 \mu\text{m}$ was determined for all the counters using the same averaging technique used for the PSL experiments. The counts were corrected for RO background count by subtracting the average RO background count $> 2 \mu\text{m}$ from the average counts/mL $> 2 \mu\text{m}$ to obtain the average corrected counts. The final corrected counts were obtained using the following equation:

$$\text{Average corrected counts} = (\text{Average counts}) - (\text{RO background counts}) \quad (2.7)$$

Measured counts per unit mass of dust. The average corrected counts > 2 µm in number per volume of suspension obtained for each experiment were converted to counts per unit mass > 2 µm using the following equation:

$$\text{Counts per unit mass of dust (\#/}\mu\text{g)} = \frac{(\text{Average corrected counts})(1000)}{(\text{Mass concentration of dust in reservoir})} \quad (2.8)$$

The average corrected counts have units of counts/mL. The concentration of dust in the reservoir is given by the following equation:

$$\text{Mass concentration of dust in reservoir (\mu g/L)} = \frac{(V)(W)}{(100)(40)} \quad (2.9)$$

where V = volume of NIST ISO medium test dust stock suspension added to reservoir (mL), W = weight of NIST ISO medium test dust used to make stock suspension (µg), 100 = volume of RO water in the stock suspension (mL), and 40 = volume of RO water in the reservoir (L).

The measured counts per unit mass were used to obtain count efficiencies using the following equation:

$$\text{Count efficiency (\%)} > 2 \mu\text{m} = \frac{(\text{Measured counts per unit mass of dust})(100)}{(9655)} \quad (2.10)$$

where 9655 = number of particles > 2 µm per µg of NIST ISO medium test dust measured by NIST using SEM and image analysis (Table 2.3)

QA/QC EXPERIMENTS

Three types of QA/QC experiments were performed:

- a. Size calibration verification
- b. Stability of size calibration over time
- c. Stability of NIST ISO medium test dust stock suspensions over time

The first two types of experiments were done using PSL suspensions and the third involved NIST (and PTI) ISO medium test dust stock suspensions. The methods and materials are described in the next several sections.

Size Calibration Verification

Size calibration verification involved determining mean particle diameters using counter measurements made in the instrument performance analysis experiments with the five “certified” PSL suspensions. The mean measured diameters were compared to the “certified” diameters using statistical analysis.

The average particle size distributions obtained from the instrument performance analysis experiments with the PSL suspensions were used to obtain “z curves” following the procedure explained in Appendix B. The z-curves were used to determine the mean measured PSL particle diameter for each experiment. The measured mean is located at $z = 0$. Table 1D in Appendix D shows the mean measured diameters obtained for each PSL experiment.

The mean measured diameters were compared with the “certified” diameters to determine if the differences between them were statistically significant. The hypotheses tested and the results of this analysis are presented in Table 2D of Appendix D. The maximum percent difference between the measured and “certified” diameters did not exceed 10%. It was concluded from this analysis that for counters A and B, the mean of the measured diameters was not significantly different from the “certified” diameter. For counters C and D, these differences were statistically significant.

Stability of Size Calibration Over Time

The less expensive “research-grade” PSL suspensions with a nominal mean diameter of 8.1 μm were used to check the stability of size calibration over time. These experiments were repeated eleven times between 5/20/98 and 6/1/99. The mean diameters measured by each of the counters with the 8.1 μm PSL suspension over this period of time were determined and a statistical analysis was used to determine if the measured mean diameters followed a significant trend with time. A typical stability check experiment is described below.

The gravity flow reservoir was filled up to the 40 L mark with RO water. The mixer was set at 350 rpm and then switched on as the reservoir was filling.

A 240 μL quantity of the “research grade” PSL suspension was pipetted following the same procedure explained in the size calibration verification experiments using the ultrasonicator, two glass beakers and micropipettes. After allowing this suspension to mix in the reservoir for 10 minutes, the suspension was allowed to flow through the four counter sensors and the measurements from the counters were recorded. This experiment was repeated on a series of dates over a time period of approximately one year.

Flow rates through sensors. The flow rates were measured for each stability check experiment using the same procedure that was used for the size calibration verification experiments. The measured flow rates were within ± 2 mL/min of manufacturer recommended flow rate.

Threshold settings. The threshold settings shown in Table 2.10 were used for all the stability check experiments.

Table 2.10
Threshold settings used in stability check experiments

Counter	Threshold settings (microns)
A	2, 4, 6, 7, 8, 9, 10, 12
B	2, 3, 4, 5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 10, 12, 15
C	2, 4, 6, 7, 8, 9, 10, 15
D	2, 3, 4, 5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 10, 12, 15

Measured data. The count vs. size data obtained from all the counters for each experiment was similar to the data measured from the instrument performance analysis experiments with PSL suspensions (Appendix A). This data was averaged in a similar manner to obtain average particle size distributions.

Mean measured diameters for each date were obtained from the average particle size distributions using the “z curves” as explained in Appendix B. These measured mean diameters for each counter were analyzed by regression analysis to determine if they followed a significant trend with time. The results of this analysis are presented in Appendix D.

It was concluded from this analysis that none of the counters showed a significant trend in measured diameters over time. Therefore, the size calibration was stable over time for all the counters.

Stability of Stock Suspensions Over Time

The instrument performance analysis experiments with NIST ISO medium test dust were conducted using the same stock suspension on different dates. It was therefore essential to determine if the stock suspensions were reasonably stable with time.

The stock suspension stability experiments were conducted by preparing grab samples of diluted stock suspensions on different dates and comparing the measured counts (counts/mL > 2 μ m) to determine if they followed a statistically significant trend with time.

A detailed description of the procedure followed and the results are provided in Appendix E. These results show that the stock suspensions were stable over time.

CHAPTER 3

RESULTS OF THE INSTRUMENT PERFORMANCE ANALYSIS TESTS

INTRODUCTION

This chapter presents the results of the instrument performance analysis (IPA) and other miscellaneous measurements used to characterize the particle counter instruments and to obtain information for developing the count performance evaluation (CPE) protocol. The material is contained in three sections:

- Sensor resolution determination with polystyrene latex suspensions.
- Instrument performance analysis with polystyrene latex suspensions.
- Instrument performance analysis with NIST ISO medium test dust.

The first section presents the resolution values that were measured using Duke certified polystyrene latex microspheres. The second shows the count efficiencies that were measured in the IPA experiments with PSL suspensions and the third section covers the count efficiencies from the IPA experiments with NIST ISO medium test dust. Results that show the effect of the dust concentration and the effect of the threshold setting on the dust count efficiency are also presented. The results of the tests used in developing the count performance evaluation (CPE) protocol are presented in Chapter 5.

SENSOR RESOLUTION DETERMINATION WITH PSL SUSPENSIONS

Equation 2.5 and the method described in Chapter 2 and Appendix B were used to determine the left and right resolution for each counter and a range of PSL microsphere diameters. Table 3.1 shows all the calculated values of R for each particle counter, for the points to the left and right of the mean, for the five PSL suspensions, and for the tests of June and August of 1998.

Table 3.1
Instrument resolutions from experiments with certified diameter PSL suspensions

COUNTER A				
Certified PSL Particle Diameter (μm)	Resolution (%)			
	June		August	
	Left	Right	Left	Right
3.063	5	10	ND	13
4.991	15	15	15	14
4.991	18	17	15	14
6.992	20	16	17	11
6.992	17	10	17	11
9.975	7	8	20	6
15.02	8	4	13	4
COUNTER B				
Certified PSL Particle Diameter (μm)	Resolution (%)			
	June		August	
	Left	Right	Left	Right
3.063	8	11	11	11
4.991	17	10	24	12
4.991	17	11	24	12
6.992	21	10	31	12
6.992	24	9	31	12
9.975	40	10	43	11
15.02	ND	10	ND	5
COUNTER C				
Certified PSL Particle Diameter (μm)	Resolution (%)			
	June		August	
	Left	Right	Left	Right
3.063	9	9	9	11
4.991	3	16	ND	8
4.991	7	8	ND	8
6.992	9	9	9	10
6.992	5	8	9	10
9.975	5	6	ND	7
15.02	4	7	6	7
COUNTER D				
Certified PSL Particle Diameter (μm)	Resolution (%)			
	June		August	
	Left	Right	Left	Right
3.063	11	9	7	7
4.991	5	14	4	3
4.991	5	4	4	3
6.992	7	6	5	5
6.992	8	7	5	5
9.975	3	4	7	3
15.02	5	4	3	4

A number of standards [e.g., USP 788 (USP 1992) and ISO 11171:1999 (ISO 1999)] suggest a maximum resolution of 10 %. In the case of the ISO standard this is for a 10- μ m-diameter particle. In general, Counters A and B had inferior resolution ($R > 10$ %) and Counters C and D had superior resolution ($R < 10$ %). For Counter A, the R-values were between 4 and 20 %, for Counter B between 5 and 43 %, for Counter C between 3 and 16 % and for Counter D between 3 and 14 %.

Counters A and B had R-values that were consistently higher (the resolution was poorer) on the left (toward smaller diameters) than on the right (toward larger diameters). This difference between the left and right values of R was greater with Counter B than Counter A. The average of the left and right R-values tended to be lower (i.e., the overall resolution was higher) for Counters C and D. For example in June, for Counter B the average R was 21 % on the left and 10 % on the right. The average R-values on the left and right for Counter C in June were 6 and 9 %, respectively. The mean and median R-values for June and August measurements combined are listed in Table 3.2.

The results in Table 3.1 were examined to determine if the R-values varied in a consistent way with the diameter of the microspheres used to make the measurements. In one case, Counter B on the left in both June and August, the magnitude of R appeared to increase as the diameter of the microspheres increased. However, in all other cases the R-value did not vary in any consistent way, i.e., the same way in both June and August, with the diameter of the test spheres.

Table 3.2
Resolution (median and mean R-values) for the four on-line counters -June and August measurements are combined

	Counter A		Counter B	
	Left	Right	Left	Right
Median	15 %	12.5 %	24 %	10 %
Mean	14.4 %	10.9 %	24.3 %	10.4 %
	Counter C		Counter D	
	Left	Right	Left	Right
Median	7 %	8.5 %	5 %	7 %
Mean	6.8 %	8.9 %	5.8 %	5.7 %

INSTRUMENT PERFORMANCE ANALYSIS USING PSL SUSPENSIONS

Count Efficiency

Count efficiencies were calculated using Equation 2.3 (Chapter 2) for all four counters using the measured counts obtained from IPA experiments with the “certified” PSL suspensions. Table 3.3 shows the measured and estimated counts along with the count efficiencies for each PSL suspension for both the June and August 1998 experiments. For all four counters the lowest count efficiency was for the 3 μm (nominal) diameter PSL, the smallest particle in the set tested. With the 3 μm PSL, Counter A had count efficiencies of 74 -78 %, Counter B had count efficiencies between 60 and 65 %, Counter C between 55 and 63 % and Counter D between 50 and 69 %.

The instruments with relatively poor resolution (Counters A and B with median R-values, left and right, of 15 and 12.5 %, and 24 and 10 %, respectively) had the higher count efficiencies with the 3 μm diameter PSL (between 60 and 78 %). The instruments with relatively good resolution (Counters C and D with median R-values, left and right, of 7 and 8.5 %, and 5 and 7 %, respectively) had somewhat lower count efficiencies with the 3 μm PSL (between 50 and 69 %). The ability of a counter to detect the 3 μm PSL particles may be related, in an inverse way, to the resolution of the sensor.

INSTRUMENT PERFORMANCE ANALYSIS USING NIST ISO MEDIUM TEST DUST MEASURED COUNTS

The measured counts (count $> 2 \mu\text{m/mL}$), corrected for the RO background concentration as explained in the "Methods" section of Chapter 2, are listed in Table 3.4. The concentration of dust in the reservoir was varied from 12 $\mu\text{g/L}$ to 1166 $\mu\text{g/L}$. The experiment at the lowest dust concentration (12 $\mu\text{g/L}$) was done twice. As expected, the measured counts (count $> 2 \mu\text{m/mL}$) for all the counters increased as the dust concentration was increased. For each dust concentration, Counter A (the instrument with the highest average R, i.e., the poorest overall resolution) showed the highest number counted except at the highest dust concentration. (Counter A also had the highest overall count efficiency for PSL particles).

Table 3.3
Count efficiencies obtained in instrument performance analysis experiments with PSL
suspensions and threshold setting of 2 μm

COUNTER A						
Certified Diameter (μm)	June		August		Count efficiency (%)	
	Mean estimated counts (#/mL)	Measured counts (#/mL)	Mean estimated counts (#/mL)	Measured counts (#/mL)	June	August
3.063	1902	1399	1902	1488	74	78
4.991	2018	2165	1009	1232	107	122
4.991	1009	1273	1009	1232	126	122
6.992	1872	1877	459	522	100	114
6.992	1872	1357	459	522	72	114
9.975	2018	1739	2018	2119	86	105
15.02	1903	2012	1903	2042	106	107
Average =					96	109
Std Dev =					20	15
COV % =					20	14
COUNTER B						
Certified Diameter (μm)	June		August		Count efficiency (%)	
	Mean estimated counts (#/mL)	Measured counts (#/mL)	Mean estimated counts (#/mL)	Measured counts (#/mL)	June	August
3.062	1902	1134	1902	1236	60	65
4.991	2018	1961	1009	912	97	90
4.991	1009	920	1009	912	91	78
6.992	1872	1698	459	359	91	78
6.992	1872	1871	459	359	100	78
9.975	2018	1618	2018	2093	80	104
15.02	1903	2025	1903	2145	106	113
Average =					89	88
Std Dev =					15	16
COV % =					17	18
COUNTER C						
Certified Diameter (μm)	June		August		Count efficiency (%)	
	Mean estimated counts (#/mL)	Measured counts (#/mL)	Mean estimated counts (#/mL)	Measured counts (#/mL)	June	August
3.063	1902	1041	1902	1190	55	63
4.991	2018	1702	1009	816	84	81
4.991	1009	809	1009	816	80	81
6.992	1872	ND	459	310	ND	68
6.992	1872	1616	459	310	86	68
9.975	2018	1233	2018	1706	61	85
15.02	1903	1488	1903	1667	78	88
Average =					74	76

Std Dev =					13	10
COV % =					18	13
COUNTER D						
Certified Diameter (μm)	June		August		Count efficiency (%)	
	Mean estimated counts (#/mL)	Measured counts (#/mL)	Mean estimated counts (#/mL)	Measured counts (#/mL)	June	August
3.063	1902	960	1902	1306	50	69
4.991	2018	1556	1009	821	77	81
4.991	1009	754	1009	821	75	81
6.992	1872	1311	459	312	70	68
6.992	1872	1516	459	312	81	68
9.975	2018	1046	2018	1707	52	85
15.02	1903	1140	1903	1714	60	90
Average =					66	77
Std Dev =					12	9
COV % =					19	12

Table 3.4
Counts obtained in instrument performance analysis experiments with NIST ISO medium test dust

Concentration of dust in the reservoir (μg/L)	Corrected count (count/mL > 2 μm)			
	Counter A	Counter B	Counter C	Counter D
12	54	33	31	31
12	61	33	30	35
39	170	101	99	116
117	429	282	265	313
441	1314	1091	993	1209
1166	2419	2875	2567	3106

Count Efficiencies

The average corrected counts (count/mL > 2 μm) were converted to counts per unit mass of dust using Equation 2.5 derived in Chapter 2 and shown below in Equation 3.1:

$$\text{Counts per unit mass of dust (\#/}\mu\text{g)} = \frac{(\text{Average corrected counts}) (1000)}{(\text{Concentration of dust in reservoir})} \quad (3.1)$$

Values from Equation 3.1 were used to determine count efficiencies using Equation 2.6, from Chapter 2, i.e.,

$$\text{Count efficiency (\%)} > 2 \mu\text{m} = \frac{(\text{Measured counts per unit mass of dust}) (100)}{(9655)} \quad (3.2)$$

where 9655 is the number of particles $> 2 \mu\text{m}$ per μg of ISO medium test dust (RM 8631) as measured by using SEM and image analysis (Table 2.4, Chapter 2).

Table 3.5 shows the counts per unit mass (number/ $\mu\text{g} > 2 \mu\text{m}$) and the count efficiencies for all the counters. According to the results listed in this table, none of the counters had count efficiencies of more than 50 %. The average counting efficiency was highest for Counter A at 37 %. The highest counting efficiencies were obtained at the lowest dust concentrations. For instance, Counters A and B had maximum counting efficiencies of 50 % and 28 % at a dust concentration of 12 $\mu\text{g/L}$.

Table 3.5
Count efficiencies obtained from instrument performance analysis experiments with NIST ISO medium test dust

Concentration of dust in ($\mu\text{g/L}$)	Counts/ $\mu\text{g} > 2 \mu\text{m}$				Count efficiency (%) $> 2 \mu\text{m}$			
	Counter A	Counter B	Counter C	Counter D	Counter A	Counter B	Counter C	Counter D
12 ¹	4792	2750	2542	2750	50	28	26	28
39	4359	2590	2538	2974	45	27	26	31
117	2667	2410	2265	2675	38	25	23	28
441	2980	2474	2252	2741	31	26	23	28
1166	2075	2466	2202	2664	21	26	23	28
Average					37	26	24	29

¹ Counts for this concentration were calculated from the average counts obtained from two trials.

Count Efficiency and Dust Concentration

The counts expressed on a unit mass basis (counts/ $\mu\text{g} > 2 \mu\text{m}$) are plotted in Figure 3.1 for the four counters and a range of dust concentrations in the reservoir. The x-axis is a log scale. The count based on mass measured by NIST with SEM and image analysis (9655/ $\mu\text{g} > 2 \mu\text{m}$) is shown as a horizontal line at the top of the figure. It is apparent that the counts/ $\mu\text{g} > 2 \mu\text{m}$ measured by Counter A decrease as the concentration of dust in the reservoir increases.

A linear regression analysis was used to determine if any of the counters showed a statistically significant increasing or decreasing trend in counts/ $\mu\text{g} > 2 \mu\text{m}$ with dust concentration. The results are shown in Table 3.6. A confidence level of 95 % was used in the

analysis. It can be concluded from the results in Table 3.6 that Counter A showed a significant decreasing trend in count performance with increasing concentration of dust in the reservoir. For the other three counters the slopes of their regression lines were not significantly different than zero at a 95 % confidence level

Table 3.6
Results of regression analysis testing the effect of dust concentration
on counter count performance

Counter	95 % Confidence Interval for Slope of Trend Line		Result
	Lower Limit	Upper Limit	
A	-3598.30	-564.28	95% confidence interval does not include zero slope, therefore effect of concentration is significantly different than zero.
B	-584.24	322.73	95 % confidence interval includes zero slope, therefore effect of concentration is not significantly different than zero.
C	-683.31	185.24	95 % confidence interval includes zero slope, therefore effect of concentration is not significantly different than zero.
D	-540.68	277.22	95 % confidence interval includes zero slope, therefore effect of concentration is not significantly different than zero.

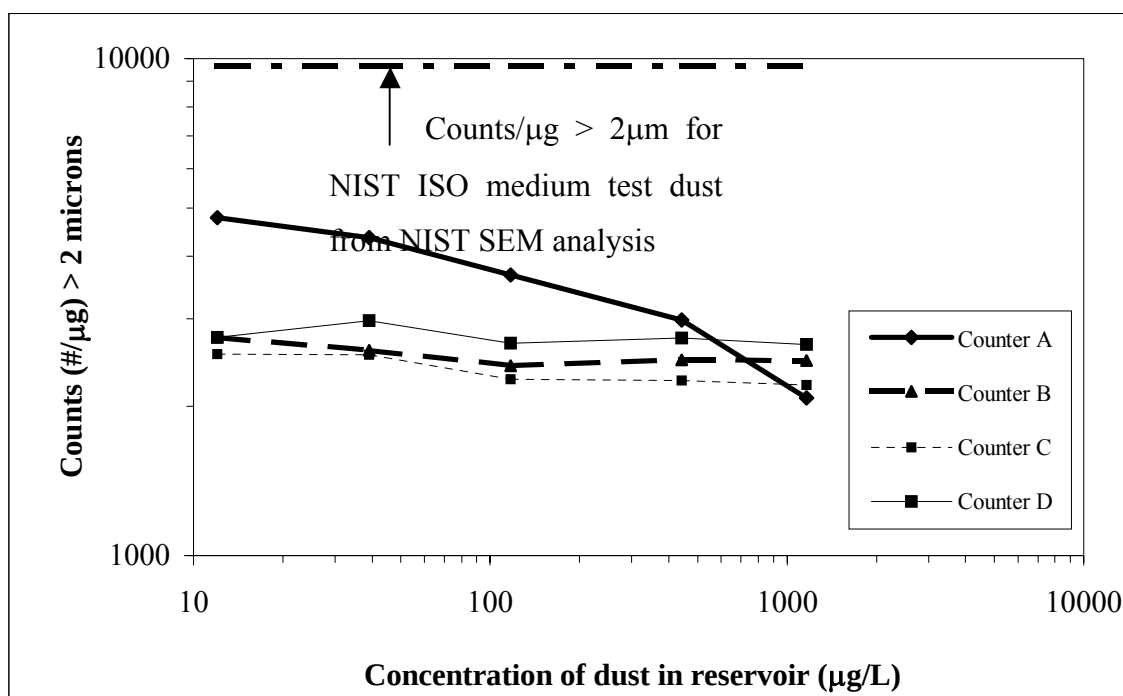


Figure 3.1 Trend in dust counts with concentration of dust in reservoir

Count Efficiency and Threshold Setting

The effect of the threshold setting on the efficiency of counting NIST ISO medium test dust was determined using four counters, three on-line (Counters A, B and D) and one grab sample type (Counter E). The count greater than the threshold setting was measured at five or more threshold settings in the range 2 to 10 μm . The number concentration measured at each threshold setting was converted to counts per unit mass of dust and this number was compared to the NIST measured counts per unit mass listed in Table 2.5 of Chapter 2. The relevant part of this table is reproduced below in Table 3.7.

Figure 3.2 shows the effect of the threshold setting on the particles counted per unit mass of dust. The count values on the y-axis are expressed as a percent of the NIST SEM values of Table 3.7. In general, as the threshold setting increased, the percent counted increased, from a minimum (in the 25 to 40 % range) at 2 μm to a maximum near a threshold setting of 6 or 7 μm . For Counter A, one of the counters with a high average value of R (inferior resolution), the maximum percentage counted was approximately 97 % at a threshold setting of 7 μm . Counter D with superior resolution (the lowest average value of R measured in the experiments with certified PSL suspensions), had a maximum percent counted of about 53 % at a threshold setting of 6 μm . However, Counter B, an instrument with relatively poor resolution, did not follow the trend suggested by Counters A and D. The maximum percent counted for Counter B was about 50 % at a threshold setting of 6 μm .

Table 3.7
NIST SEM results for NIST ISO medium test dust – number of particles per microgram of dust larger than the indicated threshold setting (taken from Table 2.5).

Threshold Setting (μm)	# > Threshold Setting/ μg
2	9655
3	4003
4	2177
5	1335
6	855
7	562
8	377
10	183

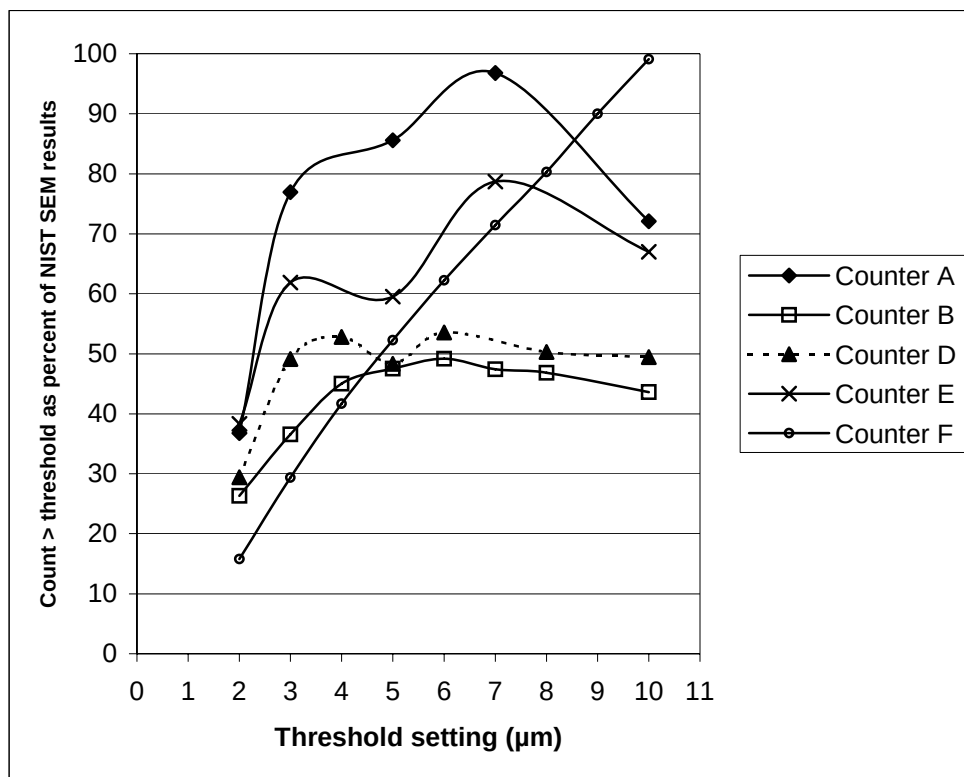


Figure 3.2 Effect of the threshold setting on fraction of dust particles counted for 3 on-line and 2 grab sample counters. Counter F is NIST's HIAC/Royco batch counter analyzing NIST ISO medium test dust in hydraulic fluid.

According to the instrument manufacturers, Counter E, the grab sample counter, has the same sensor as Counter D but different counting electronics. The maximum counting efficiency was about 79 % at 7 μm for the grab sample instrument and significantly lower, approximately 53 % at 6 μm , for the on-line instrument. This suggests that the counting electronics (possibly in addition to the sensor design) is a significant factor in determining the count efficiency of an instrument.

For both poly-disperse NIST ISO medium test dust and mono-disperse PSL suspensions the lowest count efficiencies involved the smallest particles, in the case of PSL suspensions the 3 μm (nominal) diameter microspheres and with NIST dust the 2 μm threshold setting. These results suggest that all four counters are more likely to undercount (relative to the NIST SEM measurements) when particles in the 2 to 6 μm range are counted.

For comparison Figure 3.2 shows the fraction of NIST ISO medium test dust particles counted using a HIAC/Royco batch instrument and a suspension of dust (1 $\mu\text{g/mL}$) in hydraulic

fluid. These measurements were made by NIST when they were developing the scanning electron microscope - image analysis technique for the fluid power industry (Fletcher et al. 1996). The count efficiency values plotted in Figure 3.2 (Counter F) increase monotonically from about 16 % to essentially 100 % as the threshold setting increases from 2 to 10 μm . The threshold settings in this case are based on calibration with certified polystyrene latex microspheres in hydraulic fluid. For the 2 μm threshold setting the percent counted is 10 to 20 % lower for dust in hydraulic fluid than for dust in water. At the 10 μm threshold setting the percentage counted is 30 to 60 % higher in hydraulic fluid than in water. It is possible that the results at the higher threshold settings are influenced by a reduced tendency for the larger dust particles to settle in hydraulic fluid compared to water.

CHAPTER 4

DISCUSSION OF THE IPA RESULTS

INTRODUCTION

This chapter discusses the significance of the instrument performance analysis (IPA) results presented in Chapter 3. It examines what these results tell us about the factors that determine the performance of light obscuration particle counters and, understanding these factors, what type of suspension should be used to test and compare the count performance of on-line counters. The IPA experiments were conducted in preparation for developing the count performance evaluation protocol discussed in Chapter 5.

The first parts of the chapter present a computer spreadsheet model that was developed during the study to help plan the experiments and interpret the experimental results. The model uses important instrumental parameters including counter resolution, threshold setting and threshold setting error in combination with the essential characteristics of the test suspension particle size distribution to determine, in an approximate way, the theoretical count efficiency of the instrument. The count efficiency is the particle count measured with non-ideal instrument behavior, expressed as a percentage of what the instrument would count if the counter had perfect resolution and particle detection and no error in the threshold setting.

In later sections the spreadsheet program is used with the experimental results to evaluate the significance of the various factors that affect instrument performance and to develop and apply an efficient strategy for selecting the most appropriate suspension for testing counter performance in the CPE protocol.

FACTORS THAT AFFECT COUNT PERFORMANCE

Threshold Settings and Threshold Setting Error

As a particle passes through the light beam in a light obscuration particle counter a millivolt pulse is transmitted to the counting electronics of the instrument. The magnitude of the pulse is related to a number of factors including the size of the particle.

Threshold settings are used with the instrument's counting electronics to categorize the pulses produced by the collection of particles in a suspension. After the suspension is analyzed

the electronics can be queried for quantities such as the number of pulses greater than or equal to a given magnitude or for the number of pulses with a magnitude between an upper and lower limit.

When a counting instrument is size calibrated, a relationship is established between the millivolt pulse height and the “size” of the particles in the standard suspension. The threshold settings are given specific “size” values or labels that are, to some extent, characteristic of the standard suspension. Usually the user sets the desired size thresholds or bins in the counter’s software, with certain instrument-specific requirements and limitations.

The pulse from the sensor is caused by the “shadow” that the particle casts on the light detector as it moves across the light beam. The characteristics of the shadow depend on properties of the particle, especially the refractive index. An opaque particle of a given shape and size will create a different shadow than a similar particle that transmits some light, however, these particles might look exactly the same size under a light or electron microscope. Also, particles that are not spheres like the PSL particles can have different orientations in the sensor. A flat, plate-like, particle might produce a shadow like a large sphere or a thin rod, depending on its orientation in the light beam.

As discussed previously, all the on-line counters used in this study were size calibrated by the manufacturers using PSL particles. When these instruments are used to measure particles in “real” suspensions questions regarding the interpretation of the measured sizes arise since the threshold setting – particle size relationship has been set using PSL particles, not particles with the composition, shape, etc., of the real suspension.

The difference between a threshold setting that is based on calibration with a standard particle such as PSL micro-spheres and a threshold setting that gives agreement between size distribution measurements by the counter and a reference counting and sizing method² such as visible light microscopy is the threshold setting error. When the suspension tested is spherical particles such as PSL micro-spheres this error should be close to zero (if the instrument was calibrated with spherical PSL particles) but with irregularly shaped particles or particles that have a refractive index that is significantly different than that of PSL, the threshold setting error

² Standard or reference methods are usually prepared through the voluntary standards system. There are no absolutely true particle size distributions only distributions that standard setters agree to call “true” or reasonably accurate for some application.

is likely to be greater than zero.

Resolution

The resolution of an on-line particle counter is quantified by the amount of random scatter or spread the instrument adds to the measured particle size distribution. Instruments with inferior resolution, i.e., high 'R values' (e.g., $R \geq 10\%$), add more scatter to the size distribution than instruments with superior resolution, i.e., low 'R values' ($R < 10\%$).

For example, the PSL particles in a close to mono-disperse suspension are measured with a light microscope and the standard deviation of the measured diameters is $0.05\ \mu\text{m}$. The particle size distribution is measured with a particle counter and the standard deviation is $0.08\ \mu\text{m}$. The difference between the standard deviation from the light microscope measurements ($0.05\ \mu\text{m}$) and the standard deviation from the particle counter measurements ($0.08\ \mu\text{m}$) is determined by the resolution of the counting instrument. In addition to increasing the spread of the measured particle size distribution, the resolution can significantly affect the counting performance at a given threshold setting. This problem is discussed in a subsequent section.

SPREADSHEET PROGRAM

The following spreadsheet program was prepared to demonstrate how the threshold setting, the threshold setting error and the resolution of the particle counter at the threshold setting determine the counting efficiency. The program uses the Gaussian particle size distribution for near mono-disperse suspensions and the power law relationship for poly-disperse suspensions to determine the number concentration of particles in narrow intervals of particle size across the entire particle size distribution. The threshold setting, threshold setting error and the resolution are then used to calculate the fraction of the particles that are counted in each interval of size. This fraction varies from a value of essentially zero for intervals below the threshold setting to a value of one for intervals above the setting.

The total number of particles counted (F) is determined by multiplying the fraction counted in a given interval of size (f_i) by the number of particles in that interval (ΔN_i) and summing over all the intervals (Δd_{pi}) of the particle size distribution, i.e.,

$$F = \sum_{i=1}^{\infty} f_i \Delta N_i \quad (4.1)$$

The fraction counted for each particle size interval is estimated using the threshold setting, the error in the threshold setting, and the resolution at the threshold setting in conjunction with the next two equations:

$$f_i = 1 - @normsdist(z_i) \quad (4.2)$$

where @normsdist (z_i) is an MS Excel function and z_i is given by:

$$z_i = \frac{((T)(1 + t) - d_{pi})(100)}{(R)(T)} \quad (4.3)$$

In Equation 4.3, T is the threshold setting, e.g., $2 \mu\text{m}$, d_{pi} is the particle size at the midpoint of each size interval, Δd_{pi} , and R is the counter resolution expressed as a percent of the threshold setting. (This expression assumes that the resolution is the same on both sides of the threshold setting). The quantity “ t ” is the error in the threshold setting. For example, if the true value of the threshold setting is believed to fall in the interval $\pm 10\%$ of T and T is equal to $2 \mu\text{m}$ then the limits of t are equal to $+ 0.2$ and $- 0.2 \mu\text{m}$. For calculating a minimum value of F (and the lowest counting efficiency), the positive value of t is used in Equation 4.3. (In the example calculations of this section the magnitude of the Δd_{pi} diameter interval was $0.1 \mu\text{m}$).

The number concentration of particles ΔN_i in the interval Δd_{pi} is given by the equation:

$$\Delta N_i = \left(\frac{\Delta N}{\Delta d_p} \right)_i \Delta d_{pi} \quad (4.4)$$

where $(\Delta N/\Delta d_p)_i$ is the slope of the particle size distribution function at d_{pi} .

For a Gaussian particle size distribution $(\Delta N/\Delta d_p)_i$ can be determined as a function of d_{pi} using the Excel spreadsheet function @normdist (d_{pi} , d_{pm} , σ_i , 0).

$$\left(\frac{\Delta N}{\Delta d_p} \right)_i = @normdist(d_{pi}, d_{pm}, \sigma_p, 0) \quad (4.5)$$

The parameters d_{pm} and σ_p are the mean and standard deviation of the measured particle diameters. The standard method for measuring particle diameter is generally light or scanning-electron microscopy (ASTM 1985).

For Gaussian particle size distributions the above relationships can be used to derive a simpler set of expressions for calculating the counting efficiency, E. This method is described by the next three equations:

$$E = @normsdist(Z) \quad (4.6)$$

where,

$$Z = \left[\frac{d_{pm} - (T)(1 + t)}{S} \right] \quad (4.7)$$

$$S = (\sigma_R^2 + \sigma_p^2)^{0.5} \quad (4.8)$$

and

$$\sigma_R = \frac{(R)(T)}{100} \quad (4.9)$$

For poly-disperse suspensions such as those prepared with NIST ISO medium test dust, the power law equation used by Lawler et al. (1980) for various water treatment suspensions gives a reasonable estimate of the slope $(\Delta N/\Delta d_p)_i$ of the particle size distribution at each d_{pi} ,

$$\left(\frac{\Delta N}{\Delta d_p} \right)_i = A d_{pi}^\beta \quad (4.10)$$

The counting efficiency, E , is equal to the number counted with non-zero values of R and/or t expressed as a percent of the number counted when R and threshold setting error are set equal to zero. When R and t are zero and the particle diameter is increasing the fraction counted in each diameter interval follows a step change from zero to one exactly at the particle diameter that is equal to the threshold setting.

Example Spreadsheet Calculations

The next sections describe how the spreadsheet program can be used to illustrate, in a semi-quantitative way, how factors such as the resolution and threshold setting error determine the counting efficiency, E . The first example is for a nearly mono-disperse suspension with a Gaussian particle size distribution and the second is based on a poly-disperse suspension with a size distribution that follows a power law relationship.

Near Mono-disperse Suspension with Gaussian Particle Size Distribution Function

In this example the suspension consists of polystyrene latex (PSL) particles with a narrow (i.e., near mono-disperse) Gaussian particle size distribution. The particle diameter has been measured by visible light microscopy and the mean and standard deviation are $d_{pm} = 3.05$ and $\sigma_p = 0.05 \mu\text{m}$. The counter threshold setting (T) is $2 \mu\text{m}$, the magnitude of R for diameters around this threshold setting is approximately 25 % (poor resolution) and the threshold setting error (t) is less than $\pm 0.01\%$ and therefore negligible. The threshold setting error is small because the instrument was calibrated with PSL particles; the same type of particle that the instrument is now used to count. The calculated results are listed in Table 4.1.

Table 4.1
Counting efficiency results from the spreadsheet program for a near mono-disperse Gaussian particle size distribution

Resolution (R , %)	S (μm , Eq. 4.8)	Z (Eq. 4.7)	E , counting efficiency (% , Eq. 4.6)
0	0.050	21	100.0
25	0.503	2.1	98.2

According to the results listed in Table 4.1, for this threshold setting ($T = 2 \mu\text{m}$), threshold setting error ($t \sim 0$) and resolution ($R = 25 \%$) the counting efficiency is 98.2 %. This percentage was calculated by combining Equations 4.6 to 4.9 as shown below;

$$@ \text{normsdist} \left[\frac{3.05 - 2.0}{\left[\left(\frac{(25)(2)}{100} \right)^2 + 0.05^2 \right]^{0.5}} \right] = @ \text{normsdist}(2.1) = 0.982$$

If the counter had had perfect resolution ($R = 0 \%$), the calculations would yield $S = 0.05 \mu\text{m}$, $Z = 21$, and $@ \text{normsdist} (21) = 1.0000$, which means the counting efficiency would be essentially 100.00 %.

In general, for a narrow Gaussian particle size distribution (e.g., $\text{COV} = 100 \cdot \sigma_p/d_{pm} < 2 \%$) and with Z from Equation 4.7 greater than about 3, the counting efficiency will be very close to 100 % and not affected to a significant extent by the resolution of the counter or by a small error in the threshold setting ($t < 0.1$). The effect of the magnitude of Z on the counting efficiency is shown graphically in Figure 4.1.

If the objective is to count a certain percentage of the particles in a near mono-disperse suspension the threshold setting must be lower than the mean particle diameter by an amount that depends on the standard deviation of the particle diameter and the resolution of the counter. In the above example, if the objective were to count at least 99 % of the particles in the suspension then Z must be equal to or greater than 2.33 and T must be set at or below $1.88 \mu\text{m}$. If the threshold were incorrectly set at $2.07 \mu\text{m}$, i.e., 10 % higher than the target value of $1.88 \mu\text{m}$, then the counting efficiency would be 97.4 %, slightly less than the 99.0 % objective.

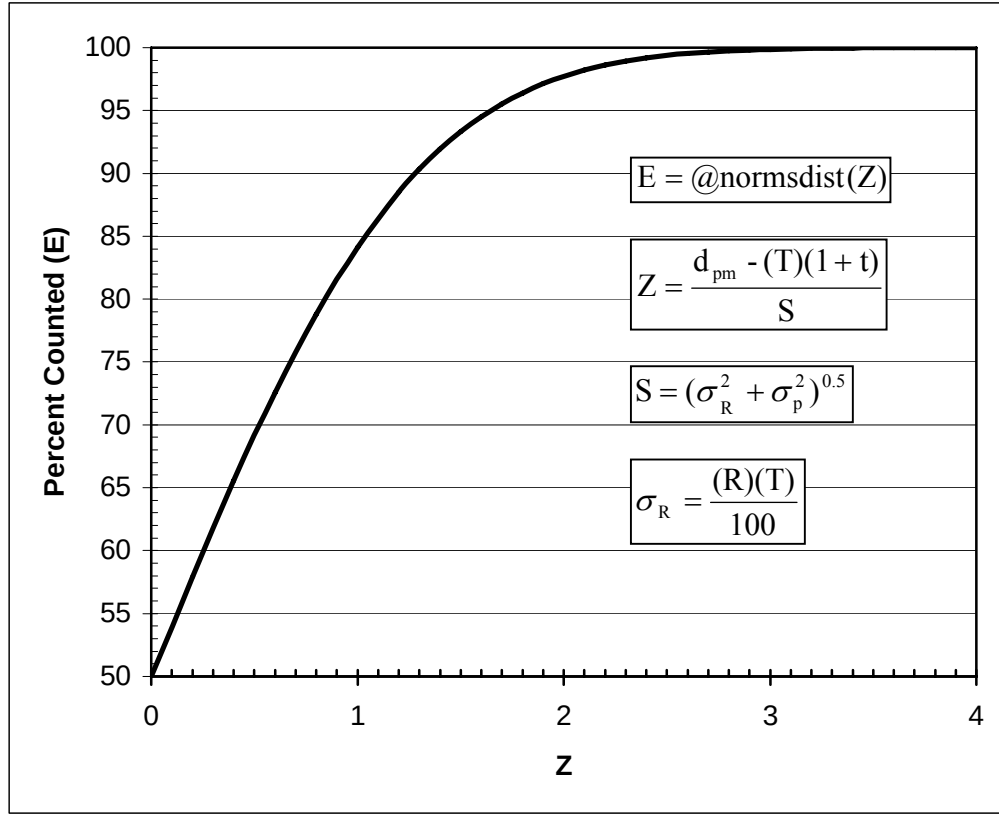


Figure 4.1 Effect of the threshold setting, threshold setting error and the counter resolution on counting efficiency for a suspension with a Gaussian particle size distribution with mean diameter d_{pm} and standard deviation σ_p .

Figure 4.2 shows how the counting efficiency varies with the counter resolution for different values of the threshold setting. As the threshold setting is moved from 2 to 3 μm and, hence, closer to the mean particle diameter (4 μm for this figure), the effect of increasing R becomes more pronounced. When the threshold is set at 2 μm (2 μm less than the mean particle diameter of 4 μm), R -values as high as 30 % have essentially no effect on the counting efficiency. At a threshold setting of 3 μm the effect of the R -value on count efficiency becomes significant as R is increased above 10 %; when R reaches 25 % the counting efficiency has decreased to about 90 %.

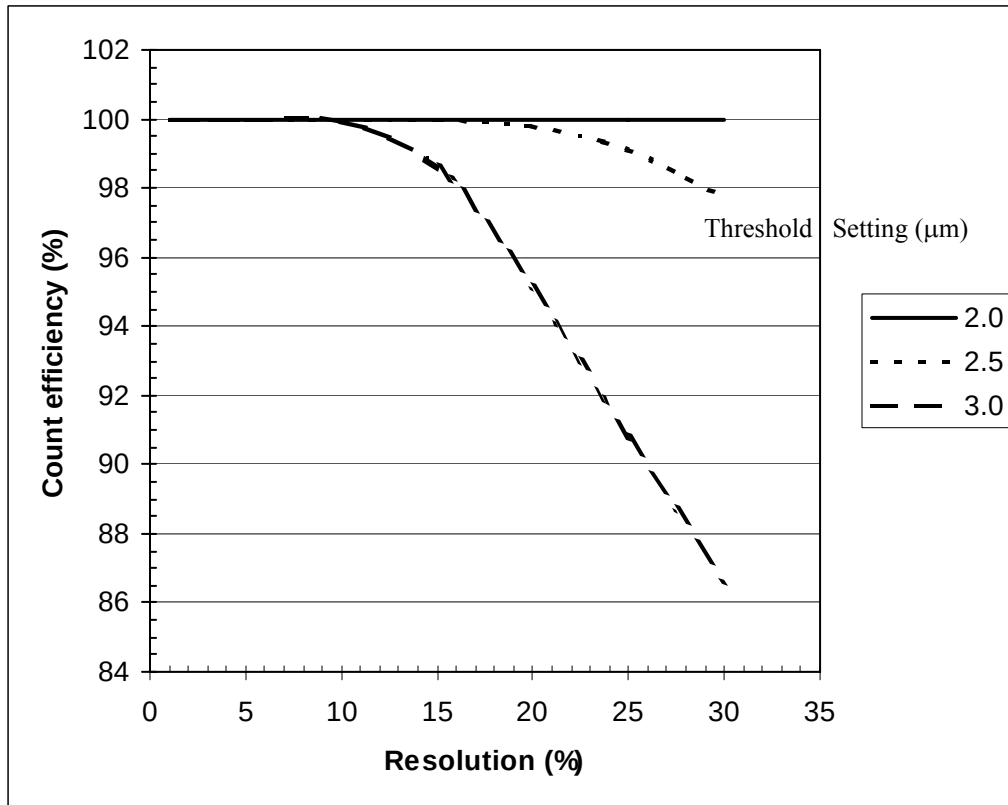


Figure 4.2 Effect of the threshold setting on the count efficiency-resolution relationship. Gaussian particle size distribution with mean of 4 μm and standard deviation of 0.08 μm .

Poly-disperse Suspension with Power Law Equation Size Distribution Function

Most on-line particle counters are used to characterize poly-disperse suspensions such as the particles in filtered water. The particle size distributions of these suspensions have been successfully modeled with power law equations. The example which follows uses the spreadsheet program to show how the threshold setting, threshold setting error and instrument resolution affect the counting performance of an instrument that is counting the particles in a poly-disperse suspension. It is first assumed that the particle counter has perfect resolution (i.e., $R = 0\%$) and then, in a subsequent section, the effect of finite instrument resolution ($R > 0\%$) is included in the analysis.

For the spreadsheet calculations the size distribution of the particles was assumed to follow the power law equation that Lawler, et al. (1980) and others have used to describe the

distribution of particles sizes in filtered water and other water treatment suspensions. This equation is given by:

$$N = \frac{A}{\beta - 1} d_p^{(1-\beta)} \quad (4.11)$$

where N is the cumulative number concentration of particles greater in size than the particle diameter d_p and A is a quantity that varies with the concentration of the suspension. This expression was fitted to part of the cumulative particle size distribution that NIST provides with their ISO medium test dust giving $A = 216,800 \mu\text{m}^2/\text{mL}$ and $\beta = 3.0$. Figure 4.3 compares the fitted power law equation with the NIST measurements. The NIST data and the power law curve are in reasonable agreement in the small particle range ($< 6 \mu\text{m}$), however, at larger particle sizes ($> 6 \mu\text{m}$) the equation predicts a greater number of large diameter particles.

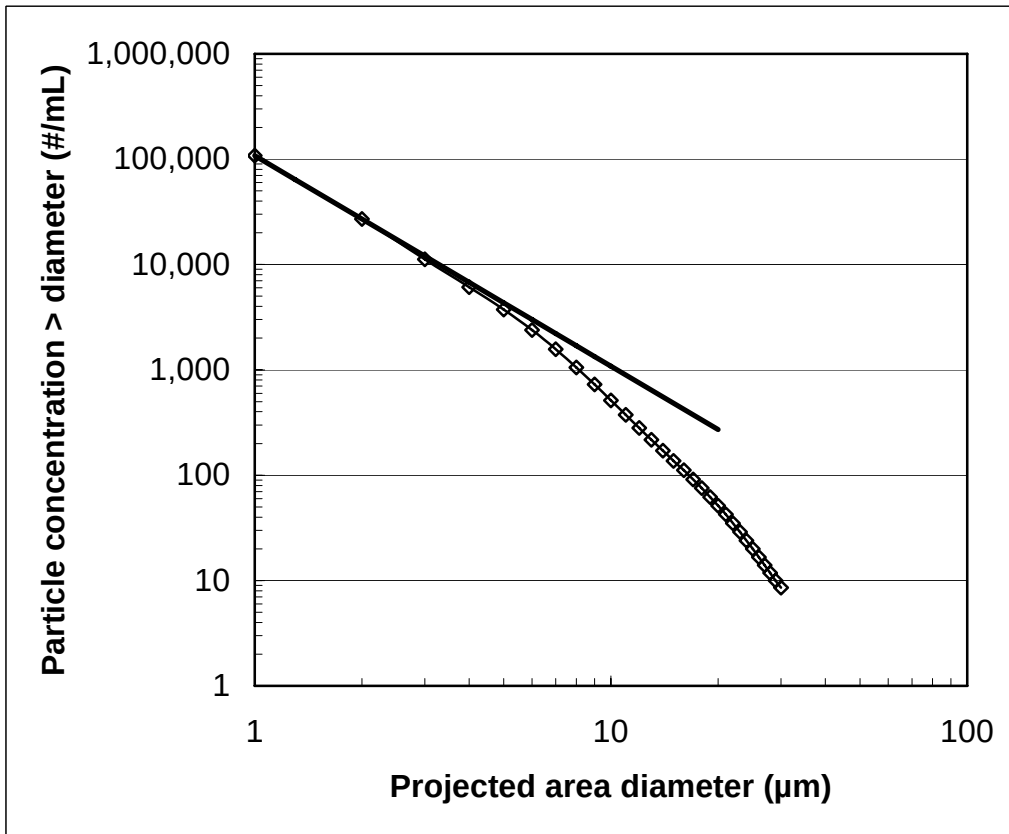


Figure 4.3 Power law equation (the straight line) fitted to part of the NIST ISO medium test dust particle size distribution

For a poly-disperse suspension the threshold setting has a significant effect on the counting efficiency. For example, if the instrument has perfect resolution ($R = 0\%$) and the threshold is set at $2\ \mu\text{m}$ (the projected area diameter of Figure 4.3), the measured total counts greater than $2\ \mu\text{m}$ should be, according to Eq. 4.11,

$$N = \frac{216,800}{3.0 - 1} 2.0^{1-3} = 27,100 / \text{mL} .$$

The number concentration greater than the threshold setting is simply N from the power law equation (Eq. 4.11) evaluated at a particle diameter equal to the threshold setting ($2\ \mu\text{m}$ in this example). It is assumed that the particles in the continuous particle size distribution are physically the same (i.e., have the same refractive index, shape, etc.) as those used to calibrate the threshold settings on the counter and therefore it can be assumed that the threshold setting error is zero.

If the threshold setting is not correct the counting efficiency can be significantly different than 100 percent. For example, if the intent is to count all particles larger than $2\ \mu\text{m}$ but the threshold is set incorrectly at $2.2\ \mu\text{m}$ then, according to the power law equation, the concentration counted will be $22,397/\text{mL}$ or 17% less than the correct value of $27,100$ for a threshold setting of $2\ \mu\text{m}$. The counting efficiency in this case is $(22,397/27,100) \times 100$ or 83% . If the threshold was set too low, e.g., at $1.8\ \mu\text{m}$ instead of $2.0\ \mu\text{m}$, then the number concentration counted would be $33,457/\text{mL}$ or 23.5% higher than the correct value of $27,100/\text{mL}$ at $T = 2\ \mu\text{m}$. The counting efficiency would be $(33,457/27,100) \times 100$ or 123.5% . In general, if the size distribution of a poly-disperse suspension is described by a power law relationship with $\beta = 3$, a threshold setting misplaced by 10% will affect the count efficiency by ± 10 to $\pm 25\%$. The exact amount depends on the resolution of the counter.

Figure 4.4 shows how the fraction counted in narrow intervals of size (from Equations 4.2 and 4.3) varies with the particle diameter for three values of the counter resolution (5 , 15 and 25%) and a threshold setting of $2\ \mu\text{m}$. As $R \rightarrow 0\%$, only particles larger than the $2\ \mu\text{m}$ threshold are counted and the count efficiency is close to 100% . If R is increased to 15% , some of the particles slightly larger than $2\ \mu\text{m}$ are not counted and some slightly smaller than $2\ \mu\text{m}$ are

counted. Given the shape of the size distribution curve of Figure 4.2 as it decreases across the 2 μm threshold, the over-counting of particles smaller than 2 μm is not balanced by the under-counting of particles larger than the threshold and the result is a significant over-count. When $R = 15\%$ the counting efficiency is 107.9 %.

As the magnitude of R increases, over-counting below the threshold becomes even more significant than under-counting above the threshold and this causes the overall count efficiency to become much greater than 100 %. For $R = 25\%$ and the size distribution of Figure 4.3 the counting efficiency is 144.1 %.

The two graphs in Figure 4.5 illustrate how the size distribution of the particles and a $\pm 10\%$ error ($\pm 0.2\ \mu\text{m}$) in the threshold setting affect the relationship between the count efficiency and the resolution. The count efficiency was plotted using 100 % efficiency when $R = 0\%$ and the threshold setting is 2 μm . The size distribution of the particles was varied by changing the magnitude of β in the power law size distribution equation (Eq. 4.11).

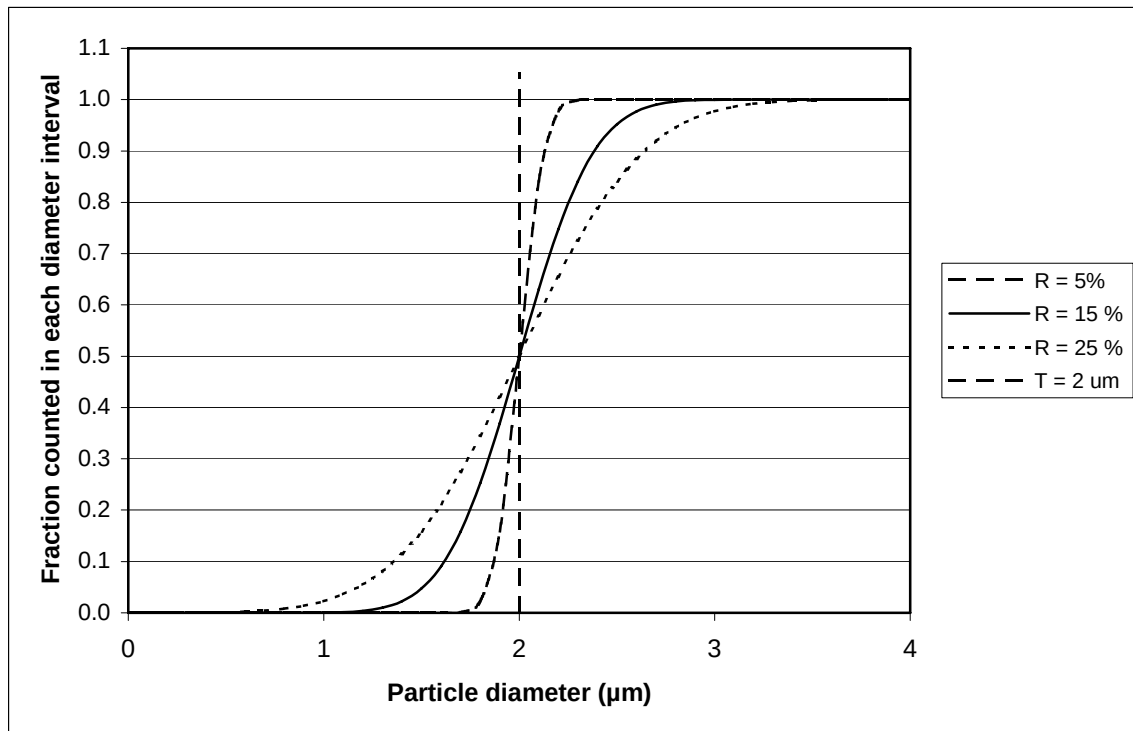


Figure 4.4 Fraction of particles counted in each diameter interval as a function of the particle diameter for threshold setting of 2 μm and three values of R : 5%, 15% and 25%.

The curves plotted in Figures 4.5 A and B show that counter performance is affected by the size distribution of the particles in a poly-disperse suspension. When β is equal to 3.5 (Figure 4.5 A) the threshold setting error and the resolution have a much greater effect on the count efficiency than when β is equal to 2.0 (Figure 4.5 B).

According to these calculated results, an instrument with relatively good resolution ($R < 10\%$) will count essentially 100 % of the particles greater than the threshold setting assuming that there is no error associated with where the threshold is placed. An instrument with relatively poor resolution ($R \geq 10\%$) will count some of the particles smaller than the threshold setting (registering their signals as greater than the threshold setting when it should not) and will detect but misplace, relative to the threshold, some of the particles that are slightly larger than the setting. Since the cumulative particle size distribution decreases across the threshold setting there are more particles to over-count below the threshold than there are particles to undercount above the threshold so the net effect tends to be an overall over-count. The magnitude of this over-count depends on the resolution and it can be decreased by a positive threshold error and increased by a negative threshold setting error.

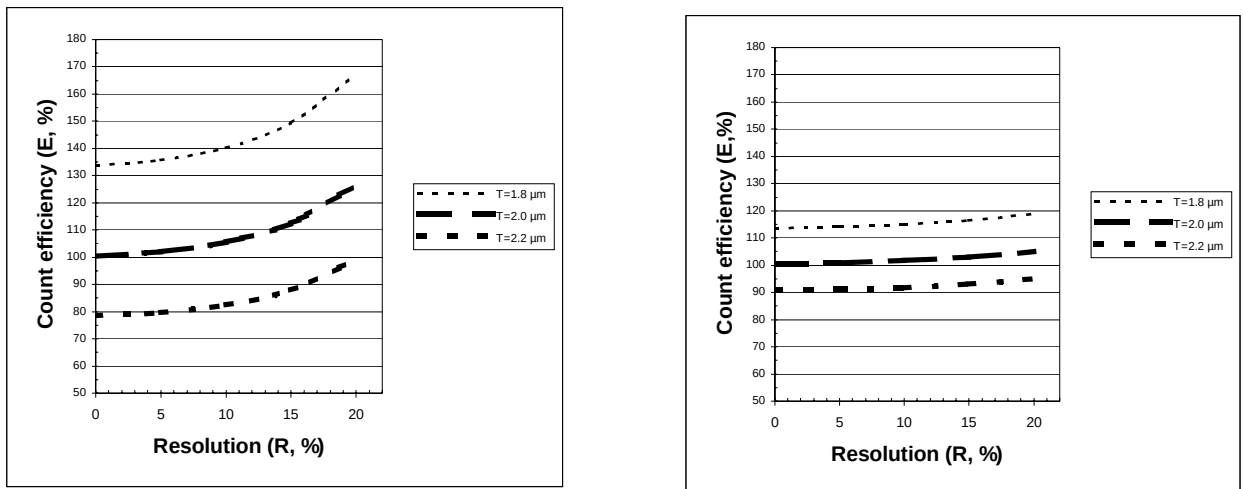


Figure 4.5 Effect of the counter resolution and threshold setting on the count efficiency for two values of the power law equation exponent, Graph A: $\beta = 3.5$ and Graph B: $\beta = 2.0$

These results indicate that the size distribution of the suspension used in count performance evaluation should resemble that of the particles the counters are used to measure in the treatment plant. If the CPE suspension does not resemble the treatment plant suspension then the relative effects of instrumental factors such as resolution and threshold setting error on the counting efficiency will be different for the two suspensions. In other words if count discrepancies are detected among an assortment of instruments using the CPE they may not accurately simulate the inter-instrument discrepancies associated with counting the particles in the treatment plant suspension.

IMPLICATIONS OF THESE RESULTS

Do the Instruments Detect All the Particles?

The results of the size calibration verification experiments (Table 2C of Appendix C) show that the maximum difference between the measured and "certified" PSL diameter for the four counters used in the study was about 10 %. Based on these results it was concluded that a threshold setting error of 10 % is a reasonable value to use in the spreadsheet calculations for near mono-disperse PSL particles.

In all the count performance evaluation experiments that used PSL particle suspensions, the threshold was set at 2 μm . Equations 4.6 – 4.9 of the spreadsheet model were used to estimate the count efficiency at a threshold setting of 2.2 μm (2 μm + 10 %). The resolution used in the calculations was the maximum value measured for each counter (Table 3.1 in Chapter 3). The spreadsheet calculated count efficiencies are shown in Tables 4.2a to 4.2d along with the measured values.

The spreadsheet-calculated count efficiencies for the 3 μm PSL particle are essentially 100 % for counters A, C and D. The measured count efficiencies for these counters are in the 50 – 77 % range and significantly less than 100 %. For counter B, the theoretical count efficiency was about 84 %, less than 100 % because the maximum value of R measured for that counter (43 %) was used in the calculations. The measured count efficiency for counter B was between 60 and 65 %.

Table 4.2a: Comparison of estimated and measured count efficiencies for Counter A

Mean diameter (microns)	Particle COV (%)	Maximum R (%)	Threshold setting (microns)	Error associated with threshold setting (%)	Fraction counted (F)	Estimated count efficiency (%)	Measured count efficiency (%)	
							June	August
3.063	1.0	20	2	10	0.9843	98.43	74	89
4.991	1.2	20	2	10	1.0000	100.00	107	122
4.991	1.2	20	2	10	1.0000	100.00	126	122
6.992	1.0	20	2	10	1.0000	100.00	100	114
6.992	1.0	20	2	10	1.0000	100.00	72	114
9.975	0.9	20	2	10	1.0000	100.00	86	105
15.02	1.0	20	2	10	1.0000	100.00	106	107

Table 4.2b: Comparison of estimated and measured count efficiencies for Counter B

Mean diameter (microns)	Particle COV (%)	Maximum R (%)	Threshold setting (microns)	Error associated with threshold setting (%)	Fraction counted (F)	Estimated count efficiency (%)	Measured count efficiency (%)	
							June	August
3.063	1.0	43	2	10	0.8420	84.20	60	65
4.991	1.2	43	2	10	0.9994	99.94	97	90
4.991	1.2	43	2	10	1.0000	99.94	91	90
6.992	1.0	43	2	10	1.0000	100.00	91	78
6.992	1.0	43	2	10	1.0000	100.00	100	78
9.975	0.9	43	2	10	1.0000	100.00	80	104
15.02	1.0	43	2	10	1.0000	100.00	106	113

Table 4.2c: Comparison of estimated and measured count efficiencies for Counter C

Mean diameter (microns)	Particle COV (%)	Maximum R (%)	Threshold setting (microns)	Error associated with threshold setting (%)	Fraction counted (F)	Estimated count efficiency (%)	Measured count efficiency (%)	
							June	August
3.063	1.0	16	2	10	0.9964	99.64	55	63
4.991	1.2	16	2	10	1.0000	100.00	84	81
4.991	1.2	16	2	10	1.0000	100.00	80	81
6.992	1.0	16	2	10	1.0000	100.00	ND	68
6.992	1.0	16	2	10	1.0000	100.00	86	68
9.975	0.9	16	2	10	1.0000	100.00	61	85
15.02	1.0	16	2	10	1.0000	100.00	78	88

Table 4.2d: Comparison of estimated and measured count efficiencies for Counter D

Mean diameter (microns)	Particle COV (%)	Maximum R (%)	Threshold setting (microns)	Error associated with threshold setting (%)	Fraction counted (F)	Estimated count efficiency (%)	Measured count efficiency (%)	
							June	August
3.063	1.0	16	2	10	0.9964	98.43	50	69
4.991	1.2	16	2	10	1.0000	100.00	77	81
4.991	1.2	16	2	10	1.0000	100.00	75	81
6.992	1.0	16	2	10	1.0000	100.00	70	68
6.992	1.0	16	2	10	1.0000	100.00	81	68
9.975	0.9	16	2	10	1.0000	100.00	52	85
15.02	1.0	16	2	10	1.0000	100.00	60	90

With the other PSL particles (the particles with certified diameters greater than 3 μm), the theoretical count efficiencies were close to 100 % for all the counters. For counters A and B, the measured count efficiencies were in the range 100 ± 20 %. For counters C and D, the measured count efficiencies for the larger particles were in the 66 –77 % range, significantly lower than 100 %.

As shown in Table 4.2 the measured count efficiencies for some counters are significantly lower than the calculated count efficiencies especially for the smallest PSL particles. Based on these differences it was concluded that sensor resolution and error associated with the threshold setting do not completely explain the low measured count efficiencies obtained with many of the near mono-disperse PSL test suspensions.

For the NIST ISO medium test dust experiments and a 2 μm threshold setting (based on size calibration with PSL micro-spheres), all the counters gave measured count efficiencies that were less than 50 %. According to the spreadsheet model, with a continuous (power law) particle size distribution, and using the measured resolutions and a 10 % threshold setting error³, the theoretical count efficiencies are all greater than 100 %. Sensor resolution and this threshold setting error do not, therefore, explain the low observed count efficiencies obtained with NIST ISO medium test dust in the IPA experiments.

³ Light extinction (Mie scattering) calculations for homogeneous spheres and light wavelengths between 400 and 700 nm were used to evaluate this threshold setting error for particles of NIST ISO medium test dust (mostly quartz with a refractive index of 1.55) and size calibration with polystyrene latex microspheres with a refractive index of 1.59. According to the results a threshold setting error of 10% at this threshold setting is a very conservative estimate. The effects of light absorption by the quartz and the different particle shapes were not considered.

The results obtained with both PSL and NIST ISO medium test dust suspensions indicate that a factor or combination of factors, other than or in addition to counter resolution and threshold setting error cause low count efficiencies. The experimental results with NIST ISO medium test dust, where counter A showed a statistically significant decreasing trend in count efficiency with increasing concentration of dust, and the results with the 3.063 μm PSL suspension, where all the counters measured low count efficiencies (less than 77 %), and the low average count efficiencies (66-77 %) measured by instruments C and D with all the PSL suspensions, suggest that the instruments do not respond to all the particles that pass through the sensor. This seems to be especially true for the smaller ($< 5 \mu\text{m}$) PSL and NIST ISO medium test dust particles. Results obtained by researchers using other particle measurement methods (see Table 1.1 in Chapter 1) have shown that particle counters do not appear to respond to all the particles that pass through their sensors.

Discussions with technical people in the particle counter industry (personal communication, 2000) support the notion that low count efficiencies may be caused in part by the counting electronics. In particle counters, the photo-detector converts changes in light intensity to electrical voltage pulses and these are sent to a signal processor or multi-channel analyzer that converts the analog signal to a digital output. If the circuitry used to count the pulses is not fast enough to process all the signals from the counter, then, as the number concentration of particles increases, the counting efficiency will decrease.

Results obtained by Chowdhury et al. (1998) (also see Chowdhury et al. 2000) also point to an inability of the instruments to detect and/or count all the particles that pass through the sensor. These investigators conducted experiments to determine if different multi-channel analyzer (MCA) cards were processing signals differently. Two grab samplers of the same make and model that had built in signal processors were connected to two different computers with two different MCA cards. It was possible to receive two sets of data from each grab sampler when counting a single sample, one set from the grab sampler and one set from the attached computer/MCA card. A single sample was counted using both grab samplers and four sets of data were obtained. The grab samplers with their own signal processors gave consistent counts but computer 1 with its MCA card gave counts almost 38 % higher than the grab samplers and computer 2 with its MCA card gave results 15 % higher than the grab samplers. The instrument

manufacturer suggested to Chowdhury et al. (2000) that the software used with the computer card might have caused the inconsistency but this was not firmly established.

It should be noted that less than 100% particle detection for particles larger than the threshold setting can make “count calibration” procedures such as those used by the fluid power industry (ISO 1999) essentially meaningless. If the instrument does not “see” all the particles of the target size range that pass through the sensor there is no “calibration” adjustment that can effectively correct for the problem. For example in the count calibration procedure when a cumulative count versus size calibration curve based on a microscopic analysis of a poly-disperse suspension like ISO medium test dust is used to label the threshold settings, the “size” labels at the smaller diameters will have essentially no relationship to the truth. An alternative to count calibration is to perform a count performance evaluation and then, if necessary and if it is feasible, have the manufacturer adjust the instrument until it achieves the desired count efficiency.

Suspensions for Count Performance Evaluation

The experimental results and calculations of this study show that count performance evaluations should be done using a well-characterized poly-disperse suspension with a size distribution that resembles, as closely as possible, the distribution that is expected for the water to be analyzed in the treatment plant. A mono-disperse suspension such as PSL micro-spheres, while appropriate for a standard size calibration, is not suitable for this purpose. As seen from the model system calculations for these suspensions, if the threshold is set well below the mean particle diameter, essentially all the particles will be counted, and the count error effects caused by sensor resolution, and errors associated with the threshold setting will not be seen. On the other hand, for a poly-disperse suspension, the effects caused by sensor resolution, and errors associated with the threshold setting and inefficient particle detection will be evident in the measured count efficiencies. The test of instrument comparability will have greater utility.

It is important that the poly-disperse suspension chosen for a count performance evaluation study have a particle size distribution that is similar to what is expected in the water treatment plant. Particle size distributions with different shapes and slopes interact with the various factors (resolution, threshold setting, etc.) differently and affect count performance in different ways. This is illustrated in Figure 4.6 where three cumulative particle size distributions

have been plotted using the power law equation (Equation 4.11 with $\beta = 2, 3$ and 4). The threshold setting is $2 \mu\text{m}$ with a 10% positive error. The count discrepancy caused by the threshold error gets larger as the slope of the distribution gets steeper. A threshold error has less of an effect on the count performance when the size distribution corresponds to $\beta = 2$ compared to when it corresponds to $\beta = 4$.

The ISO medium test dust available from NIST seems to be a reasonably suitable particle for count performance evaluation of counters that are used to measure filtered waters (See Chapter 2). The β value that was obtained by fitting the cumulative power law distribution to the size distribution results obtained by NIST using scanning electron microscopy (SEM) and image analysis for ISO medium test dust in hydraulic fluid was 3.4 . The β values that were obtained by fitting the cumulative power law distribution to the results from the count performance evaluation experiments with NIST ISO medium test dust for the four on-line counters were $3.1, 3.3, 2.6$ and 3.4 for counters A, B, C and D, respectively, and were comparable to the β value

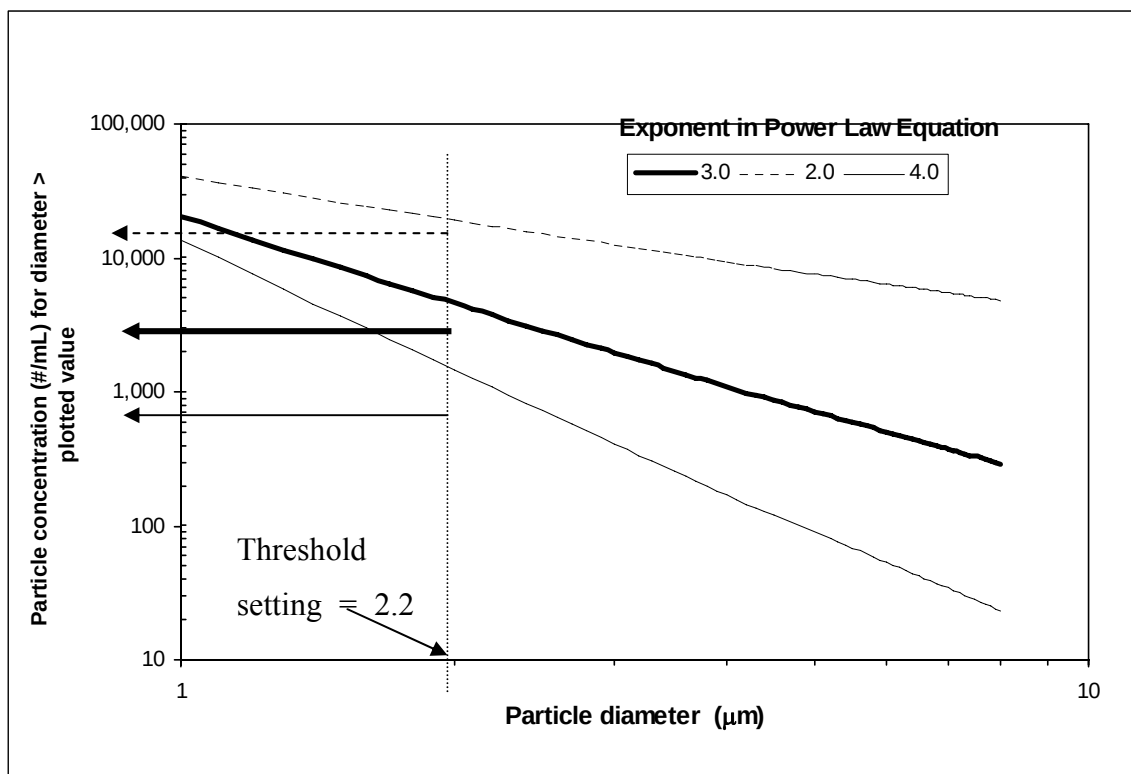


Figure 4.6 Effect of β in the power law size distribution equation and threshold setting error on count performance

obtained from the SEM results. These values are also close to the average β value obtained by fitting the cumulative power law distribution to size distributions measured by Cleasby et al. (1989) using filtered water samples from 21 plants located across the country (β values for Cleasby's filtered water samples range from 2 to 4 for particle diameters between 1 and 10 μm). Also, these values of β fall within the range of 2 to 5 reported by Lawler et al. (1980) for various water treatment plant suspensions.

CHAPTER 5

COUNT PERFORMANCE EVALUATION PROTOCOL

INTRODUCTION

The procedure or protocol presented in this chapter uses NIST ISO medium test dust (NIST MTD) to evaluate particle counter performance. NIST MTD is an irregularly shaped, naturally occurring quartz dust with a poly-disperse size distribution. The size of the particles ranges from below 1 μm to over 50 μm (as the area equivalent diameter). As discussed in Chapter 2 NIST MTD has been accurately characterized by NIST using scanning electron microscopy (SEM) and image analysis and a detailed size distribution for particles larger than 1 μm is available in the NIST documentation for this material (RM 8603)⁴. It is known, based on NIST's analysis, to contain 9,655 particles greater than 2 μm in diameter per μg of dust (see Table 2.3, Chapter 2). Another reason for selecting NIST MTD for the protocol was its size distribution resembles that of the particles in filtered water (See Chapters 2 and 4).

According to Ramaswamy (2000), and as discussed in Chapter 4, the use of NIST MTD should give count performance results that are sensitive to all of the factors that determine the count efficiency of light obscuration particle counters, including sensor resolution, threshold setting errors and inefficient particle detection. Mono-disperse suspensions, such as polystyrene latex microspheres, can be used to evaluate the instrument's ability to detect particles, but they are not useful for evaluating the effect of threshold setting errors and resolution on count performance when treatment plant suspensions are analyzed.

The count performance evaluation (CPE) protocol includes two essential parts, preparing an initial suspension of NIST MTD and diluting this suspension to make suspensions that have a dust concentration that is appropriate for counter evaluation. The initial suspension of dust is called the stock suspension and the dilutions of the stock suspension are called the working suspensions.

⁴ NIST prepared the dust particles for SEM and image analysis by filtering them from hydraulic fluid and washing them with organic solvents. This should not have altered the particle size distribution but this is an assumption that should be tested in the future.

In the CPE protocol and in the data presented to support and explain each protocol step, the particle count results are typically given as the particle count greater than the 2 μm threshold per microgram of dust in the working suspension. Normalization of the data by dividing the counts per mL by the μg of dust in each mL of working suspension (see Equation 5.1) allows the comparison of count results from working suspensions with slightly different dust concentrations and with the size distribution information prepared by NIST.

$$\text{particle count } (> 2 \mu\text{m})/\mu\text{g} = \frac{\text{particle count } (> 2 \mu\text{m/mL})}{\text{working suspension dust conc. } (\mu\text{g/mL})} \quad (5.1)$$

Count Performance Evaluation Protocol

The CPE protocol is presented as a series of proposed steps. Following the presentation of each step is a discussion of the experimental and other results that were used to make decisions about how to accomplish the step. The steps in the protocol are:

- 1) Preparation of the Stock Suspension
- 2) Confirming the Dust Concentration in the Stock Suspension (Optional)
- 3) Preparation of a Working Suspension
- 4) Feeding a Working Suspension to the Counter
- 5) Collection and Analysis of the Data

The methods and procedures used in the experiments that supported the development of the CPE are included in Appendix F.

PREPARATION OF THE STOCK SUSPENSION

Measure approximately 300 mg of NIST ISO medium test dust (RM 8603) into a pre-weighed cellulose capsule. (Note: cellulose capsules should always be handled with tweezers.) Since the dust particles in the 20 g NIST sample may have segregated by size during shipping and handling it is best to divide the entire 20 g dust sample using a micro-riffler device. Weigh the encapsulated dust and determine the actual dust weight by subtracting the empty capsule weight. Place the capsule with dust in a 100 mL plastic container with a snap-on lid and add 100 mL of “low particle water” to dissolve the capsule and suspend the dust. (Use the riffler to prepare as many dust-filled capsules as are needed to use the entire sample received from NIST, but only add the water when a fresh suspension is required.) Shake the water and capsule vigorously and store it overnight in a refrigerator. Always store the stock

suspension in a refrigerator and discard after 90 days. Express the dust concentration in the stock suspension as μg of dust per mL of water.

Low Particle Water

Count performance evaluation cannot be done without a reliable source of low particle water. Low particle water should be used to prepare all the suspensions (and to wash the glassware and other equipment and containers that comes into contact with the dust suspensions.) In this study low particle water was prepared using a reverse osmosis unit. Ninety percent of the time this device produced water with less than 10 counts per mL $> 2 \mu\text{m}$ and it always produced less than 30 counts per mL $> 2 \mu\text{m}$ while maintaining a continuous flow of approximately 500 mL per minute. There are several published techniques for the production of low particle water, as discussed below.

In a report prepared for the AWWARF, Hargesheimer, Lewis, et al. (1999) describe how to produce low particle water by recirculating water through a $0.22 \mu\text{m}$ cartridge filter using a peristaltic pump. This method produced water with less than 5 counts per mL $> 2 \mu\text{m}$. Chowdhury et al. (1998) used a technique that involved pumping distilled-deionized through successively smaller filters; $0.45 \mu\text{m}$, $0.22 \mu\text{m}$ and then $0.05 \mu\text{m}$ pore size. The production rate was approximately 10 mL per minute.

An important consideration when preparing low particle water for diluting test suspensions is the chemical makeup of the water. If filtered water from the plant must be used after supplementary membrane filtration (because there is no other source of low particle water), the effects of solution chemistry on the surface chemistry of the particles in the dust suspensions could be significant. Also the coagulant could affect the particle counts if precipitate particles form after filtration due to initial super saturation with coagulant species. Until there is more information about the effect of solution chemistry on NIST test dust suspensions the conservative approach is to use low particle water that also has low concentrations of dissolved salts, e.g., distilled-deionized water or water from a reverse osmosis unit.

Rationale for the Amount of Dust per Capsule

The amount of dust per capsule (approximately 300 mg) is based on a compromise that considered the target concentration of particles in the working suspensions (about 1000 counts/mL > 2 μ m), the capacity of the micro-riffler device, the amount of dust in the sample purchased from NIST (about 20 g), the volume of the micro-dispensers (0.1 to 5 mL), and the volume of the plastic containers that seemed to be appropriate for the type of tests to be conducted with the on-line counters.

As discussed in Chapter 2 in each cycle of its operation the Quantachrome micro-riffler divides the starting quantity of dust into 8 portions. Since the quantity of dust purchased from NIST is 20 g, the first cycle yields 8, 2.5 g portions. In the next cycles, each 2.5 g portion is divided into 8, approximately 310 mg, portions. The end result is $8 \times 8 = 64$ portions with about 310 mg of dust per portion.

It was also envisioned that low particle dilution water might be difficult to obtain in large quantities at some locations and that, therefore, it should be used efficiently. It was evident that the minimum amount of working suspension needed for one on-line counter measurement was between 1 and 2 liters. At a suspension feed rate of 100 mL/min, a volume of 1 L gives a maximum of 10 minutes of data collection. When all the factors were considered together it was concluded that 250 - 310 mg of dust in 100 mL of low particle water was a reasonable concentration for the stock suspensions.

In the ISO standard ISO 11170:1999 (ISO 1999) it is recommended that ISO medium test dust be dried in an oven at 110 – 150 °C for at least 18 hours before it is weighed and used to prepare suspensions. The pros and cons of a drying step were considered before the dust was riffled and it was decided to simply open the sealed bottle from NIST and immediately riffle its contents without drying.

Effect of Stock Suspension Age on Working Suspension Particle Count

It is recommended in this step of the proposed CPE procedure that each stock suspension be discarded after about 90 days of storage (in a refrigerator). This suggestion is based on tests conducted during the study. In these tests stock suspensions were aged and sets of working

suspensions were prepared as a function of time. The effect of stock suspension age on the average count (count/mL $> 2 \mu\text{m}$) for each set of five working suspensions was determined by linear regression analysis.

Stock suspension age experiments were conducted with 7 stock suspensions. Six, labeled B, C, D, I, J, and K, were prepared using encapsulated NIST MTD and one was prepared using dust from a large sample purchased from PTI, Inc. of Burnsville, Minnesota. (The dust from PTI was not riffled nor was it put in a cellulose capsule.) PTI dust is very similar, if not identical, to NIST MTD. The particles are the same material and both have essentially the same particle size distribution. In the early 1990s PTI, Inc. supplied the dust that NIST processed and characterized and now sells as ISO medium test dust reference material (RM 8603 in the NIST catalogue).

The working suspensions prepared from stock suspensions B, C and D were analyzed with the grab sampler and the working suspensions from stock suspensions I, J and K were analyzed using both the grab sampler and an on-line counter (Counter D) with the gear pump feed system. The PTI suspensions were analyzed with the grab sampler. The maximum age of the NIST MTD stock suspensions was 73 days (on-line counter) and the maximum age of the PTI dust stock suspension was 335 days.

The example results plotted in Figure 5.1 were obtained using stock suspensions I, J and K and Counter D with the gear pump feed system. Each point is the average count (in counts/ $\mu\text{g} > 2 \mu\text{m}$) for a set of five working suspensions and the error bars are the associated standard deviations. Each x-value plotted on the graph is the age of the stock suspension in days when that set of working suspensions was prepared.

Each set of data (average working suspension count versus stock suspension age) was fitted with a straight line by linear regression analysis. The results for all stock suspensions and count measurement methods are summarized in Table 5.1. The last three columns in this table list the best-fit slope and its 95 percent confidence limits. In every case, including the PTI dust with a maximum age of 335 days, the confidence interval brackets zero, indicating that we cannot say that the slope is significantly different than zero. Therefore, in all cases the results indicate that the age of the stock suspension does not have a statistically significant effect on the measured counts.

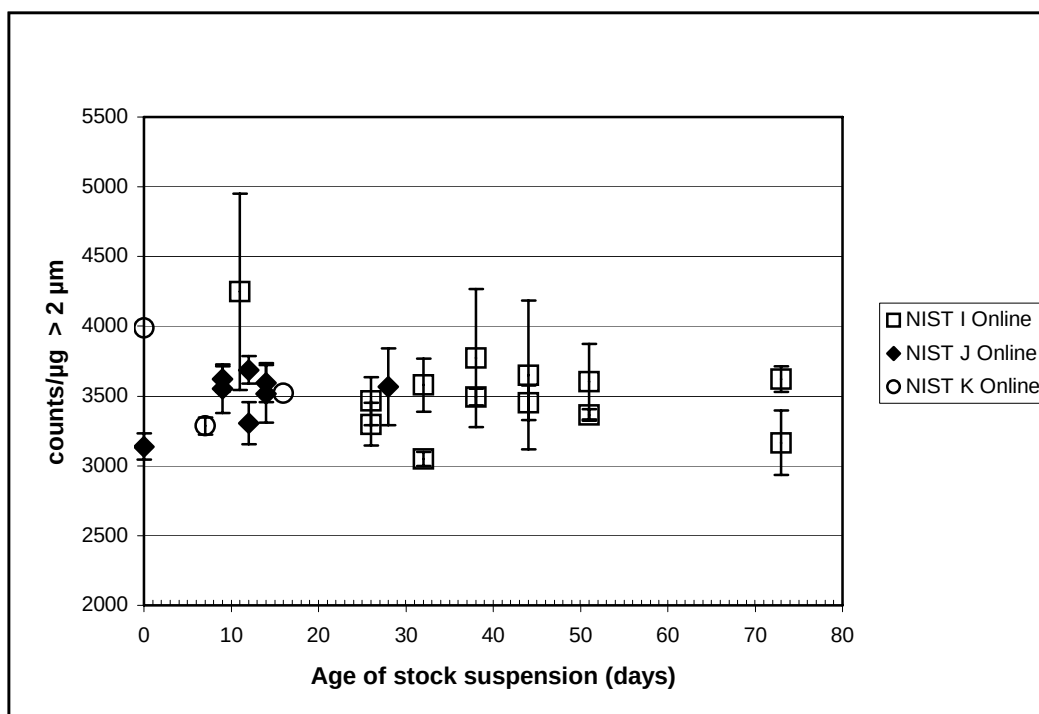


Figure 5.1 Effect of stock suspension age on working suspension counts – NIST ISO medium test dust.

Table 5.1
Summary of linear regression analysis results for the effect of stock suspension age on the concentration in working suspensions

A	B	C	D	E	F	G	H
NIST B	Grab Sampler	7	2682	0.086	-23.0	-109.0	63.1
NIST C	Grab Sampler	5	3699	0.124	-3.84	-22.5	14.9
NIST D	Grab Sampler	3	3644	0.127	7.52	-243.5	258.5
NIST I	Grab Sampler	13	3159	0.154	-6.09	-15.6	3.4
NIST J	Grab Sampler	7	3560	0.507	64.6	-8.7	137.8
NIST K	Grab Sampler	3	3872	0.623	-31.8	-346.3	282.7
ALL NIST	Grab Sampler		3422	0.008	-1.46	-6.4	3.4
NIST I	Online	13	3520	0.135	-6.15	-16.5	4.2
NIST J	Online	8	3498	0.265	12.1	-8.0	32.2
NIST K	Online	3	3598	0.359	-26.7	-480.6	427.1
ALL NIST	Online	47	3582	0.019	-2.30	-9.1	4.5
PTI Dust	Grab Sampler	8	3529	0.382	-1.39	-3.2	0.4
A – Stock Suspension B – Type of Particle Counter C – Number of Data Points Used in the Regression Analysis D – Overall Average Particle Count (Number/mL > 2 µm) E – R ² for the Best Fit Trend Line F – Slope of the Best Fit Trend Line G – H – 95% Confidence Intervals for the Slope of the Best Fit Trend Line							

For the proposed CPE protocol a conservative maximum age of 90 days was selected because, while almost 1 year of results was measured with the PTI dust suspensions, the NIST MTD suspensions were studied for less than 80 days. The 90-day limit can be revised later when additional results or experience with the method indicates that a longer storage time is reasonable.

CONFIRMING THE NIST DUST CONCENTRATION (AN OPTIONAL STEP)

Allow the stock suspension to warm to room temperature and then shake and sonicate the suspension at medium power for about 30 seconds. Weigh six aluminum pans to $\pm 0.001\text{g}$ each. Shake the stock suspension and pipette 2-mL of suspension to a weighing pan and immediately reweigh. Repeat this with two more pans. With the remaining three pans, repeat the process using 2-mL of low particle water with no dust. Place all six pans in an 80°C drying oven for 24 hours. Weigh all pans after 24 hours and return them to the oven for 2 hours. Reweigh and if the repeated weights are not within 0.001g of each other return the pans to the oven for 1 hour. Repeat until the two successive weights are within 0.001g of each other. Record the second weight and the weight of the pan plus suspension before it was dried. The dust concentration in the stock suspension is calculated using the relationships described in the following experimental example.

Gravimetric Procedure for Checking Stock Suspension Dust Concentrations

The concentration of dust in each stock suspension is determined when the suspension is prepared using the weight of dust in the capsule and the volume of low particle water added to make the suspension. However, if questions arise later about the particle count measurements or about the quality of a stock suspension (e.g., the top was left partially open for a period of time and there are concerns that some water may have evaporated) a method to check the stock suspensions dust concentration will be needed.

This proposed gravimetric procedure for checking the stock suspension dust concentration was evaluated using several NIST MTD stock suspensions. The technique used involved measuring small volumes (typically one, 2 mL quantity or 10, 0.2 mL quantities) of stock suspension using an adjustable volume micro-dispenser and dispensing them into

aluminum weighing dishes. Each dish was weighed four times with a 4-place microbalance; 1) empty, 2) with the suspension just after dispensing and before evaporation, 3) after 24 hours of evaporation and drying and, 4) after one or two additional 2-hour periods of drying. The objective of this evaluation was to determine how well the dust concentration measured with the gravimetric procedure agreed with the concentration that was expected based on the weight of dust and volume of low particle water used to prepare the stock suspension.

The results obtained in the evaluation of the stock suspension labeled NIST J are listed in Table 5.2. In this case the 2-mL volume (nominal) of stock suspension in each of the 5 dishes was measured by dispensing 0.2 mL ten times for each dish.

The stock suspension dust concentration was calculated using the following equation:

$$\text{Stock suspension dust concentration} = \frac{\text{weight of dust dispensed into pan}}{\text{volume of water dispensed into pan}} \quad (5.2)$$

The weight of dust dispensed into the weighing dish (W) is equal to the weight listed in column 4 of Table 5.2 minus the weight of the weighing dish (column 1) and minus the estimated dry weight of the cellulose capsule material in the dispensed volume (C), i.e.,

$$\text{Dry weight of dust in the dispensed volume (W)} = (\text{column 4}) - (\text{column 1}) - (C) \quad (5.3)$$

Table 5.2
Example results for the gravimetric verification of the stock suspension dust concentration
– NIST J stock suspension.

Replicate number	(1) (grams)	(2) (grams)	(3) (grams)	(4) (grams)
1	1.0014	3.0410	1.0081	1.0081
2	0.9995	2.04793	1.0071	1.0059
3	0.9995	2.7200	1.0028	1.0027
4	0.9982	2.9802	1.0051	1.0054
5	0.9982	2.9756	1.0048	1.0047
Tare weight of the aluminum pan				
Aluminum dish plus stock suspension (wet)				
Dish + suspension after 24 hours of drying				
Dish + suspension after an additional 2-hrs of drying				

The dry weight of the cellulose capsule material in the dispensed volume (C) was estimated using:

$$\text{Dry weight of cellulose capsule in the dispensed volume} \cong \frac{\text{capsule weight} \times 2 \text{ ml}}{100 \text{ ml}} \quad (5.4)$$

The weight of the capsule used to prepare the suspension (NIST J) of Table 5.1 was 89 mg and therefore, according to Equation 5.3, C is equal to 0.00178 grams.

The volume of water dispensed with the dust and capsule material is given by:

$$\text{Volume of water dispensed (ml)} = \frac{(\text{column 2}) - (\text{column 1}) - (W) - (C)}{1 \text{ gm/ml}} \quad (5.5)$$

where, “column 1” and “column 2” are the measurements listed in columns 1 and 2 of Table 5.2, W is the weight of dust dispensed into the weighing dish (from Equation 5.2) and C is the dry weight of cellulose capsule in the dispensed volume (from Equation 5.4). Equation 5.5 assumes that the density of water is 1 gm/mL. Since all the suspensions of Table 5.2 were prepared with low particle water from the RO unit the dissolved solids concentration is effectively zero. The weights from the three pans with just low particle water and no dust or capsule material confirmed this assumption.

For replicate 1 of Table 5.2 and using Equation 5.5, the volume of water dispensed (V, mL) is given by:

$$V = \frac{3.0410 - 1.0014 - 0.0049 - 0.00178}{1 \text{ gm/mL}} = 2.0329 \text{ mL} \quad \dots\dots\dots(5.6)$$

Since the weight of dust dispensed in replicate 1 is 0.0049 grams, the calculated weight concentration of dust in the stock suspension is $(0.0049 \times 1000)/2.0329 = 2.42 \text{ mg/mL}$.

For the example of Table 5.2 the average value of the stock suspension dust concentration for the five replicate measurements is 2.30 mg/mL. The concentration based on the measured weight of dust in the capsule (234 mg) and the volume of water added to the capsule to

make the stock suspension (100 mL) is 2.34 mg/mL. Therefore, the error in the example of Table 5.2 is $[(2.34 - 2.30) \times 100]/2.34 = 1.6\%$.

The experiment of Table 5.2 was repeated seven times using three different stock suspensions, NIST I, NIST J and NIST K. The results are listed in Table 5.3. The capsule weights for NIST I and NIST K were 83 mg and 86 mg, respectively.

In Table 5.3, experiments 1 and 3 were conducted using a procedure in which a 0.2 mL volume was dispensed 10 times into each weighing pan. In all the other experiments a single 2 mL volume was dispensed into each pan. Each experimental result is the average of 5 replicate measurements.

The error listed in the last row of Table 5.3 ranges from 1.5 to 27.8 %. The greatest error was observed in the first experiment conducted, experiment 2 with stock suspension NIST J, and it is, therefore, reasonable to assume that a significant portion of this error was caused by operator inexperience. The average value of the error for all 7 experiments is 8.1 %. The average error for the group of 6 experiments that excludes the first experiment conducted is 4.9 %. Based on these results it seems reasonable to assume that this procedure for verifying the stock dust concentration will yield results that are within ± 5 to 10% of the true value.

Table 5.3
Results of concentration verification tests of the stock suspensions

Stock Suspension	NIST I	NIST J	NIST J	NIST J	NIST J	NIST J	NIST K
Experiment number	1	2	3	4	5	6	7
Average experimental dust concentration (mg/mL)	2.72	1.69	2.30	2.49	2.18	2.18	2.66
Expected dust concentration (mg/mL)	2.80	2.34	2.34	2.34	2.34	2.34	2.78
Percent Error	2.9	27.8	1.5	6.3	7.0	6.8	4.3

PREPARATION OF WORKING SUSPENSIONS

Allow the stock suspension to warm to room temperature, shake gently several times and sonicate for about 30 seconds. Add exactly 20 L of low particle water to a 20 L polyethylene carboy. Place a new disposable tip on a microdispenser with a capacity of at least 2 mL. Invert the stock suspension three times and immediately draw 2 mL of suspension into the pipette. Dispense the 2-mL volume to the 20 L of low particle water in the carboy. Stir for 20 minutes using a mechanical stirrer set at a rotational speed that just begins to create a vortex. While stirring, use the carboy's spigot and a clean graduated cylinder to divide the suspension into 2 L aliquots. The aliquots can be stored for a short time (< 90 minutes) in capped 2 L plastic bottles.

The Decision to Use Microdispensers to Prepare Working Suspensions

Two methods were considered for diluting stock suspension to make the working suspensions. One approach uses serial dilutions with conventional glass pipettes and laboratory glassware and the other an adjustable-volume microdispenser and plastic containers. It was decided to use the microdispenser approach because it minimizes the need for very clean glassware and reduces the possibility of random dilution errors and contamination. Additionally, minimizing the amount of glassware reduces the amount of expensive low particle water needed to wash glassware and prepare intermediate suspensions. Microdispensers have a disadvantage; the small volumes involved make a visual check of the volume dispensed essentially impossible.

A simple gravimetric procedure was used to evaluate the accuracy and precision of the microdispensers. Low particle water from the RO unit was dispensed into aluminum weighing pans using three methods; a single 2 mL volume dispensed into each pan (column A, Table 5.4); ten, 0.2 mL volumes dispensed into each pan (column B); and one, 0.2 mL volume dispensed into one pan (column C). Ten replicates were done in each test. The results are listed in Table 5.4.

According to Table 5.4 the measured weight of water was always slightly less than the expected weight. For the test of column A (2 mL dispensed in one shot), the expected weight was 1.995 g and the mean value for the 10 replicates was 1.967 g. This indicates that the volume dispensed was less than the expected amount of 2 mL about by about 1.4 percent (See Table 5.5).

The results for columns B and C in Table 5.5 are similar; the measured mean weights of water are less than the expected values by 1.7 and 1.3 percent, respectively.

A rough indication of the variability (the precision) in the measured volume of water is given by the standard deviation and coefficient of variation (COV) of the weights listed in Table 5.4. For all three methods of delivery to the weighing pans the COV was less than 2 percent. The highest COV (1.78 %) was observed with method A, 2 mL delivered in one shot, and the lowest (0.66 %) with method B, 2 mL delivered with 10 shots of 0.2 mL per shot.

Table 5.4
Gravimetric test of microdispenser volume for three dispensing methods. Values in the table are the measured weights of water in grams.

Pan	Method and volume of water dispensed ¹		
	A	B	C
1	1.990	1.977	0.198
2	1.870	1.958	0.194
3	1.972	1.961	0.200
4	1.982	1.959	0.195
5	1.975	1.986	0.196
6	1.973	1.969	0.194
7	1.979	1.969	0.199
8	1.986	1.949	0.197
9	1.959	1.949	0.198
10	1.980	1.946	0.196
Mean	1.967	1.962	0.197
St. Dev.	0.0350	0.0130	0.0021
COV (%) ²	1.78	0.66	1.05
¹ A – target volume = 2 mL (1 shot x 2 mL/shot) B – target volume = 2 mL (10 x 0.2 mL) C – target volume = 0.2 mL (1 x 0.2 mL)			
² COV = coefficient of variation (expressed as a percent)			

Table 5.5
Percent difference between the expected and measured weights of water dispensed in the microdispenser volume test (See Table 5.4)

Test	Expected weight per pan (g)*	Average measured weight per pan (g)	% Difference between expected and measured weights
A	1.995	1.967	1.4
B	1.995	1.962	1.7
C	0.1995	0.197	1.3
* Based on water density at 23°C of 0.9975 g/mL			

Reproducibility of Working Suspensions

Ten stock suspensions were used during the study to prepare working suspensions and seven of these were analyzed with the grab sampler on the day each stock suspension was prepared. Only 1-day-old stock suspensions were used to avoid any possibility that the age of the stock suspension had an effect on the results. Table 5.6 lists the mean, standard deviation and coefficient of variation for each of these sets of results. The mean and standard deviation in Table 5.6 have the units counts/ $\mu\text{g} > 2 \mu\text{m}$.

In these tests the working suspensions were prepared by diluting 0.2 mL of stock suspension in 2 L of low particle water. Each test included the measurement of 5 replicate working suspensions. According to Table 5.6, the average count for each set of working suspensions ranged from 2841 counts/ $\mu\text{g} > 2 \mu\text{m}$ (stock suspension B1) to 3841 counts/ $\mu\text{g} > 2 \mu\text{m}$ (stock suspension C). Stock suspension B was used twice on the day it was prepared. The first time is labeled B1 in Table 5.6 and the second time is labeled B2.

Figure 5.2 is a whisker plot that shows the mean and standard deviation for each set of working suspensions. The points in Figure 5.2 show that the variability in the results was high at the beginning of this set of experiments (when stock suspension B was used) but decreased as the student became more experienced. According to Table 5.6, the COV decreased from about 28 % for stock suspension B1 to values in the range of 2 to 5 % for the later tests.

Table 5.6
Working suspension mean particle count and standard deviation values for 5 replicates
prepared when each stock suspension was fresh (< 1 day old)

Stock Suspension	No. of replicates	Mean (counts/ μg > 2 μm)	St. Dev. (counts/ μg > 2 μm)	COV (%)*
B1	5	2841	784	27.6
B2	5	3755	609	16.2
C	5	3841	328	8.5
D	5	3710	133	3.6
E	5	3592	80	2.2
F	5	3801	172	4.5
G	5	3687	193	5.2
H	5	3723	98	2.6
* COV = coefficient of variation = standard deviation/mean (as a percent)				

An analysis of variance (ANOVA) was used to determine if any of the mean working suspension particle counts was different from one or more of the other mean counts by a statistically significant amount. The results in Table 5.7 show that at a 0.05 level of significance ($p = 0.05$) and a computed p -level = 0.0063, since $0.0063 < 0.05$, at least one of the mean particle count values in Table 5.6 is significantly different than the others.

Inspection of Table 5.6 and Figure 5.2 suggests that the mean working suspension particle count for stock suspension B1 and possibly for suspension B2 are significantly different than the other mean values. *Post Hoc* analysis by the least significant difference technique was used to determine which of the mean particle count values are different than the others. The results of this analysis are listed in Table 5.8.

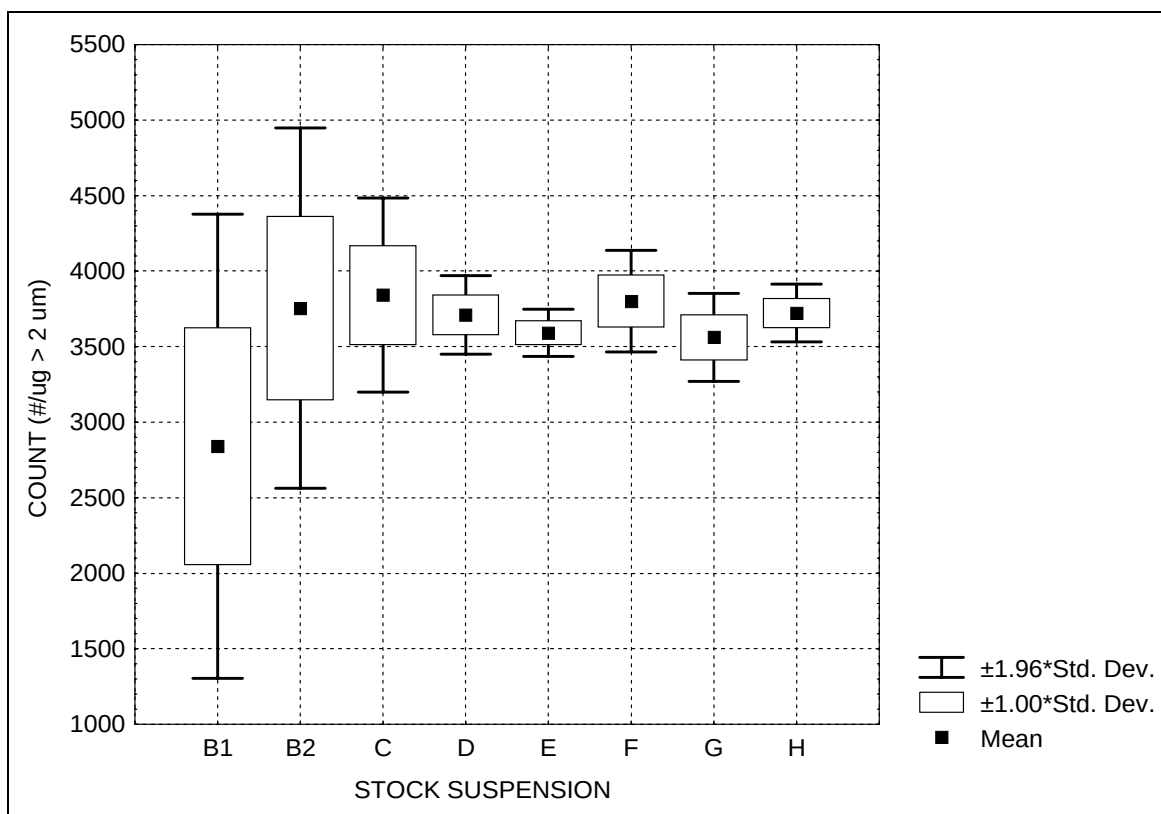


Figure 5.2 Whisker plot of the mean and standard deviation for working suspensions prepared from fresh stock suspensions.

Table 5.7
ANOVA results for working suspensions prepared using fresh stock suspensions (See Table 5.6 and Figure 5.2)

df	MS	df	MS		
Effect	Effect	Error	Error	F	p-level
7	519,987.4	32	147,174	3.533	0.0063

Table 5.8
Working suspensions from fresh stock suspensions - *post hoc* analysis by least significant difference test

Stock Suspension	B1	B2	C	D	E	F	G	H
	2841*	3755	3841	3710	3592	3801	3561	3723
B1		0.0006	0.0002	0.0011	0.0041	0.0003	0.0056	0.0009
B2	0.0006		0.7249	0.8534	0.5057	0.8511	0.4300	0.8953
C	0.0002	0.7249		0.5921	0.3116	0.8694	0.2570	0.6292
D	0.0011	0.8534	0.5921		0.6297	0.7097	0.5442	0.9575
E	0.0040	0.5056	0.3116	0.6296		0.3949	0.9004	0.5925
F	0.0003	0.8511	0.8694	0.7097	0.3949		0.3303	0.7497
G	0.0056	0.4300	0.2570	0.5442	0.9005	0.3303		0.5098
H	0.0009	0.8953	0.6292	0.9575	0.5926	0.7497	0.5098	
*Values in the second row are the mean count values for the working suspensions prepared from a given stock suspension, e.g., 2841 is the mean counts/ $\mu\text{g} > 2 \mu\text{m}$ for stock suspension B1, the first set of working suspension prepared from stock suspension B.								

The numbers in Table 5.8 are p-values, the probability that the two means being compared and corresponding to that cell could be different purely due to chance. If a computed p is less than 0.05 the difference between that pair of means is statistically significant at a 95 % level of confidence. It is apparent that the mean particle count for the working suspensions from stock suspension (B1) is different than all other mean particle counts. The mean values for the other stock suspensions (B2, C, D, E, F, G, and H) are in reasonable agreement. The overall mean particle count for all the working suspensions except those from B2 is 3730 counts/ $\mu\text{g} > 2 \mu\text{m}$ and the standard deviation is 81 counts/ $\mu\text{g} > 2 \mu\text{m}$. The overall COV is 2.2 %.

The data from these tests suggests that practice is an important factor in the preparation of working suspensions. As experience was gained during the tests over several months, the variability in the working suspensions values for a given stock suspension decreased and the agreement between mean working suspension particle count values increased.

Volume of the Working Suspensions

Selecting the volume of the working suspension was an important decision in developing the CPE protocol. If the working suspension is delivered to the particle counter by gravity flow through the flow control weir then a much larger volume is needed compared to drawing the suspension through the sensor with the gear pump (as in the portable grab sampler). For gravity flow, suspension is needed to fill the weir and tubing and to flush out the system before the particle count readings are recorded. The minimum volume needed to use the gear pump delivery method is about 2 L and the minimum volume for gravity flow is between 10 and 20 L.

Two-liter working suspensions are easier to work with; they are lighter and less cumbersome, but in the dilution process they require the use of very small volumes of stock suspension (0.2 mL). This intuitively seems more variable than a larger volume such as 2 mL. A larger suspension volume (e.g., 20 L) might produce more consistent results for two reasons: a larger pipette volume would be used in diluting the stock suspension and, assuming that the larger suspension was well mixed, the replicate aliquots sampled from it should be more similar than aliquots prepared by separate dilutions with 0.2 mL of stock suspension.

Tests were conducted using 2 L and 20 L working suspensions to assess the effect of suspension volume on particle count variability. Two stock suspensions, NIST I and NIST J, were used in these experiments. Eight trials were performed, five with NIST I and three with NIST J. Each trial involved five, 2 L replicates and one 20-L suspension from which five, 2 L aliquots were withdrawn for analysis; each 2-L working suspension was prepared using 0.2 mL of stock suspension in 2-L of low particle water and each 20-L suspension was prepared, as described above, with 2 mL of stock suspension in 20 L of low particle water.

The results of the eight trials are presented in Table 5.9. The trials performed with NIST I gave a higher overall average and coefficient of variation (COV) for the 2 L samples than did the 20 L samples between trials. The averages and COVs for the three trials performed with NIST J were much closer to one another. There was still a larger amount of variation shown by the 2 L samples, however, 4.3% COV is an acceptable amount of variation.

The t-test was used to compare the mean counts for the 2 L and 20 L working suspensions and both stock suspensions. (The t-test is used to compare two means; ANOVA is used to compare means when the set includes more than two.) For the five trials with NIST I and

a 95% confidence level, t is less than t -critical ($1.96 < 2.31$) and for the three trials with NIST J and a 95% confidence level, t is also less than t -critical ($1.30 < 2.78$). According to these results the mean counts for the 2 L and 20 L trials for both stock suspensions are not different by a statistically significant amount at a 95% confidence level.

The standard deviations listed in Table 5.9 show that the variability among replicates in a given trial and the variability among the trials for a given stock suspension were lower when the working suspension volume was 20 L compared to 2 L. For the NIST I trials the COV was 6.3 % for the 20 L working suspensions and 10.8 % for the 2 L suspensions. The overall variability in the working suspension particle counts was lower in the NIST J trials and the 20 L working suspensions again had less variability than the 2 L suspensions. For the 20 L suspensions the COV was 1.9 % and for the 2 L suspensions it was 4.6 %. This result is one of the reasons why 20 L working suspensions are recommended for the CPE protocol.

In the gravimetric test of microdispenser precision the COV for a 0.2 or 2 mL dispensed volume was 1.3 to 1.4 %. Therefore, the minimum COV in Table 5.9 for the replicates in a given trial should be greater than 1.4 %. Since most of the COV values for individual trials are between 5 and 10 % there is obviously room to reduce the variability among replicates and trials. However, the stock suspension volume measurement step should not be the only focus of an effort to make this improvement. Operator proficiency is a key consideration.

Storage and Mixing of Working Suspensions

It was learned during the experiments that it is not always possible to use each working suspension immediately after it is prepared. Since it was observed that particles settle from the concentrated stock suspensions during storage, an experiment was conducted to determine if working suspension particles that had settled in the 20 L carboy during storage could be re-suspended by mechanical mixing just before the suspension was dispensed into 2 L containers and fed to the particle counter. It is not possible to ultra-sonicate the entire volume of a 20 L working suspension with laboratory scale equipment and it is difficult (but not impossible) to manually shake a 40-lb 20 L carboy.

Table 5.9
Effect of working suspension dilution volume on measured particle counts

NIST I	Working Suspension Volume			
	2L*		20L**	
Trial	Mean	St. dev.	Mean	St. dev.
1	4247	704	3464	171
2	3298	155	3049	51
3	3579	190	3492	59
4	3773	493	3454	124
5	3651	533	3365	40
Mean	3709		3365	
St. dev.	400		211	
COV (%)	10.8		6.3	
NIST J	2L*		20L**	
	Average	St. dev.	Average	St. dev.
1	3620	91	3552	174
2	3305	151	3687	98
3	3516	204	3594	140
Mean	3480		3611	
St. dev.	161		69	
COV (%)	4.6		1.9	
* In each trial five working suspensions were prepared by diluting 0.2 mL of stock suspension in 2 L of low particle water. ** In each trial 5 working suspensions were prepared by diluting 2 mL of stock suspension in 20 L of low particle water and then analyzing 5, 2 L aliquots of the 20 L volume.				

Three, 20 L working suspensions with dust concentrations of 0.182, 0.364 and 0.728 $\mu\text{g/mL}$ were prepared and then stored. (0.364 $\mu\text{g/mL}$ is close to the concentration used in most of experiments of this study). After a period of storage each suspension was mixed with a rotating impeller (a 3-blade propeller, 5-cm in diameter) for 20 minutes immediately before three, 2 L portions were withdrawn and the grab sampler was used to measure the particle count. This process was repeated three times, on the day the suspension was prepared, after 3 days of storage and after 7 days of storage. The results are plotted (as counts/ $\mu\text{g} > 2 \mu\text{m}$) in Figure 5.2.

According to Figure 5.3 after about 3 days of quiescence it was not possible to recover the initial particle count with mechanical mixing for any of the three dust concentrations. For a storage period of less than three days the initial particle count was recovered with the 2 lowest dust concentrations (0.182 and 0.364 $\mu\text{g/mL}$) but not with the highest (0.728 $\mu\text{g/mL}$). In general, 20 L working suspensions of NIST MTD appear to exhibit a significant decrease in the particle count (for a 2 μm count threshold) if they are allowed to sit without continuous mixing for more than 24 hours. The loss is greatest at the highest dust concentration (0.728 $\mu\text{g/mL}$) possibly because of increased aggregation and deposition on the container walls.

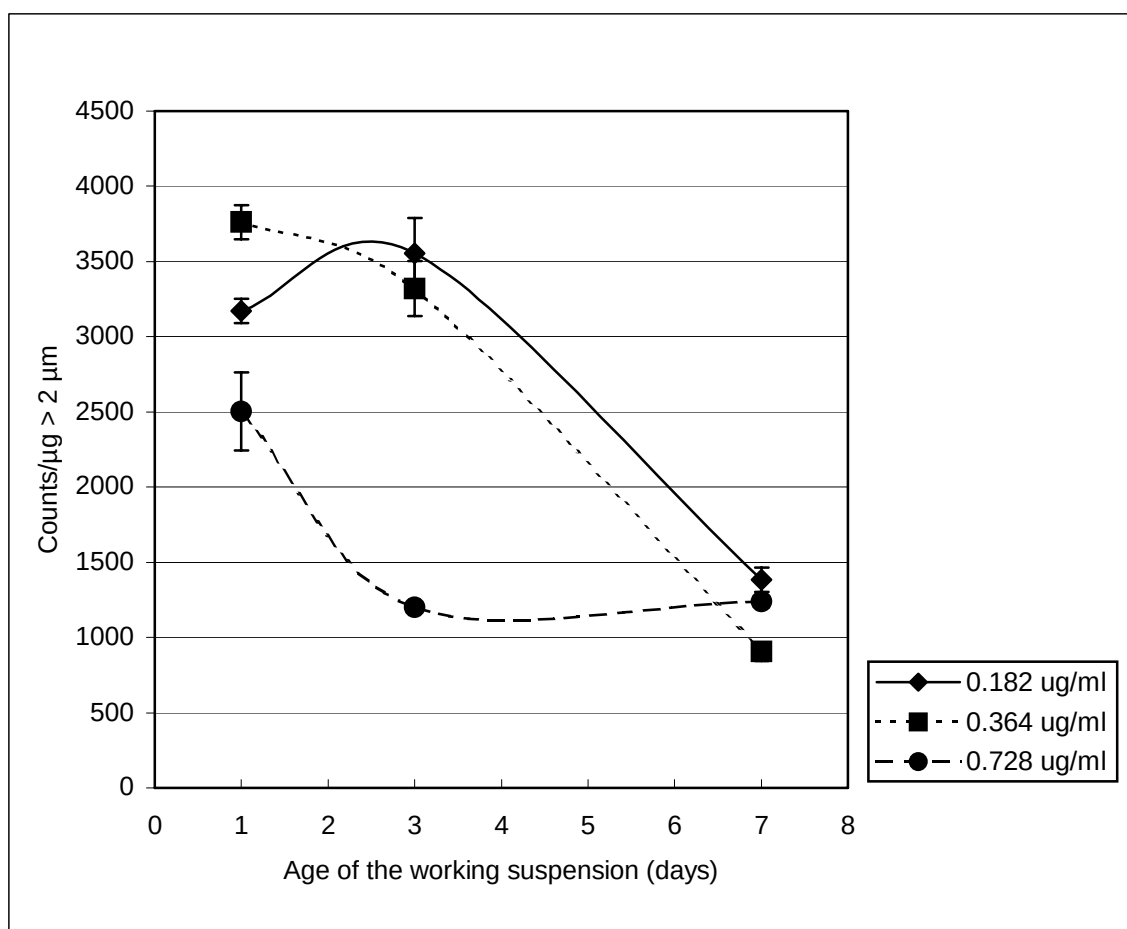


Figure 5.3 Effect of resuspension using mechanical mixing following quiescent storage on the working suspension particle count (grab sampler) for three dust concentrations. The error bars are $\pm$ one standard deviation.

The exact reason why it becomes increasingly difficult with time to resuspend particles after quiescent storage of working suspensions is not known. It is possible that settled particles and particles that were transported by Brownian diffusion to the walls of the container become attached and that this attachment becomes less reversible with time. It is also possible that irreversible aggregation of the particles occurs at the walls of the container. Stock suspensions are concentrated and losses probably occur but the fractional amount removed from the suspension is insignificant because the ratio of dust to container surface area is so large. Also, because of their relatively low volume, stock suspensions can be agitated with much greater intensity than working suspensions.

The resuspension of NIST MTD in 2 L samples removed from a 20 L working suspension was analyzed. A set of six, 2 L portions was taken from a 20 L working suspension and analyzed immediately using the grab sampler. They were allowed to stand without mixing for 90 minutes, then inverted 5 times and reanalyzed with the counter. Ninety minutes was used because it was assumed to be the maximum amount of time a 2 L portion from a 20 L working suspension might need to wait before it was fed to the particle counter.

The six, 2 L samples from the 20 L carboy gave a mean and standard deviation of 989 and 65 counts/mL $> 2 \mu\text{m}$ before storage and 1021 and 40 counts/mL $> 2 \mu\text{m}$ after storage and mixing by inversion. According to the t-test, the mean counts before and after storage are not different by a statistically significant amount at a 95 % level of confidence ($t = 1.03$ and $t\text{-critical} = 2.31$).

These results suggest that a 90 minutes quiescent period after the 2 L portions are removed from the 20 L carboy will not have a significant effect on the measured particle count. In any case, if it can be avoided, no dilute NIST dust suspension (20 L working suspension and 2 L portion) should be stored.

If working suspensions are prepared as suggested above using 2-mL of stock suspension in 20 L of low particle water and if the stock suspension has the recommended concentration of dust (approximately 300 mg in 100 mL of water) then the dust concentration in working suspensions will be approximately 0.3 mg/L or 0.3 $\mu\text{g/mL}$. For the typical light-obscuration, on-line particle counter this concentration of NIST MTD gives a particle count of about 1000/mL $> 2 \mu\text{m}$. This count level is much lower than the coincidence limit of all the particle counters tested but higher than the expected concentration of particles in filtered water. For particles larger than

the 2- μm threshold (“2- μm ” based on size calibration with PSL microspheres) the concentration of particles in most filtered water ranges from 1 to 200/mL (McTigue et al. 1998).

It was observed that the variability of the measured counts in each set of five replicate working suspensions increases as the dust concentration decreases. With a target count of 1000/mL $> 2 \mu\text{m}$ the coefficient of variation is typically less than 5 % (as discussed above) and this level of agreement can be maintained down to a count level of about 500/mL $> 2 \mu\text{m}$. Below this level it becomes increasingly difficult to prepare a consistent (low COV) set of working suspensions. Therefore, if the goal is to prepare suspensions that have count levels that are as close as possible to the levels in filtered water the amount of stock suspension added to 20-L of low particle water should not be less than 1-mL. This volume of stock suspension will give a mean working suspension count level of about 500/mL $> 2 \mu\text{m}$.

FEEDING THE WORKING SUSPENSION TO THE ON-LINE COUNTER

Invert a 2 L portion of working suspension three times and insert the counter inlet tube into the suspension bottle. Avoid having the tube touch surfaces in the bottle, especially the bottom. Turn on the gear pump and set it to deliver the required flow rate through the counter. The gear pump should always be installed after the sensor and the inlet and outlet-tubing lengths should be as short as possible. Check the flow rate by timing the collection of 50 mL of suspension flowing into a volumetric cylinder. Adjust and check the flow rate, if necessary. After flushing the tubing and sensor for at least 3 minutes, pump the suspension long enough for the counter to record 5 separate count values; a minimum of five to ten minutes will be required for most counters.

Once five count values have been recorded, invert a second 2 L aliquot of working suspension 3 times and change the counter input tube from the first sample to the second. Check and, if necessary, adjust the flow rate of the gear pump. Repeat until at least five portions of working suspension have been fed to the counter.

Gear Pump versus Gravity Feed

Two methods are available for feeding working suspension to an on-line particle counter. In the first method the working suspension flows by gravity from a container placed above the

counter. The particle counter's constant head device, typically a vertical tube with an overflow weir, controls the flowrate through the sensor. In the second method a pump is used to draw the working suspension through the sensor at a constant flowrate. A gear pump (similar to the ones used in grab sample particle counters) is preferred because this type produces a flow with negligible velocity pulsations and it is essentially self-priming.

Each method, gravity feed and gear pump, has advantages and disadvantages. A significantly larger volume of working suspension is needed for gravity feed because part of the suspension is wasted by overflow at the constant head device. The container of working suspension must be lifted to a stepladder or similar device and it must be mechanically mixed while the suspension is flowing. Also in gravity flow the suspension will invariably contact tubing and other surfaces (e.g., surfaces in the flow control device) that are not clean and may shed particles. Additional working suspension is wasted flushing out the tubing and other parts of the counter.

The gear pump approach (see Figure 5.4) requires a much smaller volume of working suspension because short sections of new, small diameter tubing can be used and there is no flow control device (such as the tube and overflow weir) between the sensor and the working suspension container. The tubing connects the gear pump to the sensor outlet and the sensor inlet to the working suspension container. With the gear pump it is easier to check and adjust the flowrate without disturbing the setup, which may release particles and can alter the flowrate. The principal disadvantages of the gear pump approach are the cost of the pump, usually between \$400 and \$800, and the need to disconnect the counter's sensor from its flow control system.

Experimental Comparison of Gravity Feed and Gear Pump

A working suspension of NIST MTD was prepared at a concentration of 0.273 $\mu\text{g/mL}$ in the 50 L overhead reservoir of the apparatus used for comparing particle counter performance (See Figure 2.1, chapter 2). Part of this suspension flowed by gravity through the distribution manifold to the Counter D and part of it was collected in 3, 2 L plastic sample bottles. The bottles were filled using one of the taps in the flow distribution manifold at the same time the suspension was flowing to the counter. After the gravity flow part of the experiment was completed the suspension in the 2 L containers was drawn through Counter D using the gear pump set at 100 mL/min.

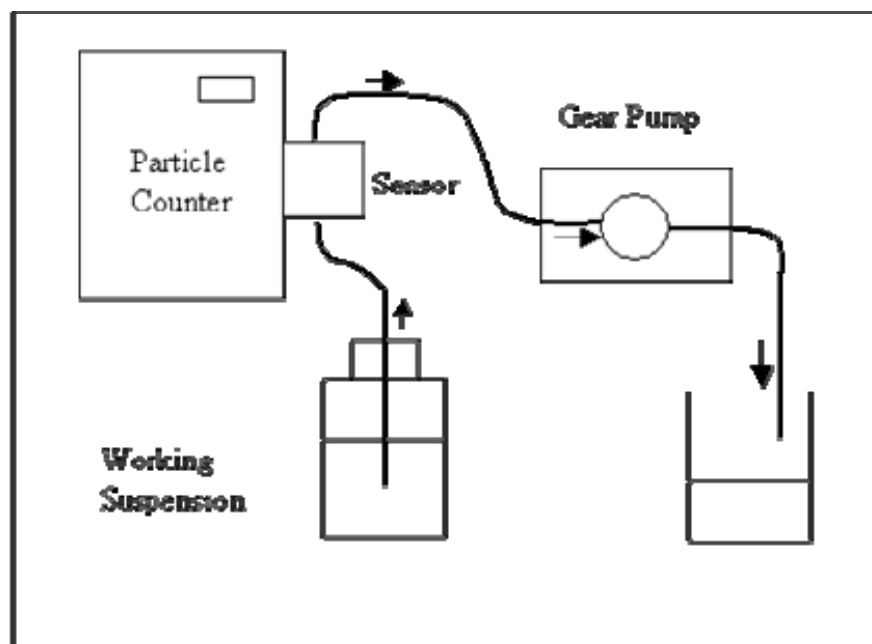


Figure 5.4 Schematic diagram of the gear pump feed system

The portion of suspension that flowed by gravity through the constant head device and counter (about 6 L) gave a particle count of 773 counts/mL $> 2 \mu\text{m}$, and the 3, 2 L portions that were drawn through the counter by the gear pump gave a particle count of 797 counts/mL $> 2 \mu\text{m}$. Since there were no true replicates measured in this experiment it was not possible to do a statistical test of the agreement of these methods. However, the particle concentration results are close (within about 3 %) and indicate that the gravity flow and gear pump feed methods give similar results.

The gear pump approach is recommended for the count performance evaluation protocol. The gravity feed technique is useful but it is more time consuming and it uses much more low particle water. There are also uncertainties about the significance of passing the working suspension through parts of the system, such as the constant head device, that are difficult to clean. Gear pumps are expensive but given the cost of labor and other resources needed to do in-plant count performance evaluation testing, the amount is not large in proportion.

COLLECTION AND ANALYSIS OF THE DATA

Before the gear pump is turned on, the data logging software should be loaded and running. Set the software to collect data as total counts/mL $> 2 \mu\text{m}$, or as a set of “bins” that

can be added to give total counts/mL $> 2 \mu\text{m}$. Each set of count data should be downloadable. Once the target flow rate through the sensor (from the manufacturer's literature) is achieved by adjusting the gear pump, record the time and the flow rate. This "time stamp" is an effective way to label the data sets so each can be matched with the corresponding portion of working suspension when the data output file is analyzed. For each portion of working suspension record at least 5 complete data sets as the suspension is pumped through the sensor. For most counters the time required to do this will be less than 10 minutes and the volume of working suspension used will be about 1 L. Toward the end of the data collection period re-measure the flow rate. The flow rate for the test is the average of the initial and this final flow rate.

After all the working suspension portions have been fed to the counter, retrieve the data output file and average the 5 or more particle count values (counts/mL $> 2 \mu\text{m}$) for each portion of working suspension. Divide the mean particle concentration (counts/mL $> 2 \mu\text{m}$) by the dust concentration in the working suspension ($\mu\text{g/mL}$) to obtain the final mean concentration in counts/ $\mu\text{g} > 2 \mu\text{m}$ for that portion. If the target flow rate and the average for the test are different correct the measured count for the flow discrepancy. Analyze the data in the count performance evaluation data set using ANOVA (Analysis of Variance) to determine if one or more particle counters are not in agreement. It is recommended that commercial statistical software (e.g., Statistica) be used to do the ANOVA.

Checking for Trends in the Data for Each Portion of Working Suspension

The particle count data logged for each 2 L portion of the working suspension (usually 5 to 15 points) should be checked to determine if the measurements decrease (or possibly increase) in a significant way with time. If, for example, the counts decrease to a significant extent with time it suggests that the particle concentration in that portion of working suspension was affected by factors such as flocculation and/or settling.

To minimize settling the working suspension portions can be mechanically mixed as the suspension is fed to the particle counter sensor. However, this increases the chance that the mixing impeller will contaminate the dust suspension and that bubbles will be formed that affect the particle count results. In experiments where the particle count and size distribution were measured as a function of time and the overall particle count decreased with time, the particle

size distribution did not change to a significant extent the way one would expect when settling removes proportionally more large ($> 10 \mu\text{m}$) than small ($2 - 3 \mu\text{m}$) particles. Given these results and our overall experience with both continuous mixing and no mixing of the working suspensions, it was concluded that the best approach (for suspension feeding periods that are less than 10 minutes in length) is to mix each portion of suspension (typically 2 L) by repeated inversion before it is pumped and to not use mechanical mixing during the feeding period.

It is suggested that the following procedure be used to check the recorded data points to determine if they vary significantly with time. The particle count values are plotted as a function of time and regression analysis is used to fit a straight line to the points. The intercept of this line on the y-axis (at time = 0) is determined along with its 95 % confidence limits. If the mean value of the measured points falls within the 95 % confidence interval for the y-axis intercept the amount of variation of the data with time is assumed to be insignificant and the mean particle count for that data set is used as a valid point. The following example illustrates the procedure.

In this example a 2 L portion of working suspension was analyzed using the grab sampler. Fifteen particle concentration points were logged on the computer as the sample was drawn through the sampler. The points are plotted versus time in Figure 5.5. The straight solid line on the graph was fitted by regression analysis and the horizontal dotted line is the mean value of all the plotted points. The y-axis intercept and its 95% confidence limits are 701 ± 21 counts/mL $> 2 \mu\text{m}$ and the mean of the plotted points is 680 counts/mL $> 2 \mu\text{m}$. Since the mean is just within the 95 % confidence interval of the intercept (680 to 722 counts/mL $> 2 \mu\text{m}$) it is assumed to be a valid point.

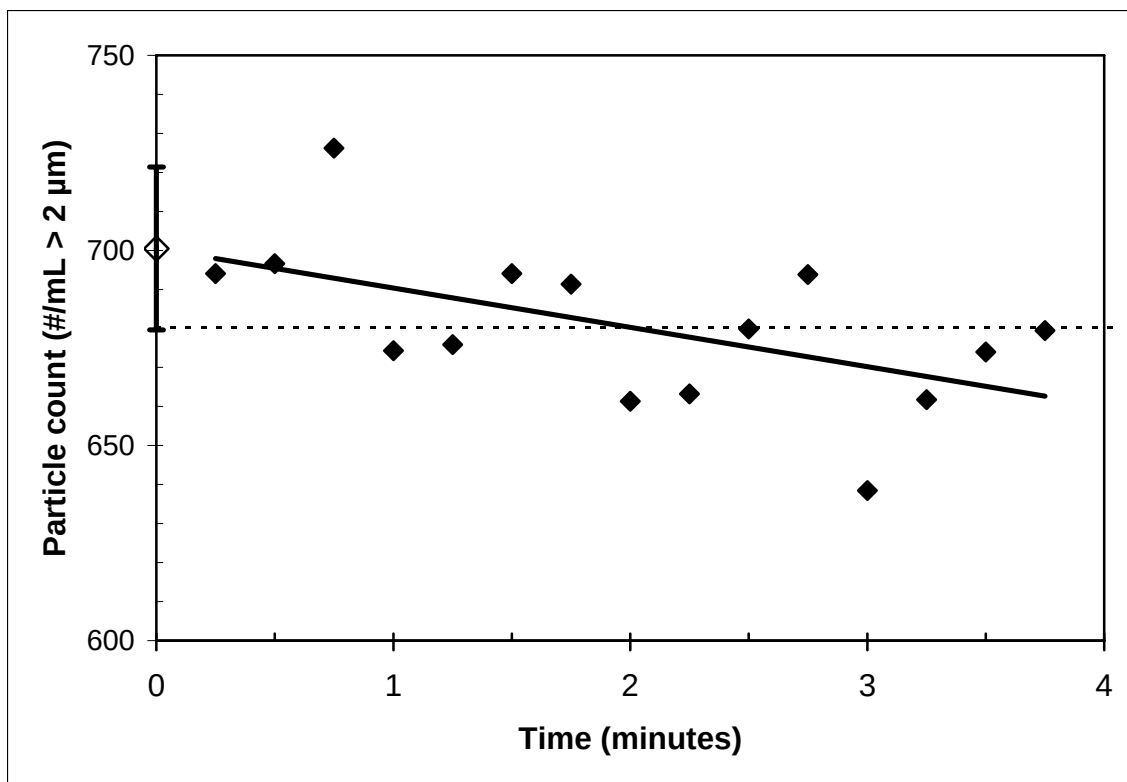


Figure 5.5 Plot of a sequence of particle count values measured during the analysis of a 2 L portion of working suspension with the grab sampler.

Based on an analysis of about 20 sets of measurements (like those in Figure 5.5) it seems that the particle count decreases by about 11.4 counts/mL > 2 μm every minute (or 1.3 % per minute) during the typical sampling period. Under conditions where it takes longer than 3 to 4 minutes to log at least 5 data points then it is possible that the y-axis intercept will be a better quantity to use than the mean to characterize the logged points. Five replicates from one set of experiments were used to determine a set of points that included both mean count and the corresponding y-axis intercept values. The coefficient of variation was slightly lower for the mean count values and the intercept count values were slightly greater than the corresponding means. This is an area that should receive additional study and analysis.

Analysis of Variance

Analysis of variance (ANOVA) is the name given to the statistical techniques that are used to compare the means of different groups of observations to determine if

there are any significant differences among the groups. ANOVA can be used to analyze the data obtained in a series of count performance evaluation (CPE) tests. The analysis reveals if the average responses of the instruments tested are different by a statistically significant amount and also if the differences, if they exist, have remained significant over time. ANOVA provides an objective method for identifying the counters that are not giving results that are consistent with the other counters in the treatment plant.

There are three steps in using ANOVA to analyze CPE test data:

- 1) Testing for homogeneity of variance
- 2) The ANOVA procedure
- 3) The post hoc analysis

The ANOVA procedure (step 2, above) includes the assumption that the values in the different groups have approximately the same variance. If the variances are widely different, e.g., the highest and lowest variances are different by a factor that is greater than 5 to 10, the ability of the F-test (the key statistical test of the ANOVA procedure) to detect differences among the group means is reduced. Testing the data to see if this assumption about the group variances is obeyed is called “testing for homogeneity of variance”. A number of statistical procedures are available for doing this test.

If the ANOVA procedure (the F-test) shows that there are significant differences among the means for the various groups of data then a *post hoc* analysis can be used to determine where the differences lie. In the analysis of CPE data the *post hoc* analysis identifies which counter or counters are giving average count results that are significantly and consistently different than those of the other counters.

These statistical procedures do not tell us why there is poor agreement between counters; they simply show that statistically significant differences in count performance are occurring among the counters. If a counter(s) produces test results that are determined to be significantly different than the results of other counters, the following should be considered:

This study has shown that consistent recoveries of NIST MTD can be accomplished using a given instrument, but it has not shown that different instruments, even from the same manufacturer, can give consistent results. As the CPE is repeated with time an instrument may

agree with the other instruments one time but not the next. In this case the stability of the instrument with time can be questioned and its performance carefully monitored.

Significant differences between and among the instruments on a given test date may be caused by factors that are not directly associated with the particle counters such as the cleanliness of the containers. The *post hoc* analysis will show if the differences persist or if agreement is reached when a new set of tests is done in the future.

Analysis of a Hypothetical CPE Data Set

The hypothetical data set of Table 5.10 assumes that a treatment plant has three particle counters, one for the raw water and one for each of the two filters. The CPE test has been done twice on all three counters, once in January and once three months later, in March. (The length of time between the tests is not a factor in the analysis). On both dates each counter was tested with five replicate working suspensions and all the working suspensions had the same dust concentration. The count values in Table 5.10 are given as counts/ μg for particles larger than the 2- μm threshold.

Table 5.10
Data for a hypothetical count performance evaluation at a treatment plant
with 3 on-line counters

count	counter	date	count	counter	date
3561	1	1	4028	1	2
3610	1	1	3080	1	2
3558	1	1	3511	1	2
3512	1	1	3353	1	2
3626	1	1	3599	1	2
3653	2	1	3513	2	2
3376	2	1	3528	2	2
3652	2	1	3599	2	2
3658	2	1	3507	2	2
3431	2	1	3790	2	2
3221	3	1	3050	3	2
3231	3	1	3265	3	2
3244	3	1	2997	3	2
3179	3	1	3120	3	2
2882	3	1	3228	3	2

In Table 5.10 each counter and analysis date has been labeled with a number. (In most statistical analysis software packages letters can also be used). The three counters are numbered 1, 2, and 3 and the dates are labeled 1 for January and 2 for March. Table 5.11 is a summary of the averages and standard deviations for the 6 data groups of Table 5.10.

The ANOVA Procedure

ANOVA tests the hypothesis that the mean count values for the 6 groups of data (3 counters and 2 dates) are equal. According to Table 5.12, Levene's test for the homogeneity of variance, indicates that for the data set of Table 5.10, the variances are significantly different ($p = 0.011 < 0.05$). Consequently the ability of the F-test (the key statistical test of the ANOVA procedure) to detect differences among the group means is reduced and the ANOVA results will have to be interpreted carefully.

Table 5.11
Mean and standard deviation for the groups of data in Table 5.10

Counter	Date	Group	Mean	Standard Deviation
1	1	{1}	3573	45.6
1	2	{2}	3514	348.5
2	1	{3}	3554	138.6
2	2	{4}	3587	119.2
3	1	{5}	3151	152.6
3	2	{6}	3132	114.3

Table 5.12
**ANOVA on absolute within-cell deviation scores for Levene's test
for homogeneity of variance**

Variable	MS Effect	MS Error	F	p-level
Count	23,217	11,278	2.06	0.011

The results of the ANOVA procedure applied to the means of the groups of data in Table 5.10 are listed in Table 5.13. They show that the differences among the three counters (with the date not considered) are very significant (in the first row the F-statistic is high and the p-level is very low) but this is not the case for between the two dates (all counters considered) and for the

counter and date in combination. In summary, the data suggests that each counter is performing consistently over time, but at least one of the counters is producing count values significantly different than the others. To find out which counter (or counters) are causing this significant difference, multiple comparisons of the means must be made using the *post hoc* analysis section in the statistical analysis software.

Post Hoc Analysis – Least Significant Difference Test

Most statistical software includes a number of methods for doing *post hoc* tests after ANOVA. In this example the least significant difference (LSD) procedure was used to illustrate the application of *post hoc* analysis when the ANOVA has shown that the difference between means is statistically significant.

The LSD test is essentially a t-test, applied to each pair of means. The means in each pair are compared and the statistical significance of the differences are computed and listed in a table. The results of the LSD analysis of the data in Table 5.10 are listed in Table 5.14.

Table 5.13
Results of F-test for the ANOVA

Effect	df Effect	MS Effect	df Error	MS Error	F	p-level
Counter	2	577,539	24	32,212	17.9	0.000017
Date	1	1713	24	32,212	0.053	0.820
Counter and Date	2	5388	24	32,212	0.167	0.847

The numbers listed in Table 5.14 are the probabilities that the difference between the means in each pair can be attributed purely to chance. For example, in the first line of Table 5.14 the probability that the difference between the mean of Group 1 (counter 1 and date 1) and the mean of Group 2 (counter 1 and date 2) can be attributed entirely to chance (i.e., to random error) is equal to 0.607 or about 61 %. This probability is significantly greater than the level of significance we are using (5 %) and we therefore conclude that the difference between the means is not statistically significant (i.e., it is probably due to chance).

According to Table 5.14 all the entry's that are associated with Groups 5 and 6 (counter 3, dates 1 and 2) have probabilities that are significantly less than 0.05, the level of significance. This strongly suggests that the mean count values for counter 3 are significantly lower than the mean count values for all the other counters on both dates.

A reasonable conclusion is that counter 3 is not in agreement with the other counters. This may be because counter 3 is not performing properly or because it simply does not count the same as the other counters. Because counter 3 is performing consistently over time a reasonable course of action would be to clean the sensor and continue to monitor the counter's performance over time, realizing that subsequent tests may show significant differences between counter 3 and one or more of the other counters. If there is any doubt about the counter's performance it should be returned to the manufacturer for testing and possibly recalibration.

Table 5.14
Probabilities for LSD test - Post hoc analysis

Counter	Date	Group	{1}	{2}	{3}	{4}	{5}	{6}
1	1	{1}		0.607	0.866	0.903	0.001	0.001
1	2	{2}	0.607		0.728	0.525	0.004	0.003
2	1	{3}	0.866	0.728		0.772	0.002	0.001
2	2	{4}	0.903	0.525	0.772		0.001	0.001
3	1	{5}	0.001	0.004	0.002	0.001		0.865
3	2	{6}	0.001	0.003	0.001	0.001	0.865	

A FIELD TEST OF THE CPE PROTOCOL

Background

The count performance evaluation (CPE) protocol that is presented above was used to test the performance of three online particle counters at the Onondaga County Water Authority (OCWA) water treatment plant in Marcellus, New York. This 25 MGD direct filtration plant is located approximately 30 minutes southwest of Syracuse, NY and provides water for the southern and western portions of Onondaga County in Central New York.

The OCWA plant operates six online particle counters, one for each of four mixed media filters, one for the filter influent and one for the combined outflow of the plant. These counters are the same make and model as the online counter selected at Syracuse University during the study to develop and test this protocol. Three of the four filter effluent counters were used in this evaluation because they could be taken offline for short periods of time to perform the evaluation; the OCWA plant personnel were very reluctant to take the filter influent and combined outflow counters off-line, even for short periods of time.

Field and Laboratory Measurements

The CPE protocol was tested two times at the OCWA plant with NIST MTD stock suspensions. The evaluations were performed three months apart, the first in March of 2000 and the second in June. During both evaluations, the online counter at the University was used as a comparison instrument. Low-particle water from the laboratory reverse osmosis unit was used to prepare all the suspensions. The photographs (Figures 5.6 – 5.8) on the next two pages show the gear pump setup at the particle counters in the filtered water pipe gallery.

During the March evaluation each counter analyzed five, 2 L portions of working suspension. Each working suspension was prepared in a 20 L carboy using a 2.34-mg/mL stock suspension. Two 20 L working suspensions had to be prepared to provide the volume needed for all 4 counters (5 portions times 2 L per portion times four counters requires 40 L of suspension). The working suspensions were prepared at Syracuse University and were transported (in about 1 hour) as 2 L aliquots to the treatment plant. The counter at the University was analyzed last and the five portions that it analyzed traveled to the water treatment plant and back over a period of 3 to 4 hours.

For the June evaluation, each counter analyzed three (instead of five) working suspension portions. The June working suspensions were prepared in 20 L carboys using a 2.78-mg/mL stock suspension. Two 20 L working suspensions were prepared, one at the University and one at the treatment plant. Only three portions were analyzed for this evaluation because the single 20 L working suspension prepared at the treatment plant limited the volume available for analysis (three portions times 2 L per portion times three counters requires 18 L of suspension). For consistency, only three portions from the 20 L quantity prepared at the University were used. The working suspensions were prepared at two locations to minimize the time each suspension was stored without mixing.



Figure 5.6 Photograph of the gear pump apparatus set up at a particle counter in the OCWA treatment plant pipe gallery.



Figure 5.7 Photograph of the students preparing to download the particle counter evaluation data from the plant computer in the laboratory manager's office.



Figure 5.8 Photograph of a student preparing to feed a 2 L portion of working suspension (in the plastic bottle) through the sensor using the gear pump. The graduated cylinder on the work surface is used for setting and checking the gear pump flowrate.

The results of the March and June tests are listed in Table 5.15. The count ranges from 3400 to 4400 counts/ $\mu\text{g} > 2 \mu\text{m}$. The highest count (4432 counts/ $\mu\text{g} > 2 \mu\text{m}$) was measured using the counter on OCWA's filter 3 in March and the lowest (3412 counts/ $\mu\text{g} > 2 \mu\text{m}$) was measured using the counter on OCWA's filter 2 in June. The standard deviations for the counters on filters 1 and 2 and the University counter were in the 150 to 350 counts/ $\mu\text{g} > 2 \mu\text{m}$ range (COV = 5 to 10 %). On both dates the counter on filter 3 gave much higher standard deviations, 1275 counts/ $\mu\text{g} > 2 \mu\text{m}$ in March and 2599 counts/ $\mu\text{g} > 2 \mu\text{m}$ in June (COV = 36 to 74 %). The results in Table 5.15 are generally consistent with the concentrations and standard deviations measured in the laboratory. The exception was the counter on filter 3 whose standard deviations were much higher than those seen in the laboratory.

Table 5.15
Results of the field test of the count performance evaluation protocol

Counter	Date of the Performance Evaluation			
	3/29/00		6/15/00	
	Mean	Standard Dev.	Mean	Standard Dev.
OCWA Filter 1	4194	152	3818	208
OCWA Filter 2	3552	348	3412	105
OCWA Filter 3	4432	1275	3663	2599
University unit	3568	275	3519	159

The counter on filter 3 was the only one at the OCWA plant whose last date of calibration was not indicated on the instrument; discussions with the chemist revealed that the counter had not been recalibrated within the year. The filter 1 counter had been re-calibrated on 10-99, the counter on filter 2 on 2-00 and the university counter on 12-99. It is not known if calibration affects instrument precision but these results suggest that the relationship should be investigated.

The March and June measurements for all four counters are shown as whisker graphs in Figure 5.9. The high standard deviations associated with the counter of filter 3 are obvious. The results for the counter on filter 1 may have been slightly higher than those of the other instruments, the counter of filter 3 included. The ANOVA statistical procedure, which follows, will show if this is the case.

Testing for Homogeneity of Variance

In preparation for using the ANOVA procedure the homogeneity of variance was tested using Levene's procedure. The results are listed in Table 5.16. When the variances are unequal to a significant extent, ANOVA can still be done but the power of the method to detect differences among the means is reduced; the risk of missing significant differences in the data increases. According to the results in Table 5.16, the p-level = 0.0019 is less than 0.05, the level of significance, and therefore the field test data variances are not homogeneous at the 95% confidence level. The ANOVA was done with the knowledge that this inhomogeneity is present and care is necessary in interpreting the results.

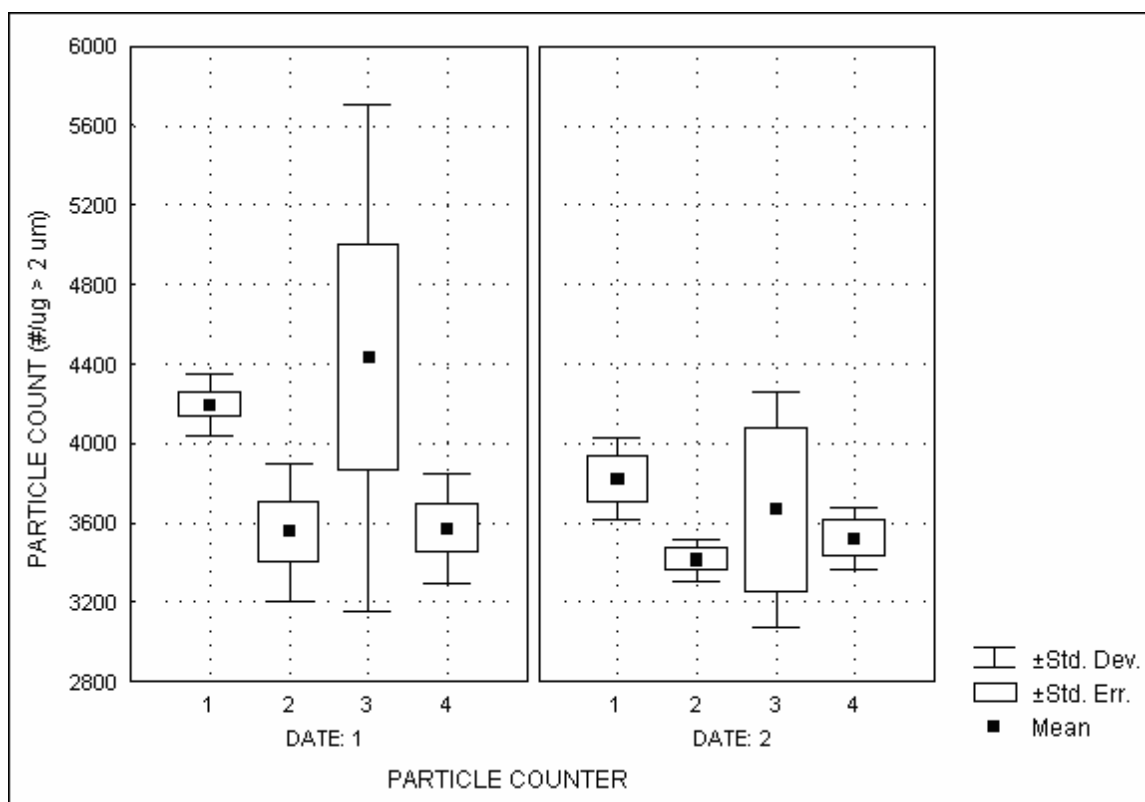


Figure 5.9 Whisker plot of the results from the protocol test at the OCWA water treatment plant. The results from date:1 are on the left and the results from date:2 are on the right. Particle counter 4 is the university unit.

Table 5.16
ANOVA on absolute within-cell deviation scores for Levene's test
for homogeneity of variance

Variable	MS Effect	MS Error	F	p-level
Count	411,114	85,471	4.81	0.0019

ANOVA Results

The ANOVA results are listed in Table 5.17. The results do not show a statistically significant differences among counters or between test dates at a 95 % confidence level, $p = 0.05$. The analysis for counters, test dates and test date combined with the counter all produce p-levels greater than 0.05, thereby indicating that any differences are likely due to chance.

A post-hoc analysis was performed using the least significant difference method to determine which pairs of means, if any, show statistically significant differences.

The post-hoc analysis using the least significant difference method yielded the results in Table 5.18. The results identify the counter of filter 3 as the only one that showed significant disagreement with other counters. By reading either across the row for Group {5} (i.e., counter 3, test date 1) or down the column for Group {5} we see that the probabilities (p-levels) are less than 0.05 for Group {5} paired with Groups {3}, {4}, {7} and {8}. For these pairings the p-level is less than 0.05, ranging from 0.026 to 0.044. The counter of filter 3 did not give results on either date that were significantly different than those of the counter of filter 1. While the counter of filter 1 on the March test date yielded a larger average value of counts/ug $> 2\mu\text{m}$, the post-hoc analysis does not show that it differs significantly from any other instrument.

It is not known if the counter of filter 3 was performing acceptably during these tests but the results suggest that its performance should be closely monitored. It seemed to be different than the other instruments, especially in how well the replicate counts/ug $> 2\mu\text{m}$ in a given set of measurements agreed. ANOVA did not identify any specific problems but it did help focus where attention should be directed. It is possible that the counter of filter 3 needs to be re-calibrated.

Table 5.17
Results of F-test for the ANOVA - field test of the CPE protocol

Effect	df Effect	MS Effect	df Error	MS Error	F	p-level
Counter	3	618,024	23	342,916	1.80	0.175
Date	1	774,109	23	342,916	2.26	0.147
Counter and Date	3	163,081	23	342,916	0.476	0.702

Table 5.18
Probabilities for LSD test - Post hoc analysis

Counter	Date	Group	{1}	{2}	{3}	{4}	{5}	{6}	{7}	{8}
1	1	{1}		0.389	0.097	0.081	0.527	0.290	0.105	0.128
1	2	{2}	0.389		0.541	0.405	0.165	0.775	0.565	0.538
2	1	{3}	0.097	0.541		0.746	0.026	0.823	0.966	0.938
2	2	{4}	0.081	0.405	0.746		0.026	0.643	0.718	0.825
3	1	{5}	0.527	0.165	0.026	0.026		0.131	0.029	0.044
3	2	{6}	0.290	0.775	0.823	0.643	0.131		0.848	0.789
4	1	{7}	0.105	0.565	0.966	0.718	0.029	0.848		0.909
4	1	{8}	0.128	0.538	0.938	0.825	0.044	0.789	0.909	

TESTING INTER-INSTRUMENT AGREEMENT USING FILTERED WATER

An experiment was performed to verify the measurements made in the field test of the count performance evaluation protocol. The on-line counter at the university that had been used to develop the protocol and in the experiments that included the counters at the OCWA treatment plant was designated the reference counter. Samples of filtered water that had been collected just after they passed through the counters at the plant were transported to the University and the particle count was measured on the reference counter. The samples were collected in 2L HDPE bottles and approximately 90 to 120 minutes elapsed between collection and analysis of the samples at the University. The samples were pumped through the University counter with the gear pump. The objective was to determine if relative differences in instrument count performance observed with the NIST MTD measurements in the protocol test would be evident when filtered water was measured on the field and University instruments.

The measured counts are listed in Table 5.19 for the counters of filters 1 and 2 (recorded at the plant) and by the on-line counter at the University using the filtered water samples. In all

cases from 2 to 5 times more particles were measured at the University than at the plant. In most cases the samples that corresponded to the lowest plant readings gave the lowest readings when they were analyzed at the university. Samples of low particle water that were transported to and from the treatment plant in sample containers did not show a significant increase in particle count when they were analyzed at the University.

Table 5.19
Particle counter comparison using filtered water samples – filtered water samples were transported to the University counter in plastic containers

		Measured Particle Count (#/mL > 2 µm)	
Location	Date	Treatment Plant	University Lab
OCWA Filter 1	4/18/00	100	196
	4/27/00	56	124
	5/2/00	27	132
	5/22/00	35	85
OCWA Filter 2	4/18/00	67	148
	5/2/00	22	147
	5/22/00	114	610

These measurements cannot be used to test the results of the count performance evaluation protocol. It is not known exactly why, in all these cases, the particle count increased when the samples were transported to the university. It could be that the plastic containers shed particles or that particles larger than 2 µm were formed by aggregation of sub-micron particles or by precipitation. Bubbles forming as the solutions increased in temperature may be part of the problem but this does not seem as likely as the aggregation and post-precipitation hypotheses. In any case, it seems that the use of filtered water samples for comparing particle counter performance is problematic. This is an area that needs additional study.

CHAPTER 6

CONCLUSIONS

The first part of this study analyzed the performance of a selection of instruments to determine the characteristics a test suspension should have to be effective for count performance evaluation. The second part developed and tested the count performance evaluation methodology using the selected test suspension.

In the instrument performance analysis experiments with polystyrene latex (PSL) micro-spheres the lowest count efficiencies (less than 77 %) were measured with the smallest mean diameter particles, 3 μm . Instruments A and B, with relatively low resolution ($R > 10\%$), had higher average count efficiencies (88 – 108 %) and instruments C and D, with relatively high resolution ($R < 10\%$), had lower average count efficiencies (66 – 76 %). Model system calculations with Gaussian particle size distributions, which took into account the sensor resolution and an error of 10% associated with the threshold setting, predicted PSL count efficiencies of $100 \pm 2\%$.

The count performance was also evaluated using NIST ISO medium test dust, a reference material from the National Institutes of Standards and Technology (NIST) in Gaithersburg, MD. At a threshold setting of 2 μm (based on size calibration with PSL micro-spheres) all the counters gave measured count efficiencies of less than 50%. With counters A, B, and C, the highest values of the count efficiency were obtained at the lowest concentrations of dust, the count efficiency decreased as the dust concentration increased. The decreasing trend was statistically significant for counter A with count efficiencies of 50%, 45%, 38%, 31% and 21% at dust concentrations of 12, 39, 117, 441, and 1166 $\mu\text{g/L}$, respectively. Model system calculations for a poly-disperse suspension (with a size distribution that resembles the NIST test dust) considered the sensor resolution and an error of 10% associated with the threshold setting and predicted count efficiencies of $100 \pm 20\%$.

The discrepancies between the measured and calculated count efficiencies for both the PSL and NIST ISO medium test dust suspensions suggest that the low measured count efficiencies were caused in part by the instruments not detecting all the particles that pass through the sensor. This result is important because it means that count calibration is not a feasible alternative and it confirms the need for a count performance evaluation protocol.

The results of the instrument performance analysis also show that count performance should be evaluated with a well-characterized poly-disperse suspension that has a particle size distribution that is similar to the size distribution of the particles that will be measured in the application of the particle counter. A mono-disperse suspension such as PSL micro-spheres is not appropriate for this purpose. According to the model system calculations for mono-disperse suspensions if the threshold is set well below the mean particle diameter, almost all the particles will be counted, and the effect of sensor resolution and errors associated with the threshold setting on the count efficiency will not be evident. On the other hand, for the appropriate poly-disperse test suspension, all the factors including sensor resolution, error associated with the threshold setting and the instrument's particle detection efficiency will potentially have an effect on the measured count efficiency. The relative significance of each source of error will be similar to what it would be with the water treatment plant suspension.

According to the model system calculations NIST's ISO medium test dust is an effective count performance evaluation suspension for particle counters used to monitor filtered water quality. The size distribution of this test dust resembles the size distribution of particles in a selection of filtered water samples.

The count performance evaluation (CPE) protocol developed in this study includes four essential parts; preparing an initial stock suspension of NIST ISO medium test dust, diluting this suspension to make working suspensions that have a dust concentration that is appropriate for counter evaluation, feeding the working suspension to the counter and collecting and analyzing the data. The statistical analysis of the data (by analysis of variance, ANOVA) gives information about inter-instrument agreement and trends in agreement with time.

APPENDIX A

EXAMPLE DATA SHEETS – INSTRUMENT PERFORMANCE ANALYSIS EXPERIMENTS WITH PSL SUSPENSIONS

Counter A

Date : 05/20/98

0.3 ml of 7 micron nominal diameter "certified" PSL sample in 50 liters of RO water

Filename : trac0520

Time	Counts/ mL between threshold settings (microns)							2-12
	2-4	4-6	6-7	7-8	8-9	9-10	10-12	
14:38	5	202	321	537	57	281	190	1593
14:39	4	192	327	539	53	285	187	1587
14:40	4	189	314	547	57	291	181	1582
14:41	4	186	305	558	58	292	189	1590
14:42	4	195	302	537	54	290	204	1586
14:43	5	191	285	550	58	297	194	1581
14:44	5	181	295	539	54	307	189	1570
14:45	5	181	290	543	59	298	201	1577
14:46	5	189	290	533	56	301	196	1570
14:47	5	181	284	536	57	303	193	1559
14:48	6	189	282	538	56	294	198	1562
14:49	5	180	278	529	57	300	193	1542
14:50	4	181	278	543	55	298	196	1555
14:51	5	177	282	537	59	302	196	1557
14:52	5	176	283	533	59	299	189	1543
14:53	4	173	284	526	57	300	200	1545
14:54	5	178	277	530	59	293	184	1525
14:55	4	180	274	542	58	296	198	1551
14:56	4	177	275	533	53	285	200	1528
14:57	4	174	285	518	57	294	196	1528
14:58	6	182	272	530	55	285	194	1523
14:59	4	170	279	527	56	284	191	1511
15:00	6	186	276	524	58	295	190	1535
15:01	6	176	277	522	58	292	188	1519
15:02	5	179	273	514	54	296	193	1514
15:03	5	179	266	528	54	286	186	1503
15:04	5	174	272	512	54	283	193	1492
15:05	4	176	283	520	55	284	191	1512
15:06	5	176	271	523	57	282	182	1495
Average =	5	182	286	533	56	293	192	1546

Counter B

Date : 05/20/98

0.3 ml of 7 micron nominal diameter "certified" PSL sample in 50 liters of RO water

File name : hach0520

Time	Counts/ mL between threshold settings (microns)													
	2.0	3.0	4.0	5.0	6.0	6.5	7.0	7.5	8.0	8.5	9.0	10.0	12.0	15.0
14:42	96	62	55	94	161	188	141	128	113	97	106	65	18	2
14:43	97	61	57	95	164	195	151	128	120	96	111	66	17	2
14:44	99	61	58	95	167	194	154	129	118	99	110	69	18	3
14:45	94	58	55	94	158	193	151	128	121	100	113	66	18	2
14:46	95	62	58	94	163	194	150	127	123	101	110	67	17	2
14:47	99	58	59	104	166	193	151	131	121	97	106	66	17	2
14:48	95	61	57	98	163	193	146	131	117	97	111	66	17	2
14:49	95	61	59	95	161	195	145	128	122	101	114	69	17	2
14:50	96	61	57	98	162	190	149	121	114	93	112	69	16	2
14:51	93	61	55	101	168	189	144	127	114	96	107	67	17	2
14:52	94	63	59	98	164	194	143	128	113	90	111	69	18	2
14:53	94	61	58	92	158	188	147	126	117	94	115	67	18	2
14:54	93	58	57	98	162	193	143	127	117	97	106	64	16	3
14:55	96	61	58	99	158	193	149	129	116	96	107	63	17	2
14:56	92	64	56	91	162	190	146	130	114	96	113	67	20	2
14:57	95	59	58	94	167	192	148	129	118	94	110	61	15	2
14:58	94	63	56	95	164	187	144	125	116	96	104	64	16	2
14:59	94	59	58	93	161	191	145	125	111	94	104	65	15	2
15:00	93	56	54	91	159	190	144	124	120	96	110	63	17	2
15:01	94	58	53	95	163	188	140	127	111	93	107	64	18	2
15:02	93	59	58	102	163	189	142	126	116	91	103	60	18	2
15:03	88	63	56	94	166	187	145	120	117	93	107	64	15	2
15:04	91	60	57	90	157	188	147	124	113	91	110	64	16	2
15:05	89	57	57	98	164	187	144	122	114	91	103	59	16	2
15:06	92	58	52	99	167	180	143	121	112	90	98	57	15	2
15:07	90	57	53	98	164	192	145	122	114	87	103	58	15	2
15:08	92	57	54	87	157	184	143	127	110	90	107	66	16	2
15:09	88	54	53	89	158	184	143	123	111	93	102	64	16	2
15:10	87	56	51	93	160	184	143	120	109	89	102	60	15	2
15:11	93	56	50	91	160	181	140	123	110	89	96	58	17	1
15:12	88	56	54	93	160	181	137	120	106	88	102	57	15	2
15:13	85	54	53	85	154	179	133	120	109	85	97	58	15	4
15:14	86	54	50	83	148	180	129	116	106	92	99	59	17	8
Average =	93	59	56	94	162	189	144	125	115	94	107	64	17	2

Counter C

Date : 05/20/98

0.3 ml of 7 micron nominal diameter "certified" PSL sample in 50 liters of RO water

File name : ibr0520

Time	Counts/ mL > threshold settings (microns)								Flow (mL/min)
	2	4	6	7	8	9	10	12	
14:44	1174	1165	884	582	225	117	28	7	60
14:44	1174	1165	884	582	225	117	28	7	60
14:46	1183	1173	896	594	227	117	29	8	60
14:47	1183	1174	897	591	229	119	29	9	60
14:47	1191	1181	903	595	230	118	30	8	60
14:47	1191	1181	903	595	230	118	30	8	60
14:49	1190	1180	901	597	228	117	28	8	60
14:50	1182	1173	898	589	227	119	30	8	60
14:51	1181	1172	895	589	228	118	29	8	60
14:51	1190	1181	902	596	231	120	30	8	60
14:51	1190	1181	902	596	231	120	30	8	60
14:53	1185	1177	900	591	225	115	29	8	59
14:54	1178	1169	900	596	232	119	30	8	59
14:54	1175	1167	894	596	231	117	29	8	59
14:56	1186	1177	905	602	234	121	30	9	59
14:57	1179	1170	902	601	233	119	30	8	59
14:58	1168	1159	895	597	230	119	29	7	59
14:58	1168	1159	895	597	230	119	29	7	59
14:59	1181	1172	905	602	232	119	29	8	59
15:00	1178	1169	905	603	235	121	31	9	58
15:01	1167	1159	898	597	229	117	30	8	58
15:01	1162	1154	896	599	233	120	31	8	58
15:01	1162	1154	896	599	233	120	31	8	58
15:03	1168	1160	901	601	231	119	30	8	58
15:04	1160	1151	898	598	230	120	30	8	58
15:04	1164	1154	900	603	232	120	31	8	58
15:04	1164	1154	900	603	232	120	31	8	58
15:06	1149	1141	885	593	227	117	30	8	57
15:07	1155	1146	896	602	232	118	29	8	57
15:08	1149	1141	894	599	230	119	29	7	57
15:08	1151	1143	898	601	228	114	28	7	57
15:08	1151	1143	898	601	228	114	28	7	57
15:10	1136	1128	889	597	231	117	30	7	57
15:11	1137	1128	891	602	233	119	29	7	56
15:11	1135	1126	888	599	230	117	29	8	56
15:11	1135	1126	888	599	230	117	29	8	56
Average =	1169	1160	897	597	230	118	29	8	58

Counter D

Date : 05/20/98

0.3 ml of 7 micron nominal diameter "certified" PSL sample in 50 liters of RO water

File name : met0520

Time	Counts/ mL between threshold settings (microns)												
	2-3	3-4	4-5	5-6	6-6.5	6.5-7	7-7.5	7.5-8	8-8.5	8.5-9	9-10	10-12	12-15
14:37	1	10	9	14	118	172	173	151	142	153	196	49	7
14:39	0	5	4	13	119	173	174	152	145	154	200	50	7
14:41	0	4	3	12	115	169	172	151	143	154	198	50	6
14:43	0	3	4	12	116	171	176	149	142	151	197	49	6
14:45	0	4	4	11	113	172	176	151	142	154	199	50	7
14:47	0	4	4	12	111	169	175	150	142	151	196	51	6
14:49	1	5	4	11	111	168	173	146	141	149	190	48	7
14:51	0	6	5	12	110	169	173	148	140	149	194	50	6
14:53	1	6	6	12	109	171	172	149	142	148	193	47	6
14:55	0	7	6	11	107	171	173	148	137	148	193	47	6
14:57	1	7	7	12	107	168	170	145	135	147	189	48	6
14:59	1	7	7	11	105	166	171	146	136	145	187	47	6
15:01	1	8	7	12	104	167	170	145	135	144	185	45	6
15:03	1	8	7	10	103	165	170	146	134	144	185	46	6
15:05	1	8	8	10	98	165	169	145	134	141	181	45	6
15:07	1	8	8	10	101	163	166	142	133	139	181	43	5
15:09	1	9	8	11	98	163	165	142	131	138	175	43	6
15:11	1	10	8	9	96	157	166	139	128	133	171	43	7
15:13	1	10	8	10	94	155	158	135	126	132	170	44	8
15:15	1	8	7	8	67	140	147	123	113	120	163	45	8
Average =	1	7	6	11	105	166	169	145	136	145	187	47	7

APPENDIX A (CONTINUED)

TYPICAL DATA SHEET FROM INSTRUMENT PERFORMANCE ANALYSIS EXPERIMENTS WITH NIST ISO MEDIUM TEST DUST

Counter B Results

Date : 06/21/99

RO background counts

Time	DC Light	Counts (#/ml) between threshold settings (mm)													
		2-2.5	2.5-3	3-3.5	3.5-4	4-5	5-6	6-7	7-8	8-9	9-10	10-12	12-15	15-20	>20
14:32	4.47	3	2	1	1	1	2	2	2	2	1	1	0	1	0
14:33	4.47	3	2	1	1	1	1	1	1	1	1	1	0	0	0
14:34	4.47	1	1	0	0	1	1	1	1	0	1	1	0	0	0
14:35	4.51	1	1	0	0	0	0	0	0	0	0	0	0	0	0
Average =	4.481	2	1	1	1	1	1	1	1	1	1	1	0	0	0

Dust counts

5.3 mL of dust stock suspension in 40L of RO water

(NIST Dust stock solution no. E : 333mg of NIST dust in 100mL RO water)

Time	DC Light	Counts (#/ml) between threshold settings (mm)													
		2-2.5	2.5-3	3-3.5	3.5-4	4-5	5-6	6-7	7-8	8-9	9-10	10-12	12-15	15-20	>20
15:40	4.51	281	175	129	97	145	103	68	37	24	15	16	9	5	2
15:41	4.51	282	176	134	100	151	104	70	41	27	17	16	8	5	1
15:42	4.51	286	175	131	101	149	101	70	39	26	16	15	10	4	1
15:43	4.51	289	176	133	102	151	102	67	38	26	16	16	9	4	2
15:44	4.51	289	178	131	99	148	101	66	40	25	15	16	9	4	1
15:45	4.51	296	179	134	101	151	101	67	37	25	15	16	9	4	1
15:46	4.51	291	179	127	98	149	100	68	37	26	14	15	8	4	1
15:47	4.51	291	177	130	98	150	95	65	38	25	16	14	8	3	1
15:48	4.47	286	174	131	101	149	96	68	35	23	14	15	7	3	1
15:49	4.47	291	178	132	98	145	95	66	36	21	14	14	8	3	1
15:50	4.47	281	175	130	97	145	95	63	36	24	13	14	7	3	1
15:51	4.47	283	175	127	93	142	97	62	36	24	14	14	7	2	1
15:52	4.47	277	178	128	97	144	92	61	36	23	15	14	6	4	1
15:53	4.47	279	168	126	93	138	94	62	34	25	12	14	8	3	1
15:54	4.51	270	168	127	92	138	93	63	37	25	14	15	8	4	1
15:55	4.51	273	170	128	92	140	94	61	38	25	16	14	8	3	1
15:56	4.51	279	165	124	97	148	90	63	38	24	14	15	7	3	1
15:57	4.51	270	168	124	95	144	99	62	38	23	14	13	7	2	1
15:58	4.51	274	167	121	96	142	94	64	38	23	14	16	7	2	1
15:59	4.51	276	169	125	95	140	95	66	37	24	14	15	7	2	0
Average =	4.50	282	174	129	97	145	97	65	37	24	15	15	8	3	1
Corrected		280	172	128	97	144	96	64	36	24	14	14	8	3	1

APPENDIX B

PLOTTING “Z” CURVES - PART OF THE METHOD USED TO DETERMINE THE MEASURED MEAN PARTICLE DIAMETER AND THE INSTRUMENT RESOLUTION

INTRODUCTION

The count concentration versus size data measured with each counter was used to calculate “f” values and these fractions were then used to determine z-curves. Z curves were used to determine the measured mean particle diameter and the instrument resolution.

THE “F” VALUES

The particle count greater than each threshold setting was determined using the count versus size data for each counter. The fraction of the total counts (f) greater than each threshold setting was determined using the count - threshold data. The denominator in this fraction was always the count greater than 2 µm because 2 µm was the lowest threshold setting available for each of the four counters used in the experiments.

Table B.1 lists measured particle count versus the threshold setting data and the corresponding “f” values for Counter B. This experiment was conducted on 6/11/98 using the 7 µm nominal diameter “certified” PSL suspension. Each value of f was determined by dividing each of the particle counts by 1698, the measured count for a threshold setting of 2 µm.

“Z-CURVES”

The “f” values were used to calculate values of the standard normal deviate (z) and these were plotted versus the threshold setting. The magnitude of z is determined by
$$z = \frac{x - \bar{x}}{\sigma}$$

where $\bar{x}$ is the mean value of the independent variable and σ is its standard deviation. Each value of “f” was converted to a value of “z” using the Excel spreadsheet function $z = @normsinv(f)$. For example, from Table B.1, the fraction of particles greater than 5 µm is 0.83, and f, therefore, is equal to 0.83, and the magnitude of z is $0.96 = @normsinv(0.83)$. If it is

assumed that the measured size distribution follows a normal (Gaussian) distribution, $z = 0.96$ tells us that the difference between the particle mean diameter and $5\ \mu\text{m}$ is equal to 0.96 times the standard deviation of the particle size distribution. The z-curve for counter B with the $7\ \mu\text{m}$ nominal diameter PSL particles from the experiment conducted on 6/11/98 is presented in Figure B.1.

Table B.1
"f" values

Threshold setting (μm)	Counts/mL > threshold setting	f
2.0	1698	1.00
4.0	1492	0.88
5.0	1411	0.83
5.5	1352	0.80
6.0	1179	0.69
6.5	644	0.38
7.0	199	0.12
7.5	108	0.06
8.0	76	0.04
9.0	40	0.02
10.0	19	0.01
15.0	0	0.00

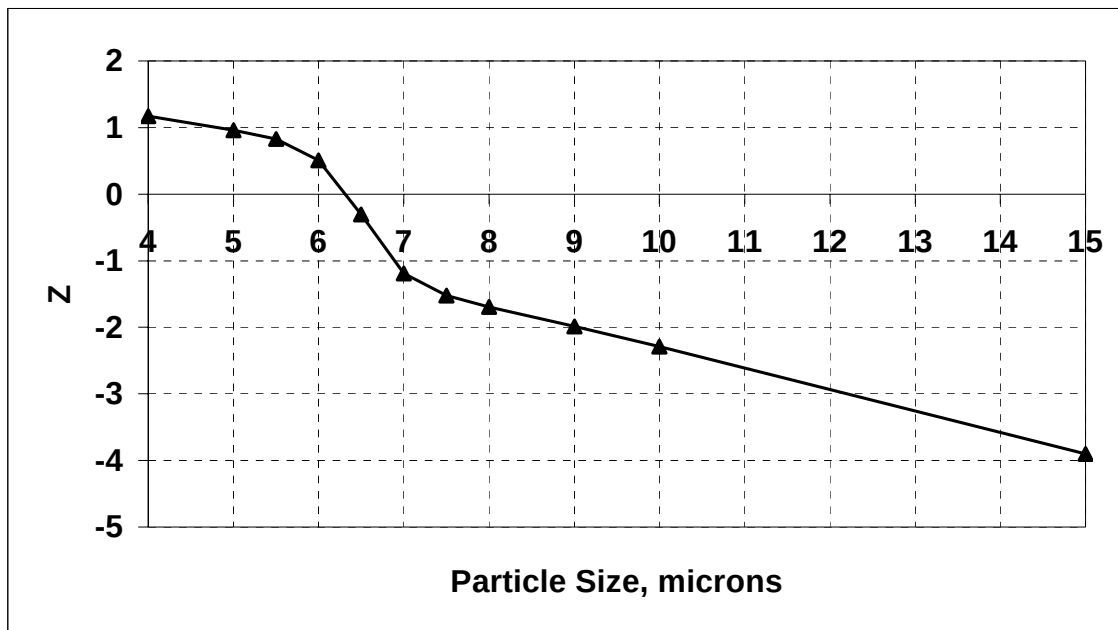


Figure B.1 A typical "z curve". This example is from an experiment conducted on 6/11/98 using the 7 μm nominal "certified" PSL particles with Counter B.

The points plotted in Figure B.1 show that the assumption of a Gaussian particle size distribution in this case is definitely approximate. For a Gaussian distribution all the points should follow a single linear trendline. There are segments of this plot that are essentially linear but the different segments have somewhat different slopes.

The z-curves were used to determine the mean measured PSL particle diameter for each experiment. The measured mean is located at $z = 0$. From Figure B.2, the particle size at $z = 0$ is 6.3 μm . This was the mean diameter measured by counter B for the experiment conducted on 6/11/98.

The standard deviation of the measured particle size distribution is the difference between the diameter at $z = 0$ and the diameter at $z = 1$. In this example the standard deviation on the right is equal to $6.3 - 4.4 \mu\text{m} = 1.9 \mu\text{m}$ and the standard deviation on the left is $6.9 - 6.3 \mu\text{m} = 0.6 \mu\text{m}$. The left and right standard deviations are used to calculate the resolution (the R-values) on the left and right.

APPENDIX C

SIZE CALIBRATION VERIFICATION

INTRODUCTION

Size calibration verification involved comparing the measured mean diameters of the PSL particles used in the instrument performance analysis experiments with the manufacturer's "certified" diameters.

MEASURED MEAN DIAMETER

The z curves discussed in Appendix B were used to determine the mean particle diameter for each suspension. The mean diameters measured in June and August are listed in Table C.1 below. The experiments with 4.991 and 6.992 μm particles were done twice in August and June.

COMPARISON OF MEASURED AND "CERTIFIED" DIAMETERS

The measured and "certified" diameters were compared statistically using the "t" test for the difference between two sample means. The level of significance used in the test was 0.01, (i.e., $p < 0.01$). In this analysis the differences were always expressed as a percent of the certified mean diameter. For example for Counter C and the 3 μm nominal diameter particle, the percent difference between the June mean measured diameter and the certified mean diameter is $[(3.063 - 2.8) \times 100] / 3.063 = 8.6\%$. For all the tests the greatest negative difference was -8.3% for Counter A, in August with the 10 μm particle and the greatest positive difference was $+9.9\%$ for Counter C in June with the 7 μm particle. The average percent difference between the measured and certified mean diameters (the average is for the seven experiments in each set) varied from $+0.24\%$ for Counter A in August to $+9.3\%$ for Counter C in June (see Table C.2 below). The following hypotheses were tested.

Hypotheses tested:

- A: The mean PSL diameter measured in June is significantly different than the certified mean particle diameter of Table C.1 ($p < 0.01$). (The difference is expressed as a percent of the certified mean diameter).
- B: The mean PSL diameter measured in August is significantly different than the certified mean particle diameter ($p < 0.01$). (The difference is expressed as a percent of the certified mean diameter).

Table C.1
Measured diameters

Certified diameter (μm)	Measured diameters (μm)							
	Counter A		Counter B		Counter C		Counter D	
	August	June	August	June	August	June	August	June
3.063	3.1	3.4	2.7	2.6	2.7	2.8	2.9	2.8
4.991	5.1	5.2	4.9	5	4.5	5.4	4.8	4.8
4.991	5.1	4.5	4.9	5	4.5	4.4	4.8	4.7
6.992	6.5	6	6.1	6.4	6.2	6.3	6.6	6.5
6.992	6.5	6.8	6.1	6.3	6.2	6.3	6.6	6.5
9.975	10.8	9.7	9.4	9.5	9.4	9.7	9.7	9.6
15.02	14.8	14.8	14	14.3	14.2	14.4	14.7	14.4

Table C.2

COUNTER A

Certified diameter (xC)	Measured diameter		% difference for hypothesis A (xC -xA)100 / xC	% difference for hypothesis B (xC -xJ)100 / xC
	August (xA)	June (xJ)		
3.063	3.1	3.3	-1.21	-7.74
4.991	5.1	5.2	-2.18	-4.19
4.991	5.1	4.5	-2.18	9.84
6.992	6.5	6.0	7.04	14.19
6.992	6.5	6.8	7.04	2.75
9.975	10.8	9.7	-8.27	2.76
15.02	14.8	14.8	1.47	1.47
Average % difference			0.24	2.26
Std. Dev. % difference			5.48	8.35
Calculated t			0.11	0.66

From t statistical tables, at level of significance = 0.01 , and 6 degrees of freedom, $T < -3.707$ and $T > 3.707$

Outcome : The % difference between the certified diameter and August diameter is not significant
The % difference between the certified diameter and June diameter is not significant

Testing of hypotheses A and B

COUNTER B

Certified diameter (μm) (xC)	Measured diameter (μm)		% difference for hypothesis A (xC -xA)100 / xC	% difference for hypothesis B (xC -xJ)100 / xC
	August (xA)	June (xJ)		
3.063	2.8	2.8	8.59	9.57
4.991	4.9	5.0	1.82	-0.18
4.991	4.9	5.0	1.82	-0.18
6.992	6.4	6.3	8.47	9.90
9.975	9.4	9.5	5.76	4.76
15.02	14.0	14.3	6.79	4.79
Average % difference			7.65	6.10
Std. Dev. % difference			4.87	5.53
Calculated t			3.85	2.70

From t statistical tables, at level of significance = 0.01 , and 6 degrees of freedom, $T < -3.707$ and $T > 3.707$

Outcome : The % difference between the certified diameter and August diameter is significant
The % difference between the certified diameter and June diameter is not significant

(continued)

COUNTER C

Certified diameter (μm) (xC)	Measured diameter (μm)		% difference for hypothesis A (xC -xA)100 / xC	% difference for hypothesis B (xC -xJ)100 / xC
	August (xA)	June (xJ)		
3.063	2.80	2.80	8.59	8.59
4.991	4.50	5.40	9.84	-8.20
4.991	4.50	4.50	9.84	9.84
6.992	6.30	6.30	9.90	9.90
9.975	9.40	9.70	5.76	2.76
15.02	14.20	14.40	5.46	4.13
Average % difference			9.34	5.56
Std. Dev. % difference			2.66	6.90
Calculated t			8.60	1.97

From t statistical tables, at level of significance = 0.01, and 6 degrees of freedom, $T < -3.707$ and $T > 3.707$

Outcome : The % difference between the certified diameter and August diameter is significant
 The % difference between the certified diameter and June diameter is not significant

COUNTER D

Certified diameter (μm) (xC)	Measured diameter (μm)		% difference for hypothesis A (xC -xA)100 / xC	% difference for hypothesis B (xC -xJ)100 / xC
	August (xA)	June (xJ)		
3.063	2.9	2.8	5.32	8.59
4.991	4.8	4.8	3.83	3.83
4.991	4.8	4.7	3.83	5.83
6.992	6.6	6.5	5.61	7.04
9.975	9.7	9.6	2.76	3.76
15.02	14.7	14.4	2.13	4.13
Average % difference			4.15	5.74
Std. Dev. % difference			1.37	1.99
Calculated t			7.24	7.41

From t statistical tables, at level of significance = 0.01, and 6 degrees of freedom, $T < -3.707$ and $T > 3.707$

Outcome : The % difference between the certified diameter and August diameter is significant

Table C.2 (continued)

According to Table C.2, the mean diameters measured by Counter A in June and August were not different than the certified mean diameters by a statistically significant amount. For Counter D, these differences were significant.

The percent differences (between the measured and certified mean diameters and the difference between the June and August mean diameters) that were calculated for Counter A were as high as those calculated for other counters. However, in the case of Counter A, there were several particle sizes for which the measured mean diameters were larger than the certified diameters. (For most of the other counters the measured diameters were consistently smaller than the certified diameters). Having both positive and negative differences in the statistical analysis for Counter A tended to reduce the overall percent difference and favored the conclusion that the overall difference was not statistically significant.

For Counter D, the average percent difference between the mean diameters measured in June and August and the corresponding certified mean diameters was between 5 and 10%. Essentially the same range of percent difference values was obtained with counters, B and C. However, for Counter D the percent differences were more consistent among the different size particles tested and this made the standard deviation low. With a low standard deviation the “t” test indicated that the average percent difference for this counter was significantly different than zero.

APPENDIX D STABILITY OF SIZE CALIBRATION WITH TIME

MEAN MEASURED DIAMETERS

The average particle diameter for each “research-grade” 8.1 μm PSL suspension was determined using “z-curves” as described in Appendix B. The measured mean is located at $z = 0$. Table D.1 shows the measured mean diameters for the experiments conducted eleven times over a period of more than a year.

STATISTICAL ANALYSIS OF MEAN MEASURED DIAMETERS – TREND OVER TIME

The measured mean diameters were analyzed to determine if they followed a significant increasing or decreasing trend with time. The hypothesis tested and the results obtained are listed in Table D.2. Table D.3 summarizes the results of the analysis. The results show that the measured mean diameters did not follow a statistically significant trend with time. Therefore, it was concluded that the size calibration of each instrument was reasonably stable with time.

Table D.1

Date	Nominal diameter (microns)	Measured particle diameter (microns)			
		Counter A	Counter B	Counter C	Counter D
5/20/1998	8.1	7.60	7.00	7.00	7.90
6/8/1998	8.1	7.70	7.00	7.20	7.50
1/25/1999	8.1	7.50	7.00	7.10	7.90
2/15/1999	8.1	7.50	6.95	7.20	7.80
3/8/1999	8.1	9.30	7.40	8.20	8.00
3/16/1999	8.1	7.50	6.70	7.40	7.80
3/16/1999	8.1	7.50	6.70	7.40	8.00
3/17/1999	8.1	7.50	6.80	7.30	7.80
4/13/1999	8.1	7.50	6.60	7.40	7.80
5/13/1999	8.1	7.50	6.95	7.40	7.80
6/1/1999	8.1	7.50	6.95	7.20	7.80

Measured mean diameters for “research-grade” PSL suspensions

Table D.2
Results of statistical analysis
Hypothesis tested: The measured mean “research-grade” PSL diameters do not follow a significant trend with time.

Counter	95% confidence interval for slope of trend line		Result
	Lower 95%	Upper 95%	
A	-0.003	0.003	Confidence interval includes zero slope. So, trend with time is not significantly different from zero.
B	-0.002	0.001	Confidence interval includes zero slope. So, trend with time is not significantly different from zero.
C	-0.001	0.003	Confidence interval includes zero slope. So, trend with time is not significantly different from zero.
D	-3.70E-04	0.001	Confidence interval includes zero slope. So, trend with time is not significantly different from zero.

Table D.3

Hypothesis:	The measured mean diameters do not follow a significant trend with time.
Instrument	Hypothesis
A	yes
B	yes
C	yes
D	yes

Summary of statistical analysis results

APPENDIX E

STABILITY OF PTI MEDIUM TEST DUST STOCK SUSPENSIONS WITH TIME

The results of the stability experiments with ISO medium test dust conducted over a period of two months (from 5/26/99 to 7/15/99) are shown in Table E.1. Two of the stock suspensions (B and C) were tested. Table E.1 shows the mass concentrations obtained by averaging the counts/μg from the five replicate working suspensions on each date. The standard deviations and coefficient of variations (COVs) for each set of measurements are also listed.

Table E.1

Stock suspension B: 357 mg of NIST dust in 100 mL of RO water				
Stock suspension C: 325 mg of NIST dust in 100 mL of RO water				
The average (counts/ug) for each date is based on 5 replicates.				
Date	Average			
	Stock suspension no.	Counts/ug	COV, %	Std. Dev.
5/26/1999	B	2841	27.6	784
5/26/1999	B	3755	16.2	609
5/27/1999	B	2923	13	380
5/27/1999	B	3094	12.1	374
6/3/1999	B	2672	7.2	191
6/7/1999	B	2747	9.3	256
6/8/1999	B	2929	5.5	161
	Overall average	2994		
	Std. Dev	362		
	COV	0.121		
Date	Average			
	Stock suspension no.	Counts/ug	COV, %	Std. Dev.
6/4/1999	C	3841	8.5	328
6/7/1999	C	3946	8	315
6/8/1999	C	3496	2.2	76
6/16/1999	C	3588	5.7	206
7/15/1999	C	3624	10.2	368
	Overall average	3699		
	Std. Dev	211		
	COV	0.057		

Stability with time results for NIST ISO medium test dust stock suspensions

Stock suspension B was tested seven times (7 dates) over a period of two months and stock suspension C was tested five times (5 dates). The overall average mass concentration from all the stock suspension B stability experiments was 2994/ μg with a COV of 12.1 %. The overall average mass concentration from stock suspension C stability experiments was 3699/ μg with a COV of 5.7%.

Figure E.1 shows the variation of the average mass concentrations from stock suspensions B and C with time. The error bars shown indicate the standard deviation for the five replicates on each date.

The average mass concentrations in Table E.1 for stock suspensions B and C were analyzed statistically. The hypothesis that the average mass concentrations did not follow a significant trend with time was tested (i.e. the slope of the linear regression line fitted to the average mass concentrations plotted versus time (Figure E.1) was not significantly different from zero). The confidence level used in this test was 95%. The detailed results of this analysis are presented in Table E.2.

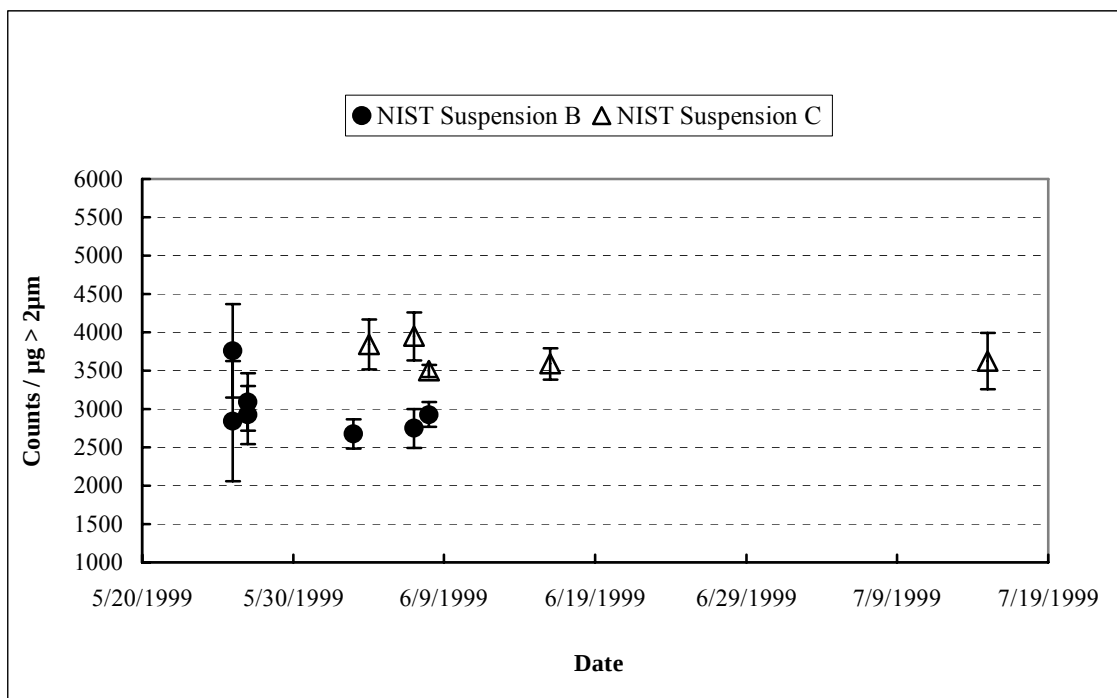


Figure E.1 Stability with time of NIST ISO medium test dust stock suspensions

Table E.2
Statistical analysis - stability of NIST ISO medium test dust stock suspension with time
Hypothesis: The counts/ug do not show a significant trend with time

Stock suspension no.	95% confidence interval for slope of trend line		Result
	Lower 95%	Upper 95%	
B	-92.92	30.46	Confidence interval includes zero slope. So, trend with time is not significantly different from zero.
C	-23.1	15.17	Confidence interval includes zero slope. So, trend with time is not significantly different from zero.

From this analysis it was concluded that since for both stock suspensions B and C, the confidence interval included a zero slope, both the stock suspensions were essentially stable with time.

APPENDIX F

MATERIALS AND PROCEDURES FOR THE PROTOCOL DEVELOPMENT EXPERIMENTS OF CHAPTER 5

PREPARING QA/QC SUSPENSIONS OF RESEARCH GRADE POLYSTYRENE LATEX MICROSPHERES

The overhead reservoir was filled with 50-L of RO water. A 10-L quantity was allowed to flush through the on-line counters. The stock suspension of research-grade 8.1 μm polystyrene latex spheres (PSL) from Duke Scientific was taken from the refrigerator, warmed to room temperature and ultra-sonnicated (in a Fischer Scientific 8 oz bath) for 30 seconds. Two 100-mL beakers were washed three times with RO water and five drops of PSL suspension were added to one beaker. The other beaker was filled with 40-mL of RO water, and 230- μL of PSL suspension from beaker 1 was pipetted using an adjustable 200- μL pipette into beaker 2. Beaker 2 was covered with parafilm and sonnicated for approximately 40 seconds.

Beaker 2 was then poured into the 40-L remaining in the overhead reservoir; the beaker was rinsed with RO water two times and the rinse water was also poured into the reservoir. The stirrer was set at 40% power, and the suspension was mixed for 15 minutes. The stopcocks to the counters were then opened and 30-L of suspension was allowed to flow through the counters. This took approximately 15 minutes. The data was logged on the PC during the period.

PREPARING STOCK SUSPENSIONS OF NIST ISO MEDIUM TEST DUST

A cellulose gelcap with a known weight of NIST ISO medium test dust was placed in a 100 mL Corning Snap-Seal polypropylene container. A 100-mL graduated cylinder was rinsed three times with RO water and filled to the 100-mL mark. The volume was poured into the container that was stored in a refrigerator overnight to allow the gelcap to dissolve and the dust to disperse.

The stock suspension was prepared for use by taking it out of the refrigerator and warming the contents to room temperature. The entire container was placed in the sonication bath for sixty seconds. Before quantities were pipetted from the stock suspension, the container was shaken vigorously three times.

GRAVIMETRIC CHECK OF THE STOCK SUSPENSION DUST CONCENTRATION

The mass concentration of dust in the stock suspension was verified by gravimetric analysis. The tare weights of at least six aluminum pans were determined using an Allied/Fischer Scientific 2100 Microbalance. Each pan was marked with a number.

The stock suspension was vigorously shaken three times and a 2-mL quantity of suspension was pipetted into each weighing pan using an adjustable 5-mL Wheaton micropipettor. The pan and suspension was quickly weighed and the weight recorded. This was repeated two more times with different pans.

The remaining three pans were prepared in the same manner but RO dilution water was used instead of suspension. A new pipette tip was used for the RO dilution water to avoid dust carryover. These pans were used as method blanks.

All six pans were placed in a desiccator until they could be transferred to the 80°C drying oven where they remained overnight. The next day the dry samples were weighed and their weights recorded. The pans with sample were returned to the oven for one hour and then weighed again. The samples were assumed to be dry when the difference between two successive weights was less than 0.001 g.

GRAVIMETRIC CHECK OF THE WHEATON MICROPIPETTORS

A gravimetric analysis was used to check the volume dispensed by the Wheaton adjustable micropipettors. In these tests 2-mL volumes of suspension were added to the aluminum weighing pans using two methods. In the first method, a 2-mL micropipettor was used to dispense one 2-mL quantity of water into each of three pans. In the second another triplicate set of pans was prepared by adding a series of ten 200 μ L volumes of suspension pipetted with a Wheaton 200 μ L adjustable micropipettor. All the other steps were the same.

WASHING THE CONTAINERS

Three types of containers were used to prepare dust working suspensions: 2-L rectangular HDPE from Nalgene, 2-L round glass from Cole Parmer, and 20-L MDPE wide mouth carboys with spigot from Cole Parmer. These containers were used repeatedly and washed between analyses.

The containers were filled with tap water and a drop of concentrated liquid dish soap and 20-mL of 0.02N NaOH were added. The containers were covered, shaken and allowed to stand for ten minutes. The water was then drained and the container was refilled with tap water, shaken and drained two times. Then, the containers were filled with RO water, capped and shaken. The RO rinse was repeated 3 more times. If any soap bubbles were apparent at this point, RO rinses were continued until no bubbles were observed.

The bottles were spot checked for background particle count using the Met One grab sampler. At least one bottle was checked for every three bottles washed. If the RO water from the last rinse, measured before it was drained from the container, gave less than 15 counts >2 μm per mL, then the washing procedure was assumed to be acceptable. Containers were stored with the cap on and filled with RO water. Before use, each container was drained of its contents.

PREPARING 2-LITER VOLUME WORKING SUSPENSIONS DIRECTLY IN THE 2-LITER CONTAINERS

In preparation for each experiment the stock suspension was taken from the refrigerator, warmed to room temperature and ultra-sonnicated. Each washed container was then filled with 2-L of RO water measured with a 1-L volumetric flask. The stock suspension was shaken vigorously 3 times and 0.2-mL was immediately pipetted from it with the adjustable 200- μL pipette. The 0.2-mL volume was dispensed into the 2-L of RO water in each container. Finally, each container was covered and inverted (not shaken) 5 times.

PREPARING 2-LITER VOLUME WORKING SUSPENSIONS USING THE 20-LITER CONTAINER

The stock suspension was taken from the refrigerator, warmed and sonnicated. Each 20-L container was filled with 20-L of RO water measured using 20 steps with a 1-L volumetric flask. The stock suspension was shaken vigorously 3 times. 2-mL of suspension was immediately pipetted from it with the adjustable 5-mL pipette and dispensed to the 20-L of RO water that had just been prepared. Because of its large volume, the diluted sample was stirred not shaken. A Cole Parmer Stir-Pak laboratory mixer with stainless steel shaft that included Cole Parmer bottom and mid-shaft turbine propellers was inserted into the sample through a round hole in the

carboy lid. The mixer motor was set at the 20% power setting and mixing continued for 20 minutes; a vortex was always present during mixing.

After 20 minutes, with the mixer still turning, the desired number of 2-L portions of suspension was dispensed into washed 2-L containers through the spigot. Each 2-L container was covered and inverted 5 times just prior to use with the gear pump.

TESTING COUNT PERFORMANCE USING THE GEAR PUMP

A Cole Parmer Pump Drive gear pump was used to draw NIST dust suspensions through the on-line counters. The pump was attached to the sensor outlet with approximately 1 meter of Tygon™ tubing and the suspension was drawn through by suction. This tended to minimize the effect of bubbles and particles from the pump and tubing on counter performance. A short length (0.5-meter) of Tygon™ tubing connected the sensor inlet to the bottle that contained the dust suspension. This tube was fitted with a quick connect on one end for attaching to the counter, the other end was open. About 2 meters of Tygon™ tubing was used to direct the suspension leaving the pump to waste.

The following procedure was used to analyze each NIST dust suspension. The 2-L container with the working suspension was inverted 5 times and the tube from the counter inlet was inserted. The gear pump was turned on and the flowrate through the sensor was set at 100 mL/min (± 2 mL/min) by adjusting the controller attached to the pump. The pump was allowed to purge the sensor and tubes of bubbles and then run an additional 60 seconds to ensure that the flow rate had become steady. A graduated cylinder and stopwatch were used to time the collection of 50 mL of suspension from the gear pump output tube in a 30-second time period. The pump was allowed 30 seconds to reach equilibrium after any power changes before the flow was re-measured.

The particle count data was collected on a desktop computer using proprietary software supplied by the instrument manufacturers. Each of the four systems was turned on prior to starting the gear pump and the software was loaded. As soon as the gear pump was set at 100 mL/min, the time was noted as the start of sampling for this 2-L of suspension. The pumping continued until at least 5 discrete values were recorded by the data collection utility for this suspension, or approximately 10 minutes. (The Met One software logged counts to memory every 2 minutes). The data collection utility logged counts according to the following particle

size thresholds: $>2\ \mu\text{m}$, $>15\ \mu\text{m}$, $2\text{-}3\ \mu\text{m}$, $3\text{-}4\ \mu\text{m}$, $4\text{-}5\ \mu\text{m}$, $5\text{-}6\ \mu\text{m}$, $6\text{-}6.5\ \mu\text{m}$, $6.5\text{-}7\ \mu\text{m}$, $7\text{-}7.5\ \mu\text{m}$, $7.5\text{-}8\ \mu\text{m}$, $8\text{-}8.5\ \mu\text{m}$, $8.5\text{-}9\ \mu\text{m}$, and $9\text{-}10\ \mu\text{m}$.

If additional samples were to be analyzed, the next sample was inverted 5 times after the data for the first sample had been recorded. The sensor inlet tube was taken from sample 1 and placed in sample 2 without turning off or adjusting the power setting on the gear pump. The gear pump plus tubing was allowed to purge for 60 seconds and the flow rate was checked and adjusted, as previously described, to a target value of 100 mL/min. The time at which sample 2 measurements started was marked as before. Subsequent samples were handled in this same way. The time that had been recorded when each sample analysis was started was used with the time stamp on the data collection utility file to locate the data that corresponded to the sample analyzed.

PREPARING THE OVERHEAD RESERVOIR FOR COUNT PERFORMANCE EXPERIMENTS

The overhead reservoir is a rectangular, covered, HDPE container fitted with a mechanical stirrer. There are two, 5-cm diameter, holes cut in the cover, one for the impellor and one to provide access for adding particle suspensions. The access hole was plugged with a rubber stopper.

With the impellor stirring at the 40% power setting, the reservoir was filled to the 40-L mark with RO water and drained. It was then rinsed one more time with 10 L of RO water and drained. Some of the 10 L of RO was collected in a washed 2-L container and checked for background counts using the Met One grab sampler. The reservoir was assumed to be acceptably clean if the counts for particles greater than $2\ \mu\text{m}$ were less than 15/mL.

To prepare the suspension the overhead reservoir was filled with 43-L of RO water, and 4.3-mL of stock suspension was added using the adjustable 5-mL capacity pipette. The suspension was stirred for 15 minutes.

After stirring, the stopcocks below the manifold were opened and the suspension was allowed to flow through the counters for 20 minutes. Data collection began as soon as the stopcocks were opened. Four, 2-L samples of the suspension in the overhead reservoir were collected in washed 2-L containers using one of the unused taps on the flow distribution manifold.

Data was logged for 20 minutes, the stopcocks were closed, and the gear pump was attached to the Met One online counter. The Met One grab sampler was used to analyze one sample, while the Met One online counter with the gear pump was used to analyze the other three.

WORKING SUSPENSION STABILITY EXPERIMENT

A series of experiments was conducted to determine the best way to prepare and store the working suspensions of NIST medium test dust prepared in the 20-L carboy. Unlike working suspensions prepared in 2-L containers, the suspensions in the 20-L carboy might not be used in one analysis. Therefore, the length of time the suspension might be stored and still produce consistent results was of interest.

Four 20-L samples were prepared. The containers were washed and filled with RO water by the standard procedure. The stock suspension was warmed to room temperature and sonicated and dispensed to the four, 20-L, containers of RO using the 5 mL adjustable pipette. Container 1 received 2.6 mL, container 2 received 1.3 mL, container 3 received 5.2 mL (added as two 2.3 mL volumes), and container 4 received 2.6 mL as a series of 20, 133 μ L volumes.

The samples were shaken 3 times, and then stirred for fifteen minutes prior to each analysis. The impellor used for these samples was a small, three-blade propeller. The power setting was 60%. Three 2-L aliquots from each 20-L suspension were collected in washed 2-L containers and measured with the Met One, Hach, and Chemtrac online counters using the gear pump procedure.

The remaining 14-L of suspension was capped and stored in the lab at room temperature for 48 hours (2 days) at which time it was shaken and stirred again. Triplicate samples were taken and analyzed. The remaining 8-L was again capped and stored for 120 hours (5 days). At the end of this period the analysis process was repeated a third time.

PROCEDURE USED TO TEST THE CPE PROTOCOL IN THE FIELD

The count performance evaluation (CPE) protocol was tested at the Onondaga County Water Authority's Otisco Lake (OL) water treatment plant in Marcellus, NY. The OL plant uses six, Met One, on-line counters, one for the raw water, one for the finished water, and four for the

filtered water. All six counters are identical to the one used in the laboratory study at the University. Five counters were used to test the CPE protocol, three of the filtered water counters at the OL plant, the Met One on-line counter at the University and the Met One grab sampler that had been used as a referee counter for most of the study.

Two methods were compared for preparing working suspensions. In the first method the working suspensions were prepared at the University (in 2, 20-L, carboys) and hauled (in about 90 minutes) to the OL plant for the field test. After the field-testing was finished, five, 2-L portions, of the working suspensions were brought back to the University for analysis with the Met One on-line and grab sample counters. Five replicate measurements were made with each counter. A variation on this method was also performed that had five replicates analyzed at the University before the remaining samples were hauled to the field.

In the second method the stock suspension and RO water were brought to the OL plant and the working suspensions were prepared (in the 20-L carboy) in the plant's laboratory just before they were used in the field test. The working suspensions tested at the University were prepared in the University laboratory (in a 20-L carboy with 2-L portions) just before they were used. Three replicate measurements were made with each counter.

In the first test method, one (out of the five) 2-L working suspension for each counter was analyzed using the Met One grab sampler. In the second test method, every 2-L working suspension was tested on-site using the Met One grab sampler.

Two working suspensions were tested; one had a low dust concentration and the other a high dust concentration. The low dust concentration gave a count/mL ($> 2 \mu\text{m}$) of about 100/mL and the high dust concentration gave about 1000/mL ($> 2 \mu\text{m}$). These concentrations bracket the expected values for the filtered water at the OL plant. The low dust concentration was prepared by adding 200 μL of stock suspension to 20 L of RO water in a 20-L carboy.

In this CPE test, all the on-line counters, the four at the OL plant and the one at the University, were operated using the gear pump technique that had been developed in the laboratory phase of the study. The Met One grab sampler was used to make measurements for testing consistency, both among the suspensions prepared for this set of experiments and with suspensions made and tested in the earlier parts of the study.

ANALYZING SAMPLES OF FILTERED WATER FROM THE OTISCO LAKE TREATMENT PLANT

In this part of the study samples of filtered water were collected at the OL treatment plant (from the filters used in the CPE test). The samples were driven to the University and analyzed on the laboratory Met One on-line counter using the gear pump technique. The objective was to compare these measurements with those recorded at the treatment plant to determine if the results obtained in the CPE test with the NIST dust suspensions help explain any discrepancies.

At the OL plant the 2-L filtered water samples were collected at the overflows from the on-line turbidimeters that were mounted next to the on-line counters. The overflow tubes on the counters could not be used because they were too short and manipulating them would likely have caused particles to be released. While the filtered water samples were being collected, the count data being logged by the OL plant computer was recorded.

APPENDIX G
MINUTES OF THE MEETING AT NIST – JANUARY 1999

To: Distribution
From: Chris Schulz
Subject: AWWARF Project 454: NIST Meeting Minutes
Date: January 15, 1999

A meeting was held at the offices of the National Institute for Standards and Technology (NIST) in Gaithersburg, MD, on January 14, 1999. The purpose of the meeting was to discuss NIST's potential involvement in developing a test suspension for the drinking water industry for the count evaluation of on-line particle counters. The following individuals attended the meeting:

- Robert Fletcher (RF), Research Physicist, Surface and Microanalysis Science Division, NIST
- David Simons (DS), Group leader, Surface and Microanalysis Science Division, NIST
- Richard Cavanagh, Division Chief, Surface and Microanalysis Science Division, NIST
- Robert Gettings (RG), Project Manager, Standards Reference Materials Program
- Chris Schulz (CS), Co-Principal Investigator, CDM
- Ray Letterman (RL), Co-Principal Investigator, Syracuse University
- Steve Via (SV), Regulatory Engineer, AWWA

The meeting was around two hours long and generally covered the issues listed in the attached agenda. Following the meeting, Chris Schulz and Ray Letterman continued discussions with Bob Fletcher and Dave Simons over lunch. We believe the meeting was a success and will move the current AWWARF research project (and potential follow-up projects) along a defined pathway towards the development of a new reference material for the evaluation of the count performance of on-line particle counters for the drinking water industry. The key discussion points from the meeting are summarized below.

These minutes were reviewed by Messrs. Letterman and Fletcher, and their comments have been incorporated into the final version.

Drinking Water Industry Needs

1. CS and RL presented an overview of the AWWARF particle counter project, which included a discussion of project objectives, project participants (including utility involvement in the project), and laboratory work completed to date by Syracuse University.
2. SV, CS and RL discussed the role of particle counting in the drinking water industry, regulatory drivers (ESWTR, LTESWTR), and the need to develop a reliable method for the count performance evaluation of online particle counters and to achieve reasonable inter-particle counter agreement within a treatment plant setting and across the drinking water industry.
3. CS mentioned two particle counter calibration standards developed for other industries that can be viewed as reasonable models for developing a test suspension for the drinking water industry. These include the ANSI/NFPA standard for calibrating liquid automatic particle counters for the U.S. Fluid Power Industry (FPI), and the Japanese industrial standard for light scattering automatic particle counters for liquids, developed by the Japanese Standards Association. NIST developed the standard reference material (ISO medium test dust in oil) for the FPI. The selected reference material for the Japanese standard is polystyrene latex spheres (PSLs).
4. Based on the foregoing discussion, the NIST representatives (DS, RF, and RG) obtained a clearer understanding of the need for a standard reference material (SRM) or reference material (RM) for the drinking water industry. As defined by NIST, an SRM is a well-characterized material produced in quantity by NIST that is certified for specific chemical or physical properties and is issued by NIST with a certificate that reports the characterization. An RM is a material whose property values are sufficiently homogenous and stable to allow for the test of repeatability and reproducibility of measurement by an apparatus; RMs are issued by NIST with a report of investigation (instead of a certificate).

Development of a NIST Reference Material for Particle Counter Calibration

5. In principal, NIST is willing to pursue the development of an SRM and/or RM for the drinking water industry, in a collaborative effort with AWWA, USEPA and other interested parties. Assuming agreement by all parties, this collaborative effort should be pursued as a follow-on project to the current AWWARF project. DS mentioned that the needs of the drinking water industry meet the following criteria established by NIST for developing SRMs or RMs for a new industry:
 - Industry must present a legitimate need
 - The SRM or RM must support existing technology
 - There needs to be an Industry/NIST compatibility with regard to the proposed reference material
 - Monetary funds from outside of NIST must be available to support the work; internal funding is usually not adequate
6. The development of an SRM is funded primarily through future sales of the SRM and funding from collaborating industries. Research leading up to production is often partially funded by industrial consortia or other agencies, according to RG. A market assessment is performed by NIST over a 7-10 year supply cycle before reaching a final decision on whether an SRM will be developed. Based on some general statistics provided by SV, RG mentioned that NIST's market assessment for the drinking water industry should be straightforward.
7. RF gave an overview of how an SRM (medium test dust in oil) was developed by NIST for the FPI. The FPI approached NIST to develop an SRM after considerable industry research had been completed over a 10-year period. This research involved use of PSLs and test dust (air cleaner fine test dust, ACFTD) for calibration of particle counters. NIST was asked to develop an SRM for medium test dust in oil, which took about three years to accomplish. The SRM development process can be summarized as follows:
 - Powder Technology, Inc. (PTI) classifies raw dust using a large jet milling device to produce a large batch of medium test dust with a size distribution as measured by Coulter Counter. The production batch produced for PTI was on the order of several tons of dust, 600 pounds of which were made available to NIST.

- For the RM, a commercial laboratory, Laboratory Quality Services International (South Holland, Illinois) subdivides under NIST supervision a portion of the PTI dust batch into 20 gram quantities using a large riffler (147 bottle capacity). The dry dust is stored in sealed bottles and shipped to NIST for QA/QC testing.
 - NIST tests the bottles to verify homogeneity of the spin riffling process and thus the resulting RM
 - The bulk of the dust batch is sent to Fluid Technologies (private company) where the dust is blended with oil. This mixture is divided into bottles and continuously monitored by a particle counter. The sealed bottles are shipped to NIST for characterization by NIST. Bottle-to-bottle homogeneity testing was conducted by NIST and FTI using optical particle counters calibrated to the same ISO 4402 standard.
 - NIST performs scanning electron microscope/image analysis (SEM/IA) of selected bottles to characterize the SRM. The bottles are stored in temperature-controlled warehouse for sale to the FPI. The batch of SRMs are 4-5 years old with no indication of degradation over that period.
8. RF gave RL a draft copy of a paper which summarizes NIST's involvement with FPI in developing a dust-in-oil SRM. The paper includes detailed discussion on the statistics used by NIST for assessing the data. RL will review the statistical information for possible use in the AWWARF project.
 9. RF and DS emphasized that the development of a new SRM for an industry is no simple task and could take several years, although a dust-in-water SRM could build upon the FPI work already completed with dust in oil. A reasonable approach would be to first develop an RM for the drinking water industry that would meet industry objectives of improving inter-particle counter agreement. An SRM could then be developed, if necessary, to support a future regulation dealing with particle counts.

Development of a Dust-in-Water Reference Material

10. RL and CS discussed several options for developing a dust-in-water RM for the drinking water industry. These include the following:

- Use of medium fine test dust, which, from a particle counting perspective, is similar to filtered water particles, based on work completed by RL and others.
 - Packaging exact quantities of dry dust in dissolving pills, which would be dissolved and dispersed in measured volumes of particle-free water in the water treatment plant (WTP) lab to create stock suspensions for routine calibration purposes.
 - Packaging dry dust in capped and sealed plastic or glass containers, which would be shipped to the WTP lab to prepare stock suspensions. Particle-free water would be added to the container and blended with the dry dust. This method was preferred over shipment of dust-in-water containers which would be subject to physical and chemical changes during storage and shipment, likely affecting the characterization of the dust.
 - Use of a master counter, owned and calibrated by NIST, and used to characterize the ISO medium test dust. A size distribution chart would be provided by NIST for each RM shipped to WTP laboratories. The master counter could be NIST's existing HIAC/ Royco counter, a coulter counter, or possibly SEM/IA.
11. RG and RF stated that the ideas discussed under Item 10 were not beyond imagination and that NIST has the in-house expertise to develop and make available such RMs for the drinking water industry.
 12. CS, RL and RF discussed the accuracy of laser particle counters. RF has data which shows that a high-end particle counter (HIAC/Royco) will underestimate the true count of a medium test dust suspension in oil (total particles greater than 2 microns). The SEM/IA reading for the SRM was approximately 27,000/mL whereas the HIAC/Royco reading was much lower. Therefore, it was concluded that particle counter technology is not currently capable of accurately measuring true cumulative counts (SEM-measured) counts greater than 2 microns. However, RF thought that the count agreement between SEM/IA and laser counters could possibly be much improved if the count threshold was increased to 3 or 5 microns.
 13. Given the variability of readings between SEM/IA and laser counters, and the high cost of performing SEM/IA, it may be preferred to use a laser or coulter counter for

developing an RM for the drinking water industry. While less accurate than SEM/IA this would still meet the objective of improving inter-particle counter agreement.

Alternative Reference Materials for Particle Counters

14. CS requested any information from NIST on alternate calibration materials for laser particle counters. RG and RF mentioned the following NIST SRMs and RMs:
 - Glass Beads - 10-60 micron, polydispersed, characterized size distribution, refractive index does not match silica/clay type particles; research underway on lower size range (< 8 microns).
 - Silicon Nitride - 0.2 to 10 microns, polydispersed, size distribution is skewed towards particles less than 2 microns which is not size of interest for filtered water particles or laser counters.
 - Silica Sand - Starting material for glass beads, does not meet low end particle size requirements (>8 microns)
15. CS and RL suggested that the ISO medium test dust appears to be the best available material for the drinking water industry for the following reasons:
 - The size distribution of dust is similar to filtered water particles (1 to 10 micron range)
 - The refractive index and other physical properties of dust should be similar to filtered water particles.
 - Dust-in-water suspensions are reasonably stable and homogenous, based on results of RL experiments.
 - Dust RM will be relatively inexpensive compared to other materials.
 - A significant amount of work has already been done by NIST (and others) on particle counter calibration in the fluid power industry using medium test dust. This information has the potential to be very valuable for particle counter calibration in the drinking water industry.

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Abbreviations

$\mu\text{g/mL}$	Micrograms per milliliter
σ_m	Measured Standard Deviation
μm	Microns
σ_p	Standard Deviation
ACFTD	Air Cleaner Fine Test Dust
ANOVA	Analysis of Variance
ASTM	American Society for Testing Materials
AWWARF	American Water Works Association Research Foundation
COV	Coefficient of Variation
CPE	Count Performance Evaluation
d	Measured Mean Particle Diameter
EC	Estimated Concentration
HDPE	High Density Polyethylene
IPA	Instrument Performance Analysis
MCA	Multi-channel Analyzer
MTD	Medium Test Dust
NA	Not Applicable
ND	Not Disclosed
NFPA	National Fluid Power Association
NIST	National Institute of Standards and Technology
OCWA	Onondaga County Water Authority
PSL	Polystyrene Latex
QA/QC	Quality Assurance / Quality Control
R	Magnitude of Resolution
RM	Reference Material
RO	Reverse Osmosis
SEM	Scanning Electron Microscopy
SRM	Standard Reference Material
USP	United States Pharmacopoeia